

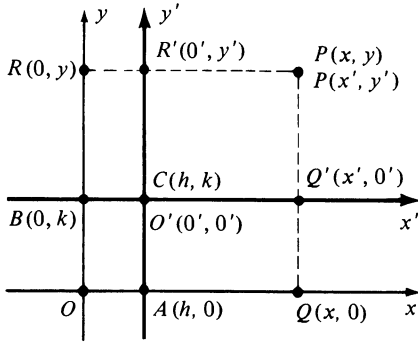


FIGURE 12.6

is illustrated in Figure 12.6 where, for simplicity, we have placed C in the first quadrant. We shall use primes on letters to denote coordinates of points in the $x'y'$ -coordinate system to distinguish them from coordinates with respect to the xy -coordinate system. Thus the point $P(x, y)$ in the xy -system will be denoted by $P(x', y')$ in the $x'y'$ -system. If we label projections of P on the various axes as indicated in Figure 12.6, and let A and B denote projections of C on the x - and y -axes, respectively, then using directed distances we obtain

$$x = \overline{OQ} = \overline{OA} + \overline{AQ} = \overline{OA} + \overline{O'Q'} = h + x'$$

$$y = \overline{OR} = \overline{OB} + \overline{BR} = \overline{OB} + \overline{O'R'} = k + y'.$$

This proves the following result.

Theorem (12.3)

If (x, y) are the coordinates of a point P relative to an xy -coordinate system, and if (x', y') are the coordinates of P relative to an $x'y'$ -coordinate system with origin at the point $C(h, k)$ of the xy -system, then

$$(i) \quad x = x' + h, \quad y = y' + k$$

$$(ii) \quad x' = x - h, \quad y' = y - k.$$

The formulas in Theorem (12.3) enable us to go from either coordinate system to the other. Their major use is to change the form of equations of graphs. To be specific, if, in the xy -plane, a certain collection of points is the graph of an equation in x and y , then to find an equation in x' and y' that has the same graph in the $x'y'$ -plane, we may substitute $x' + h$ for x and $y' + k$ for y in the given equation. Conversely, if a set of points in the $x'y'$ -plane is the graph of an equation in x' and y' , then to find the corresponding equation in x and y we substitute $x - h$ for x' and $y - k$ for y' .

As a simple illustration of the preceding remarks, the equation

$$(x')^2 + (y')^2 = r^2$$

has, for its graph in the $x'y'$ -plane, a circle of radius r with center at the origin O' . Using Theorem (12.3), an equation for this circle in the xy -plane is

$$(x - h)^2 + (y - k)^2 = r^2,$$

which is in agreement with the formula for a circle of radius r with center at $C(h, k)$ in the xy -plane.

As another illustration, we know that

$$(x')^2 = 4py'$$

is an equation of a parabola with a vertex at the origin O' of the $x'y'$ -plane. Using (12.3), we see that

$$(x - h)^2 = 4p(y - k)$$

is an equation of the same parabola in the xy -plane. Since the vertex is $V(h, k)$, comparison with (i) of Theorem (12.2) shows that the focus is $F(h, k + p)$ and directrix is $y = k - p$. Similarly, starting with the equation $(y')^2 = 4px'$, and using Theorem (12.3) gives us $(y - k)^2 = 4p(x - h)$. The next theorem summarizes this discussion.

Theorem (12.4)

The graph of each of the following equations is a parabola with vertex $V(h, k)$ and having the indicated focus and directrix.

- (i) $(x - h)^2 = 4p(y - k)$: focus $F(h, k + p)$, directrix $y = k - p$.
- (ii) $(y - k)^2 = 4p(x - h)$: focus $F(h + p, k)$, directrix $x = h - p$.

In each case the axis of the parabola is parallel to a coordinate axis. The parabola having equation (i) opens upward or downward, whereas the parabola in (ii) opens to the right or left. Typical graphs are sketched in Figure 12.7.

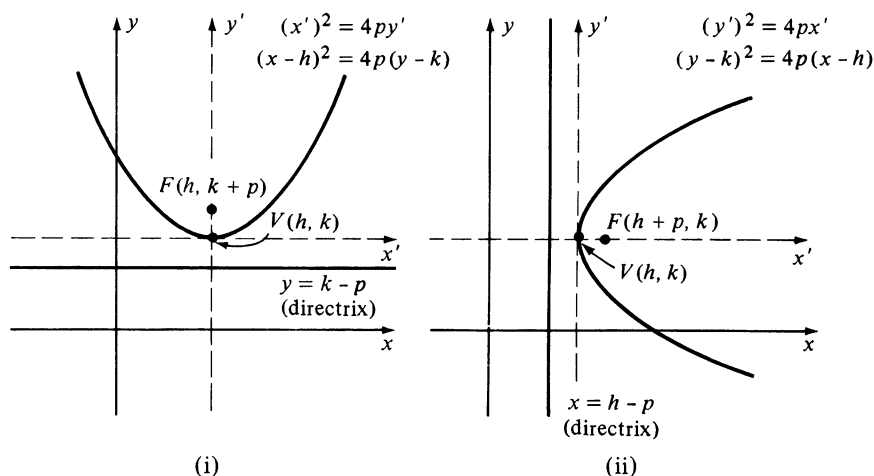


FIGURE 12.7

Squaring the left side of the equation in (i) of Theorem (12.4) and simplifying leads to an equation of the form

$$y = ax^2 + bx + c$$

where a , b , and c are real numbers. Conversely, given such an equation, we may complete the square in x to arrive at the form in (12.4). This technique will be illustrated in Example 3. Consequently, if $a \neq 0$, then the graph of $y = ax^2 + bx + c$ is a parabola with a vertical axis.

Example 3 Discuss and sketch the graph of $y = 2x^2 - 6x + 4$.

Solution The graph is a parabola with vertical axis. To obtain the proper form we begin by writing the equation as

$$y - 4 = 2x^2 - 6x = 2(x^2 - 3x).$$

Next we complete the square for the expression $x^2 - 3x$. Recall that to complete the square for any expression of the form $x^2 + qx$, we add the square of half the coefficient of x , that is, $(q/2)^2$. Thus, for $x^2 - 3x$ we must add $(-\frac{3}{2})^2$, or $\frac{9}{4}$. However, if we add $\frac{9}{4}$ to $x^2 - 3x$ in the equation $y - 4 = 2(x^2 - 3x)$, then because there is a factor 2 outside the parentheses, this amounts to adding $\frac{9}{2}$ to the right side of the equation. Hence we must compensate by adding $\frac{9}{2}$ to the left side. This gives us

$$\begin{aligned} y - 4 + \frac{9}{2} &= 2\left(x^2 - 3x + \frac{9}{4}\right) \\ y + \frac{1}{2} &= 2\left(x - \frac{3}{2}\right)^2 \end{aligned}$$

or equivalently

$$\left(x - \frac{3}{2}\right)^2 = \frac{1}{2}\left(y + \frac{1}{2}\right).$$

The last equation is in form (i) of (12.4) with $h = \frac{3}{2}$, $k = -\frac{1}{2}$, and $4p = \frac{1}{2}$, or $p = \frac{1}{8}$. Hence the vertex of the parabola is $V(\frac{3}{2}, -\frac{1}{2})$. The focus is $F(h, k + p)$, or $F(\frac{3}{2}, -\frac{3}{8})$, and the directrix is

$$y = k - p = -\frac{1}{2} - \frac{1}{8}, \quad \text{or} \quad y = -\frac{5}{8}.$$

As an aid to sketching the graph we note that the y -intercept is 4. To find the x -intercepts we solve $2x^2 - 6x + 4 = 0$ or the equivalent equation $(2x - 2)(x - 2) = 0$, obtaining $x = 2$ and $x = 1$. The graph is sketched in Figure 12.8. ■

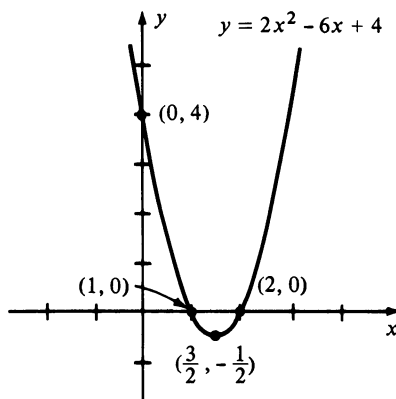


FIGURE 12.8

The equation in (ii) of (12.4) may be written as

$$x = ay^2 + by + c$$

where a , b , and c are real numbers. Conversely, the preceding equation can be expressed in form (12.4) by completing the square in y as illustrated next, in Example 4. Hence, if $a \neq 0$, the graph of $x = ay^2 + by + c$ is a parabola with a horizontal axis.

Example 4 Discuss and sketch the graph of the equation

$$2x = y^2 + 8y + 22.$$

Solution By our previous remarks, the graph is a parabola with a horizontal axis. Writing

$$y^2 + 8y = 2x - 22$$

we complete the square on the left by adding 16 to both sides. This gives us

$$y^2 + 8y + 16 = 2y - 6.$$

The last equation may be written

$$(y + 4)^2 = 2(x - 3)$$

which is in form (ii) of (12.4) with $h = 3$, $k = -4$, and $4p = 2$, or $p = \frac{1}{2}$. Hence the vertex is $V(3, -4)$. Since $p = \frac{1}{2} > 0$, the parabola opens to the right with focus at $F(h + p, k)$, that is, $F(\frac{7}{2}, -4)$. The equation of the directrix is $x = h - p$, or $x = \frac{5}{2}$. The parabola is sketched in Figure 12.9. ■

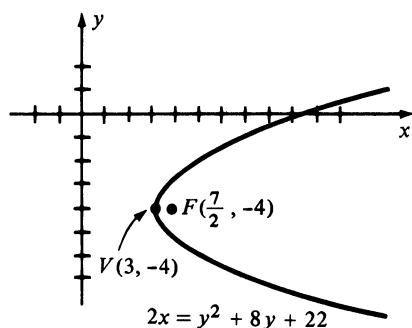


FIGURE 12.9

Example 5 Find an equation of the parabola with vertex $(4, -1)$, with axis parallel to the y -axis, and which passes through the origin.

Solution By (12.4) the equation is of the form

$$(x - 4)^2 = 4p(y + 1).$$

If the origin is on the parabola, then $(0, 0)$ is a solution of this equation, and hence $(0 - 4)^2 = 4p(0 + 1)$. Consequently $16 = 4p$ and $p = 4$. The desired equation is, therefore,

$$(x - 4)^2 = 16(y + 1).$$

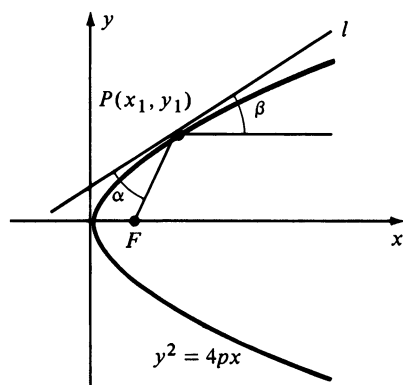


FIGURE 12.10

We shall conclude this section by pointing out an important property of parabolas. Suppose l is the tangent line at a point $P(x_1, y_1)$ on the graph of $y^2 = 4px$, and let F be the focus. As in Figure 12.10, let α denote the angle between l and the line segment FP , and let β denote the angle between l and the indicated horizontal half-line with endpoint P . In Exercise 34, you are asked to prove that $\alpha = \beta$. This property has many practical applications. For example, the shape of the mirror in a searchlight is obtained by revolving a parabola about its axis. If a light source is placed at F , then by a law of physics a beam of light will be reflected along a line parallel to the axis. The same principle is employed in the construction of mirrors for telescopes or solar ovens, where a beam of light coming toward the parabolic mirror, and parallel to the axis, will be reflected into the focus. Antennas for radar systems and field microphones used at football games also make use of this property.

12.2 Exercises

In Exercises 1–16 find the vertex, focus, and directrix of the parabola with the given equation and sketch the graph.

- 1 $x^2 = -12y$
- 2 $y^2 = (1/2)x$
- 3 $2y^2 = -3x$
- 4 $x^2 = -3y$
- 5 $8x^2 = y$
- 6 $y^2 = -100x$
- 7 $y^2 - 12 = 12x$
- 8 $y = 40x - 97 - 4x^2$
- 9 $y = x^2 - 4x + 2$
- 10 $y = 8x^2 + 16x + 10$
- 11 $y^2 - 4y - 2x - 4 = 0$
- 12 $y^2 + 14y + 4x + 45 = 0$
- 13 $4x^2 + 40x + y + 106 = 0$
- 14 $y^2 - 20y + 100 = 6x$
- 15 $x^2 + 20y = 10$
- 16 $4x^2 + 4x + 4y + 1 = 0$

- 17 Describe a method for using a derivative to locate the vertex of a parabola having the equation $y = ax^2 + bx + c$. Using this technique to find the vertices in Exercises 8–10. Describe a similar method for $x = ay^2 + by + c$.
- 18 Describe how a second derivative may be used to determine whether the parabola $y = ax^2 + bx + c$ opens upward or downward. Illustrate this technique with Exercises 8–10.

In Exercises 19–24 find an equation for the parabola that satisfies the given conditions.

- 19 Focus $(2, 0)$, directrix $x = -2$
- 20 Focus $(0, -4)$, directrix $y = 4$
- 21 Focus $(6, 4)$, directrix $y = -2$
- 22 Focus $(-3, -2)$, directrix $y = 1$
- 23 Vertex at the origin, symmetric to the y -axis, and passing through the point $A(2, -3)$
- 24 Vertex $V(-3, 5)$, axis parallel to the x -axis, and passing through $A(5, 9)$

- 25 A searchlight reflector is designed so that a cross section through its axis is a parabola and the light source is at the focus. Find the focus if the reflector is 3 ft across at the opening and 1 ft deep.
- 26 One section of a suspension bridge has its weight uniformly distributed between twin towers which are 400 ft apart and which rise 90 ft above the horizontal roadway (see figure). A cable strung between the tops of the towers has the shape of a parabola and its center point is 10 ft above the roadway. Suppose coordinate axes are introduced as shown in the figure.
 - (a) Find an equation for the parabola.
 - (b) Set up an integral that gives the length of the cable.
 - (c) If nine equispaced vertical cables are used to support the bridge (see figure), find the total length of these supports.

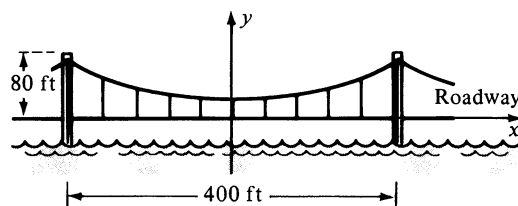


FIGURE FOR EXERCISE 26

- 27 Find an equation of the parabola with a vertical axis that passes through the points $A(2, 3)$, $B(-1, 6)$, and $C(1, 0)$.
- 28 Prove that the point on a parabola that is closest to the focus is the vertex.
- 29 Prove that the equation of the tangent line to the parabola $x^2 = 4py$ at the point $P(x_1, y_1)$ is $x_1x - 2py - 2py_1 = 0$.
- 30 Prove that the parabola with equation $y = ax^2 + bx + c$ has no points of inflection.

In Exercises 31 and 32 let R denote the region in the xy -plane that is bounded by the given equation.

- (a) Find the area of R .
- (b) If R is revolved about the y -axis, find the volume of the resulting solid.
- (c) If R is revolved about the x -axis, find the volume of the resulting solid.
- 31 The parabola $x^2 = 4y$ and the line l through the focus that is perpendicular to the axis of the parabola.

- 32 The graphs of $y^2 = 2x - 6$ and $x = 5$.
- 33 A vertical gate in a dam has the shape of the parabolic region of Exercise 31. If the line l lies along the surface of the water, find the force exerted on the gate by the water.
- 34 Prove that $\alpha = \beta$ in Figure 12.10.
- 35 A line segment that passes through the focus of a parabola and has its endpoints on the parabola is called a **focal chord**. If AB is a focal chord, prove that the tangent lines at A and B are perpendicular.
- 36 If AB is a focal chord of a parabola (see Exercise 35), prove that the tangent lines at A and B intersect on the directrix.
- 37 Prove that there is exactly one line of a given slope m which is tangent to the parabola $x^2 = 4py$, and that its equation is $y = mx - pm^2$.
- 38 Prove that there is exactly one line of a given slope $m \neq 0$ which is tangent to the parabola $y^2 = 4px$, and that its equation is $y = mx + (p/m)$.
- 39 Prove that any tangent line to a parabola at a point P intersects the axis of the parabola at a point Q such that the midpoint of the line segment PQ is on the line tangent to the parabola at its vertex.
- 40 Suppose a tangent line to a parabola at a point P intersects the directrix at a point Q . If F is the focus, prove that angle PFQ is a right angle.

12.3 Ellipses

An ellipse may be defined as follows.

Definition (12.5)

An **ellipse** is the set of all points in a plane, the sum of whose distances from two fixed points in the plane (the **foci**) is constant.

It is known that the orbits of planets in the solar system are elliptical, with the sun at one of the foci. This is only one of many important applications of ellipses.

There is an easy way to construct an ellipse on paper. We may begin by inserting two thumbtacks in the paper at points labeled F and F' and fastening the ends of a piece of string to the thumbtacks. If the string is now looped around a pencil and drawn taut at point P , as in Figure 12.11, then moving the pencil and at the same time keeping the string taut, the sum of the distances $d(F, P)$ and $d(F', P)$ is the length of the string, and hence is constant. The pencil will, therefore, trace out an ellipse with foci at F and F' . By varying the positions of F and F' but keeping the length of string fixed, the shape of the ellipse can be made to change considerably. If F and F' are far apart, in the sense that $d(F, F')$ is almost the same as the length of the string, then the ellipse is quite flat. On the other hand, if $d(F, F')$ is close to zero, the ellipse is almost circular. Indeed, if $F = F'$, a circle is obtained.

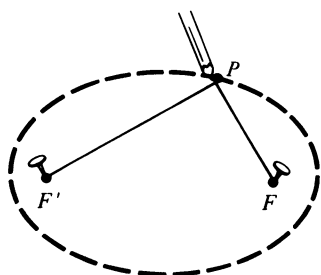


FIGURE 12.11

By introducing suitable coordinate systems we may derive simple equations for ellipses. Let us choose the x -axis as the line through the two foci F and F' , with the origin at the midpoint of the segment $F'F$. This point is called the **center** of the ellipse. If F has coordinates $(c, 0)$, where $c > 0$,

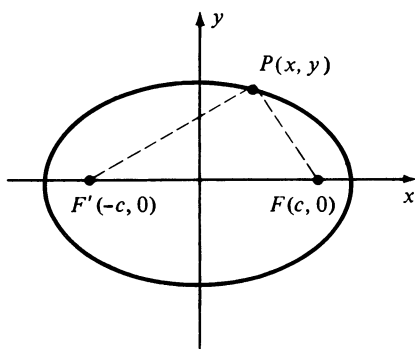


FIGURE 12.12

then, as shown in Figure 12.12, F' has coordinates $(-c, 0)$ and hence the distance between F and F' is $2c$. Let the constant sum of the distances of P from F and F' be denoted by $2a$, where in order to get points that are not on the x -axis we must have $2a > 2c$, that is, $a > c$. (Why?) By Definition (12.5), $P(x, y)$ is on the ellipse if and only if

$$d(P, F) + d(P, F') = 2a$$

or, by the Distance Formula,

$$\sqrt{(x - c)^2 + (y - 0)^2} + \sqrt{(x + c)^2 + (y - 0)^2} = 2a.$$

Writing the preceding equation as

$$\sqrt{(x - c)^2 + y^2} = 2a - \sqrt{(x + c)^2 + y^2}$$

and squaring both sides, we obtain

$$x^2 - 2cx + c^2 + y^2 = 4a^2 - 4a\sqrt{(x + c)^2 + y^2} + x^2 + 2cx + c^2 + y^2$$

which simplifies to

$$a\sqrt{(x + c)^2 + y^2} = a^2 + cx.$$

Squaring both sides gives us

$$a^2(x^2 + 2cx + c^2 + y^2) = a^4 + 2a^2cx + c^2x^2$$

which may be written in the form

$$x^2(a^2 - c^2) + a^2y^2 = a^2(a^2 - c^2).$$

Dividing both sides by $a^2(a^2 - c^2)$ leads to

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1.$$

For convenience, we let

$$b^2 = a^2 - c^2 \quad \text{where } b > 0$$

in the preceding equation, obtaining

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Since $c > 0$ and $b^2 = a^2 - c^2$ it follows that $a^2 > b^2$ and hence $a > b$.

We have shown that the coordinates of every point (x, y) on the ellipse in Figure 12.12 satisfy the equation $x^2/a^2 + y^2/b^2 = 1$. Conversely, if (x, y) is a solution of this equation, then by reversing the preceding steps we see that the point (x, y) is on the ellipse.

The x -intercepts may be found by setting $y = 0$. Doing so gives us $x^2/a^2 = 1$, or $x^2 = a^2$, and consequently the x -intercepts are a and $-a$. The corresponding points $V(a, 0)$ and $V'(-a, 0)$ on the graph are called the

vertices of the ellipse, and the line segment $V'V$ is referred to as the **major axis**. Similarly, letting $x = 0$ in the equation of the ellipse, we obtain $y^2/b^2 = 1$, or $y^2 = b^2$. Hence the y -intercepts are b and $-b$. The segment from $M'(0, -b)$ to $M(0, b)$ is called the **minor axis** of the ellipse. Note that the major axis is longer than the minor axis, since $a > b$.

Applying tests for symmetry we see that the ellipse is symmetric to both the x -axis and the y -axis. It is also symmetric with respect to the origin since substitution of $-x$ for x and $-y$ for y does not change the equation.

The preceding discussion may be summarized as follows.

Theorem (12.6)

The graph of the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where $a^2 > b^2$ is an ellipse with vertices $(\pm a, 0)$. The endpoints of the minor axis are $(0, \pm b)$. The foci are $(\pm c, 0)$, where $c^2 = a^2 - b^2$.

Example 1 Discuss and sketch the graph of the equation

$$4x^2 + 18y^2 = 36.$$

Solution To obtain the form in Theorem (12.6) we divide both sides of the given equation by 36 and simplify. This leads to

$$\frac{x^2}{9} + \frac{y^2}{2} = 1$$

which is in the proper form with $a^2 = 9$ and $b^2 = 2$. Thus $a = 3$, $b = \sqrt{2}$, and hence the endpoints of the major axis are $(\pm 3, 0)$ and the endpoints of the minor axis are $(0, \pm \sqrt{2})$. Since

$$c^2 = a^2 - b^2 = 9 - 2 = 7, \quad \text{or} \quad c = \sqrt{7},$$

the foci are $(\pm \sqrt{7}, 0)$. The graph is sketched in Figure 12.13. ■

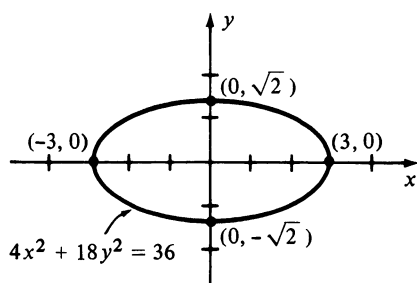


FIGURE 12.13

Example 2 Find an equation of the ellipse with vertices $(\pm 4, 0)$ and foci $(\pm 2, 0)$.

Solution Using the notation of Theorem (12.6), $a = 4$ and $c = 2$. Since $c^2 = a^2 - b^2$, we see that $b^2 = a^2 - c^2 = 16 - 4 = 12$. This gives us the equation

$$\frac{x^2}{16} + \frac{y^2}{12} = 1.$$

Multiplying both sides by 48 leads to $3x^2 + 4y^2 = 48$. ■

It is sometimes convenient to choose the major axis of the ellipse along the y -axis. If the foci are $(0, \pm c)$, then by the same type of argument used to prove Theorem (12.6) we obtain the following.

Theorem (12.7)

The graph of the equation

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

where $a^2 > b^2$ is an ellipse with vertices $(0, \pm a)$. The endpoints of the minor axis are $(\pm b, 0)$. The foci are $(0, \pm c)$, where $c^2 = a^2 - b^2$.

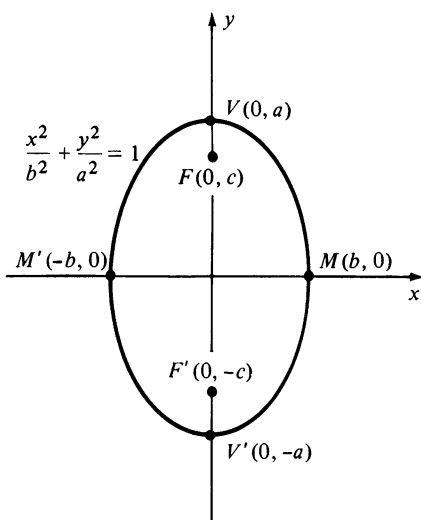


FIGURE 12.14

A typical graph is sketched in Figure 12.14.

The preceding discussion shows that an equation of an ellipse with center at the origin and foci on a coordinate axis can always be written in the form

$$\frac{x^2}{p} + \frac{y^2}{q} = 1 \quad \text{or} \quad qx^2 + py^2 = pq$$

where p and q are positive. If $p > q$ the major axis lies on the x -axis, whereas if $q > p$ the major axis is on the y -axis. It is unnecessary to memorize these facts, since in any given problem the major axis can be determined by examining the x - and y -intercepts.

Example 3 Sketch the graph of the equation $9x^2 + 4y^2 = 25$.

Solution The graph is an ellipse with center at the origin and foci on one of the coordinate axes. To find the x -intercepts, we let $y = 0$, obtaining $9x^2 = 25$, or $x = \pm \frac{5}{3}$. Similarly, to find the y -intercepts, we let $x = 0$, obtaining $4y^2 = 25$, or $y = \pm \frac{5}{2}$. This enables us to sketch the ellipse (see Figure 12.15). Since $\frac{5}{3} < \frac{5}{2}$, the major axis is on the y -axis. ■

Example 4 Find the area of the region bounded by an ellipse whose major and minor axes have lengths $2a$ and $2b$, respectively.

Solution By (12.6), an equation for the ellipse is $x^2/a^2 + y^2/b^2 = 1$. Solving for y gives us

$$y = (\pm b/a)\sqrt{a^2 - x^2}.$$

The graph of the ellipse has the general shape shown in Figure 12.12 and hence, by symmetry, it is sufficient to find the area of the region in the first quadrant and multiply the result by 4. Using 5.12,

$$A = 4 \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx.$$

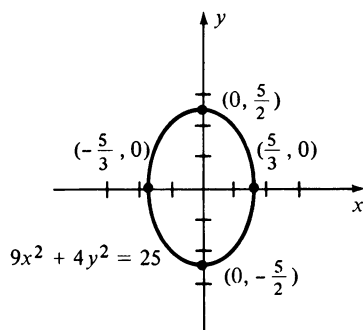


FIGURE 12.15

If we make the trigonometric substitution $x = a \sin \theta$, then

$$\sqrt{a^2 - x^2} = a \cos \theta \quad \text{and} \quad dx = a \cos \theta \, d\theta.$$

Moreover, the values of θ that correspond to $x = 0$ and $x = a$ are $\theta = 0$ and $\theta = \pi/2$, respectively. Consequently,

$$\begin{aligned} A &= 4 \frac{b}{a} \int_0^{\pi/2} a^2 \cos^2 \theta \, d\theta = 4ab \int_0^{\pi/2} \frac{1 + \cos 2\theta}{2} \, d\theta \\ &= 2ab \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/2} = 2ab[\pi/2] = \pi ab. \end{aligned}$$

As a special case, if $b = a$ the ellipse is a circle and $A = \pi a^2$. ■

By using the translation of axes formulas stated in Theorem (12.3) we can extend our work to an ellipse with center at any point $C(h, k)$ in the xy -plane. For example, since the graph of

$$\frac{(x')^2}{a^2} + \frac{(y')^2}{b^2} = 1$$

is an ellipse with center at O' in an $x'y'$ -plane (see Figure 12.16), then according to (12.3), its equation relative to the xy -coordinate system is

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1.$$

Squaring the indicated terms in the last equation and simplifying gives us an equation of the form

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

where the coefficients are real numbers and A and C are both positive. Conversely, if we start with such an equation, then by completing squares we can obtain a form that displays the center of the ellipse and the lengths of the major and minor axes. This technique is illustrated in the next example.

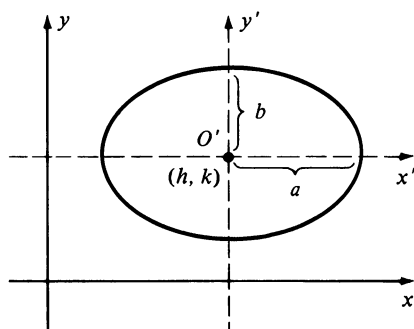


FIGURE 12.16

Example 5 Discuss and sketch the graph of the equation

$$16x^2 + 9y^2 + 64x - 18y - 71 = 0$$

Solution We begin by writing the equation in the form

$$16(x^2 + 4x) + 9(y^2 - 2y) = 71.$$

Next, we complete the squares for the expressions within parentheses, obtaining

$$16(x^2 + 4x + 4) + 9(y^2 - 2y + 1) = 71 + 64 + 9.$$

Note that by adding 4 to the expression within the first parentheses we have added 64 to the left side of the equation and hence must compensate by adding 64 to the right side. Similarly, by adding 1 to the expression within the second parentheses, 9 is added to the left side and consequently 9 must also be added to the right side. The last equation may be written

$$16(x + 2)^2 + 9(y - 1)^2 = 144.$$

Dividing by 144 we obtain

$$\frac{(x + 2)^2}{9} + \frac{(y - 1)^2}{16} = 1$$

which is of the form

$$\frac{(x')^2}{9} + \frac{(y')^2}{16} = 1$$

with $x' = x + 2$ and $y' = y - 1$. This corresponds to letting $h = -2$ and $k = 1$ in the translation of axes formulas (12.3). Since the graph of $(x')^2/9 + (y')^2/16 = 1$ is an ellipse with center at the origin O' in the $x'y'$ -plane, it follows that the given equation is an ellipse with center $C(-2, 1)$ in the xy -plane and with axes parallel to the coordinate axes. The graph is sketched in Figure 12.17. ■

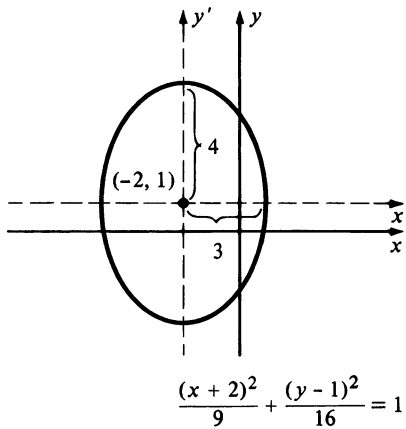


FIGURE 12.17

12.3 Exercises

In Exercises 1–14 sketch the graph of the equation and give coordinates of the vertices and foci.

- 1 $\frac{x^2}{9} + \frac{y^2}{4} = 1$
- 2 $\frac{x^2}{25} + \frac{y^2}{16} = 1$
- 3 $4x^2 + y^2 = 16$
- 4 $y^2 + 9x^2 = 9$
- 5 $5x^2 + 2y^2 = 10$
- 6 $\frac{1}{2}x^2 + 2y^2 = 8$
- 7 $4x^2 + 25y^2 = 1$
- 8 $10y^2 + x^2 = 5$
- 9 $4x^2 + 9y^2 - 32x - 36y + 64 = 0$
- 10 $x^2 + 2y^2 + 2x - 20y + 43 = 0$
- 11 $9x^2 + 16y^2 + 54x - 32y - 47 = 0$
- 12 $4x^2 + 9y^2 + 24x + 18y + 9 = 0$

13 $25x^2 + 4y^2 - 250x - 16y + 541 = 0$

14 $4x^2 + y^2 = 2y$

In Exercises 15–20 find an equation for the ellipse satisfying the given conditions.

- 15 Vertices $V(\pm 8, 0)$, foci $F(\pm 5, 0)$
- 16 Vertices $V(0, \pm 7)$, foci $F(0, \pm 2)$
- 17 Vertices $V(0, \pm 5)$, length of minor axis 3
- 18 Foci $F(\pm 3, 0)$, length of minor axis 2
- 19 Vertices $V(0, \pm 6)$, passing through $(3, 2)$
- 20 Center at the origin, symmetric with respect to both axes, passing through $A(2, 3)$ and $B(6, 1)$

In Exercises 21 and 22 find the points of intersection of the graphs of the given equations. Sketch both graphs on the same coordinate axes, showing points of intersection.

- 21 $\begin{cases} x^2 + 4y^2 = 20 \\ x^2 + 2y = 6 \end{cases}$
- 22 $\begin{cases} x^2 + 4y^2 = 36 \\ x^2 + y^2 = 12 \end{cases}$

- 23 An arch of a bridge is semi-elliptical with major axis horizontal. The base of the arch is 30 ft across and the highest part of the arch is 10 ft above the horizontal roadway. Find the height of the arch 6 ft from the center of the base.
- 24 The **eccentricity** of an ellipse is defined as the ratio $(\sqrt{a^2 - b^2})/a$. If a is fixed and b varies, describe the general shape of the ellipse when the eccentricity is close to 1 and when it is close to zero.
- 25 Find an equation of the tangent line to the ellipse $5x^2 + 4y^2 = 56$ at the point $P(-2, 3)$.
- 26 Prove that an equation of the tangent line to the ellipse $x^2/a^2 + y^2/b^2 = 1$ at the point $P(x_1, y_1)$ is
- $$xx_1/a^2 + yy_1/b^2 = 1.$$
- 27 If tangent lines to the ellipse $9x^2 + 4y^2 = 36$ intersect the y -axis at the point $(0, 6)$, find the points of tangency.
- 28 If tangent lines to the ellipse $b^2x^2 + a^2y^2 = a^2b^2$ intersect the y -axis at the point $(0, d)$, where $d > b$, find the points of tangency.
- 29 Find the volume of the solid obtained by revolving the region bounded by the ellipse $b^2x^2 + a^2y^2 = a^2b^2$ about the x -axis.
- 30 Find the volume of the solid obtained by revolving the region bounded by the ellipse $b^2x^2 + a^2y^2 = a^2b^2$ about the y -axis.
- 31 The base of a solid is a plane region bounded by an ellipse with major and minor axes of lengths 16 and 9, respectively. Find the volume of the solid if every cross section by a plane perpendicular to the major axis is (a) a square; (b) an equilateral triangle.
- 32 Generalize Exercise 31 to the case where the major and minor axes of the ellipse have lengths $2a$ and $2b$, respectively.
- 33 Find the dimensions of the rectangle of maximum area that can be inscribed in an ellipse of semi-axes a and b , if two sides of the rectangle are parallel to the major axis.
- 34 A cylindrical tank whose cross sections are elliptical with axes of lengths 6 ft and 4 ft, respectively, is lying on its side. If the tank is half full of water, find the force exerted by the water on one end of the tank.

- 35 Let l denote the tangent line at the point P on an ellipse with foci F' and F , as illustrated in the figure. If α is the angle between $F'P$ and l , and if β is the angle between FP and l , prove that $\alpha = \beta$. (This is analogous to the reflective property of the parabola illustrated in Figure 12.10.)

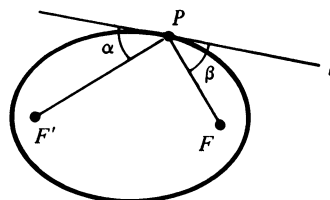


FIGURE FOR EXERCISE 35

- 36 The base of a right elliptic cone has major and minor axes of lengths $2a$ and $2b$, respectively. Find the volume if the altitude is h .
- 37 Find the centroid of the semi-elliptical region bounded by the graph of $y = (b/a)\sqrt{a^2 - x^2}$ and the x -axis.
- 38 The region bounded by the upper half of the ellipse $b^2x^2 + a^2y^2 = a^2b^2$ and the x -axis is revolved about the y -axis. Find the centroid of the resulting solid.
- 39 A line segment of length $a + b$ moves with its endpoints A and B attached to the coordinate axes, as illustrated in the figure. Prove that if $a \neq b$, then the point P traces an ellipse.

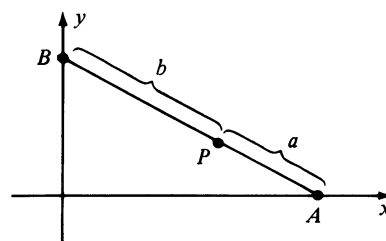


FIGURE FOR EXERCISE 39

- 40 Consider the ellipse $px^2 + qy^2 = pq$, where $p > 0$ and $q > 0$. Prove that if m is any real number, there are exactly two lines of slope m that are tangent to the ellipse, and that their equations are $y = mx \pm \sqrt{p + qm^2}$.

12.4 Hyperbolas

The definition of a hyperbola is similar to that of an ellipse. The only change is that instead of using the *sum* of distances from two fixed points we use the *difference*.

Definition (12.8)

A **hyperbola** is the set of all points in a plane, the difference of whose distances from two fixed points in the plane (the **foci**) is a positive constant.

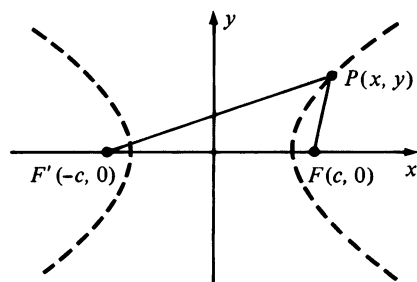


FIGURE 12.18

To find a simple equation for a hyperbola, we choose a coordinate system with foci at $F(c, 0)$ and $F'(-c, 0)$, and denote the (constant) distance by $2a$. Referring to Figure 12.18, we see that a point $P(x, y)$ is on the hyperbola if and only if either one of the following is true:

$$d(P, F') - d(P, F) = 2a$$

$$d(P, F) - d(P, F') = 2a.$$

For hyperbolas (unlike ellipses) we need $a < c$ in order to obtain points on the hyperbola that are not on the x -axis, for if P is such a point, then from Figure 12.18 we see that

$$d(P, F) < d(F', F) + d(P, F')$$

because the length of one side of a triangle is always less than the sum of the lengths of the other two sides. Similarly,

$$d(P, F') < d(F', F) + d(P, F).$$

Equivalent forms for the previous two inequalities are

$$d(P, F) - d(P, F') < d(F', F)$$

$$d(P, F') - d(P, F) < d(F', F).$$

Since the differences on the left both equal $2a$, and since $d(F', F) = 2c$, the last two inequalities imply that $2a < 2c$, or $a < c$.

The equations $d(P, F') - d(P, F) = 2a$ and $d(P, F) - d(P, F') = 2a$ may be replaced by the single equation

$$|d(P, F) - d(P, F')| = 2a.$$

It then follows from the Distance Formula that an equation of the hyperbola is given by

$$|\sqrt{(x-c)^2 + (y-0)^2} - \sqrt{(x+c)^2 + (y-0)^2}| = 2a.$$

Employing the type of simplification procedure used to derive an equation for an ellipse, we arrive at the equivalent equation

$$\frac{x^2}{a^2} - \frac{y^2}{c^2 - a^2} = 1.$$

For convenience, let

$$b^2 = c^2 - a^2 \quad \text{where } b > 0$$

in the preceding equation, obtaining

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

We have shown that the coordinates of every point (x, y) on the hyperbola in Figure 12.18 satisfy the last equation. Conversely, if (x, y) is a solution of that equation, then by reversing the preceding steps we see that the point (x, y) is on the hyperbola.

By the Tests for Symmetry this hyperbola is symmetric with respect to both axes and the origin. The x -intercepts are $\pm a$. The corresponding points $V(a, 0)$ and $V'(-a, 0)$ are called the **vertices**, and the line segment VV' is known as the **transverse axis** of the hyperbola. There are no y -intercepts, since the equation $-y^2/b^2 = 1$ has no solutions.

The preceding discussion may be summarized as follows.

Theorem (12.9)

The graph of the equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

is a hyperbola with vertices $(\pm a, 0)$. The foci are $(\pm c, 0)$, where $c^2 = a^2 + b^2$.

If the equation in Theorem (12.9) is solved for y , we obtain

$$y = \pm \frac{b}{a} \sqrt{x^2 - a^2}.$$

Hence there are no points (x, y) on the graph if $x^2 - a^2 < 0$, that is, if $-a < x < a$. However, there *are* points $P(x, y)$ on the graph if $x \geq a$ or $x \leq -a$.

The line $y = (b/a)x$ is an asymptote for the hyperbola in the sense that the distance $d(x)$ between the point $P(x, y)$ on the hyperbola and the corresponding point $P'(x, y_1)$ on the line approaches zero as x increases without bound. To prove this we note that if $x > 0$, then

$$d(x) = \frac{b}{a}x - \frac{b}{a}\sqrt{x^2 - a^2} = \frac{b}{a}(x - \sqrt{x^2 - a^2}).$$

Since

$$x - \sqrt{x^2 - a^2} = \frac{a^2}{x + \sqrt{x^2 - a^2}}$$

we have

$$\lim_{x \rightarrow \infty} d(x) = \lim_{x \rightarrow \infty} \frac{b}{a} \frac{a^2}{x + \sqrt{x^2 - a^2}} = 0.$$

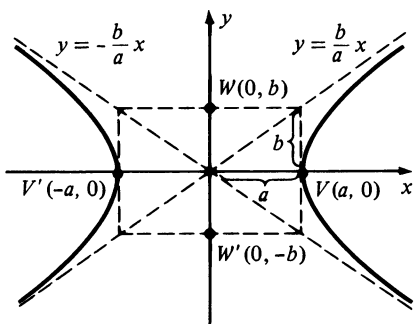


FIGURE 12.19

Similarly, $d(x)$ has the limit zero as x becomes negatively infinite. It can also be shown that the line $y = (-b/a)x$ is an asymptote for the hyperbola in Theorem (12.9).

The asymptotes serve as excellent guides for sketching the graph. A convenient way to sketch the asymptotes is first to plot the vertices $V(a, 0)$, $V'(-a, 0)$ and the points $W(0, b)$, $W'(0, -b)$ (see Figure 12.19). The line segment $W'W$ of length $2b$ is called the **conjugate axis** of the hyperbola. If horizontal and vertical lines are drawn through the endpoints of the conjugate and transverse axes, respectively, then the diagonals of the resulting rectangle have slopes b/a and $-b/a$. Hence, by extending these diagonals we obtain lines with equations $y = (\pm b/a)x$. The hyperbola is then sketched as in Figure 12.19, using the asymptotes as guides. The two curves that make up the hyperbola are called the **branches** of the hyperbola.

Example 1 Discuss and sketch the graph of the equation $9x^2 - 4y^2 = 36$.

Solution Dividing both sides by 36, we have

$$\frac{x^2}{4} - \frac{y^2}{9} = 1,$$

which is in the form of Theorem (12.9) with $a^2 = 4$ and $b^2 = 9$. Hence $a = 2$ and $b = 3$. The vertices $(\pm 2, 0)$ and the endpoints $(0, \pm 3)$ of the conjugate axis determine a rectangle whose diagonals (extended) give us the asymptotes. The graph of the equation is sketched in Figure 12.20. The equations of the asymptotes, $y = \pm \frac{3}{2}x$, can be found by referring to the graph or to the equations $y = (\pm b/a)x$. From (12.9) we have $c^2 = a^2 + b^2 = 4 + 9 = 13$, and consequently the foci are $(\pm\sqrt{13}, 0)$. ■

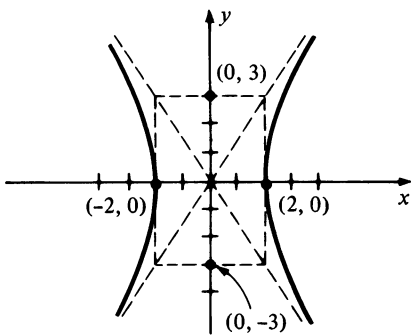


FIGURE 12.20

The preceding example indicates that for hyperbolas it is not always true that $a < b$, as was the case for ellipses. Indeed, we may have $a < b$, $a > b$, or $a = b$.

Example 2 Find an equation, the foci, and the asymptotes of a hyperbola that has vertices $(\pm 3, 0)$ and passes through the point $P(5, 2)$.

Solution Substituting $a = 3$ in Theorem (12.9), we obtain the equation

$$\frac{x^2}{9} - \frac{y^2}{b^2} = 1.$$

If $(5, 2)$ is a solution of this equation, then

$$\frac{25}{9} - \frac{4}{b^2} = 1.$$

This gives us $b^2 = \frac{9}{4}$ and hence the desired equation is

$$\frac{x^2}{9} - \frac{4y^2}{9} = 1,$$

or equivalently, $x^2 - 4y^2 = 9$.

From (12.9), $c^2 = a^2 + b^2 = 9 + \frac{9}{4} = \frac{45}{4}$ and, therefore, the foci are $(\pm\frac{3}{2}\sqrt{5}, 0)$. Substituting for b and a in $y = \pm(b/a)x$ and simplifying, we obtain equations $y = \pm\frac{1}{2}x$ for the asymptotes. ■

If the foci of a hyperbola are the points $(0, \pm c)$ on the y -axis, then by the same type of argument used to prove Theorem (12.9) we obtain the following.

Theorem (12.10)

The graph of the equation

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

is a hyperbola with vertices $(0, \pm a)$. The foci are $(0, \pm c)$, where $c^2 = a^2 + b^2$.

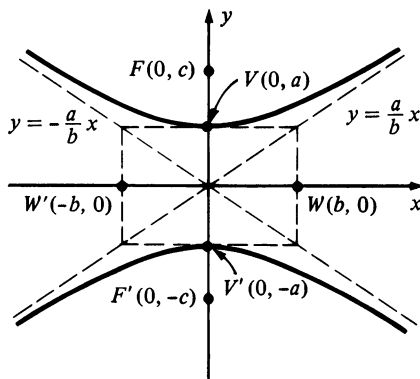


FIGURE 12.21

In Theorem (12.10) the endpoints of the conjugate axis are $W(b, 0)$ and $W'(-b, 0)$. The asymptotes are found, as before, by using the diagonals of the rectangle determined by these points, the vertices, and lines parallel to the coordinate axes. The graph is sketched in Figure 12.21. The equations of the asymptotes are $y = (\pm a/b)x$. Note the difference between these equations and the equations $y = (\pm b/a)x$ for the asymptotes of the hyperbola given by (12.9).

Example 3 Discuss and sketch the graph of the equation $4y^2 - 2x^2 = 1$.

Solution The form in Theorem (12.10) may be obtained by writing the equation as

$$\frac{y^2}{\frac{1}{4}} - \frac{x^2}{\frac{1}{2}} = 1.$$

Thus $a^2 = \frac{1}{4}$, $b^2 = \frac{1}{2}$, and $c^2 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$. Consequently $a = \frac{1}{2}$, $b = \sqrt{2}/2$, and $c = \sqrt{3}/2$. The vertices are $(0, \pm\frac{1}{2})$ and the foci are $(0, \pm\sqrt{3}/2)$. The graph has the general appearance of the graph shown in Figure 12.21. ■

As was the case for ellipses, we may use translations of axes to generalize our work. The following example illustrates this technique.

Example 4 Discuss and sketch the graph of the equation

$$9x^2 - 4y^2 - 54x - 16y + 29 = 0.$$

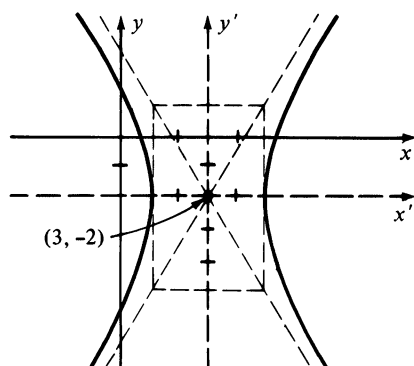


FIGURE 12.22

Solution We arrange our work as follows:

$$\begin{aligned}
 9(x^2 - 6x) - 4(y^2 + 4y) &= -29 \\
 9(x^2 - 6x + 9) - 4(y^2 + 4y + 4) &= -29 + 81 - 16 \\
 9(x - 3)^2 - 4(y + 2)^2 &= 36 \\
 \frac{(x - 3)^2}{4} - \frac{(y + 2)^2}{9} &= 1.
 \end{aligned}$$

which is of the form

$$\frac{(x')^2}{4} - \frac{(y')^2}{9} = 1$$

with $x' = x - 3$ and $y' = y + 2$. By translating the x - and y -axes to the new origin $C(3, -2)$ we obtain the sketch shown in Figure 12.22. ■

The results of the last three sections indicate that the graph of every equation of the form

$$Ax^2 + Cy^2 + Dx + Ey + F = 0$$

is a conic, except for certain degenerate cases in which points, lines, or no graphs are obtained. Although we have only considered special examples, our methods are perfectly general. If A and C are equal and not zero, then the graph, when it exists, is a circle or, in exceptional cases, a point. If A and C are unequal but have the same sign, then by completing squares and properly translating axes we obtain an equation whose graph, when it exists, is an ellipse (or a point). If A and C have opposite signs, an equation of a hyperbola is obtained, or possibly, in the degenerate case, two intersecting straight lines. Finally, if either A or C (but not both) is zero, the graph is a parabola or, in certain cases, a pair of parallel straight lines.

12.4 Exercises

In Exercises 1–18 sketch the graph of the equation, find the coordinates of the vertices and foci, and write equations for the asymptotes.

1 $\frac{x^2}{9} - \frac{y^2}{4} = 1$

2 $\frac{y^2}{49} - \frac{x^2}{16} = 1$

3 $\frac{y^2}{9} - \frac{x^2}{4} = 1$

4 $\frac{x^2}{49} - \frac{y^2}{16} = 1$

5 $y^2 - 4x^2 = 16$

6 $x^2 - 2y^2 = 8$

7 $x^2 - y^2 = 1$

8 $y^2 - 16x^2 = 1$

9 $x^2 - 5y^2 = 25$

10 $4y^2 - 4x^2 = 1$

11 $3x^2 - y^2 = -3$

12 $16x^2 - 36y^2 = 1$

13 $25x^2 - 16y^2 + 250x + 32y + 109 = 0$

14 $y^2 - 4x^2 - 12y - 16x + 16 = 0$

15 $4y^2 - x^2 + 40y - 4x + 60 = 0$

16 $25x^2 - 9y^2 - 100x - 54y + 10 = 0$

17 $9y^2 - x^2 - 36y + 12x - 36 = 0$

18 $4x^2 - y^2 + 32x - 8y + 49 = 0$

In Exercises 19–26 find an equation for the hyperbola satisfying the given conditions.

19 Foci $F(0, \pm 4)$, vertices $V(0, \pm 1)$

20 Foci $F(\pm 8, 0)$, vertices $V(\pm 5, 0)$

21 Foci $F(\pm 5, 0)$, vertices $V(\pm 3, 0)$

- 22 Foci $F(0, \pm 3)$, vertices $V(0, \pm 2)$
 23 Foci $F(0, \pm 5)$, length of conjugate axis 4
 24 Vertices $V(\pm 4, 0)$, passing through $P(8, 2)$
 25 Vertices $V(\pm 3, 0)$, equations of asymptotes $y = \pm 2x$
 26 Foci $F(0, \pm 10)$, equations of asymptotes $y = \pm \frac{1}{3}x$

In Exercises 27 and 28 find the points of intersection of the graphs of the given equations and sketch both graphs on the same coordinate axes, showing points of intersection.

27 $\begin{cases} y^2 - 4x^2 = 16 \\ y - x = 4 \end{cases}$ 28 $\begin{cases} x^2 - y^2 = 4 \\ y^2 - 3x = 0 \end{cases}$

- 29 The graphs of the equations

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{and} \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$$

are called **conjugate hyperbolas**. Sketch the graphs of both equations on the same coordinate system with $a = 2$ and $b = 5$. Describe the relationship between the two graphs.

- 30 Find an equation of the hyperbola with foci $(h \pm c, k)$ and vertices $(h \pm a, k)$, where $0 < a < c$ and $c^2 = a^2 + b^2$.
 31 Find an equation of the tangent line to the hyperbola $2x^2 - 5y^2 = 3$ at the point $P(-2, 1)$.
 32 Prove that an equation of the tangent line to the graph of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ at the point $P(x_1, y_1)$ is

$$\frac{x_1 x}{a^2} - \frac{y_1 y}{b^2} = 1.$$

- 33 If tangent lines to the hyperbola $9x^2 - y^2 = 36$ intersect the y -axis at the point $(0, 6)$, find the points of tangency.
 34 If tangent lines to the hyperbola $b^2x^2 - a^2y^2 = a^2b^2$ intersect the y -axis at the point $(0, d)$, find the points of tangency.
 35 Find an equation of a line through $P(2, -1)$ that is tangent to the hyperbola $x^2 - 4y^2 = 16$.
 36 Find equations of the tangent lines to the hyperbola $8x^2 - 3y^2 = 48$ that are parallel to the line $2x - y = 10$.
 37 The region bounded by the hyperbola $b^2x^2 - a^2y^2 = a^2b^2$ and a vertical line through a focus is revolved about the x -axis. Find the volume of the resulting solid.
 38 If the region described in Exercise 37 is revolved about the y -axis, find the volume of the resulting solid.
 39 Find the area of the region bounded by the hyperbola $b^2x^2 - a^2y^2 = a^2b^2$ and a line through a focus perpendicular to the transverse axis.
 40 The region in the xy -plane that is bounded by the hyperbola $b^2y^2 - a^2x^2 = a^2b^2$ and the line $y = c$, where $c > a$, is revolved about the y -axis. Find the centroid of the resulting solid.
 41 Let l denote the tangent line at the point P on the hyperbola illustrated in Figure 12.18. If α is the angle between $F'P$ and l , and if β is the angle between FP and l , prove that $\alpha = \beta$. (This is analogous to the reflective property of the ellipse described in Exercise 35 of Section 12.3.)
 42 Let l denote the tangent line at the point P on the hyperbola illustrated in Figure 12.18. If Q and R are the points where l intersects the asymptotes, prove that P is the midpoint of the line segment QR .

12.5 Rotation of Axes

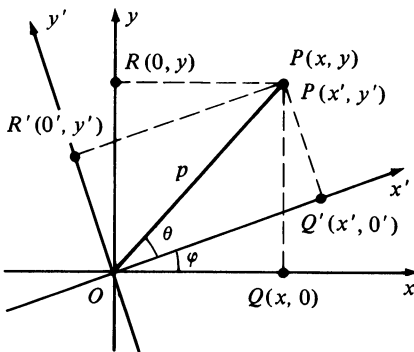


FIGURE 12.23

The $x'y'$ -coordinate system used in a translation of axes may be thought of as having been obtained by moving the origin O of the xy -system to a new position $C(h, k)$ while, at the same time, not changing the positive directions of the axes or the units of length. We shall next introduce a new coordinate system by keeping the origin O fixed and rotating the x - and y -axes about O to another position denoted by x' and y' . A transformation of this type will be referred to as a **rotation of axes**.

Consider a rotation of axes as shown in Figure 12.23 and let φ denote the angle through which the positive x -axis must be rotated in order to coincide with the positive x' -axis. If (x, y) are the coordinates of a point P relative to the xy -plane, then (x', y') will denote its coordinates relative to the new $x'y'$ -coordinate system.

Let the projections of P on the various axes be denoted as in the figure and let θ denote angle POQ' . If $p = d(O, P)$, then

$$\begin{aligned}x' &= p \cos \theta, & y' &= p \sin \theta, \\x &= p \cos (\theta + \varphi), & y &= p \sin (\theta + \varphi).\end{aligned}$$

Applying the addition formulas for the sine and cosine we see that

$$\begin{aligned}x &= p \cos \theta \cos \varphi - p \sin \theta \sin \varphi \\y &= p \sin \theta \cos \varphi + p \cos \theta \sin \varphi.\end{aligned}$$

Using the fact that $x' = p \cos \theta$ and $y' = p \sin \theta$ gives us (i) of the next theorem. The formulas in (ii) may be obtained from (i) by solving for x' and y' .

Theorem (12.11)

If the x - and y -axes are rotated about the origin O , through an angle φ , then the coordinates (x, y) and (x', y') of a point P in the two systems are related as follows:

- (i) $x = x' \cos \varphi - y' \sin \varphi, \quad y = x' \sin \varphi + y' \cos \varphi$
- (ii) $x' = x \cos \varphi + y \sin \varphi, \quad y' = -x \sin \varphi + y \cos \varphi.$

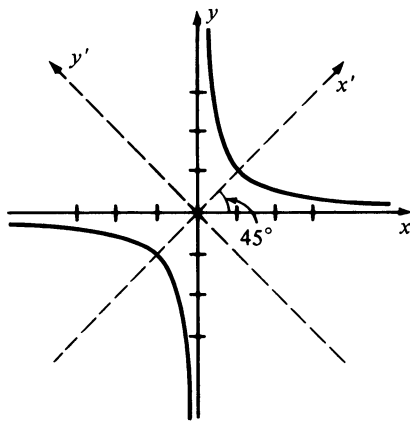


FIGURE 12.24

Example 1 The graph of the equation $xy = 1$, or equivalently $y = 1/x$, is sketched in Figure 12.24. If the coordinate axes are rotated through an angle of 45° , find the equation of the graph relative to the new $x'y'$ -coordinate system.

Solution Letting $\varphi = 45^\circ$ in (i) of Theorem (12.11), we obtain

$$x = x' \left(\frac{\sqrt{2}}{2} \right) - y' \left(\frac{\sqrt{2}}{2} \right) = \left(\frac{\sqrt{2}}{2} \right) (x' - y')$$

$$y = x' \left(\frac{\sqrt{2}}{2} \right) + y' \left(\frac{\sqrt{2}}{2} \right) = \left(\frac{\sqrt{2}}{2} \right) (x' + y').$$

Substituting for x and y in the equation $xy = 1$ gives us

$$\left(\frac{\sqrt{2}}{2} \right) (x' - y') \left(\frac{\sqrt{2}}{2} \right) (x' + y') = 1.$$

This reduces to

$$\frac{(x')^2}{2} - \frac{(y')^2}{2} = 1$$

which is an equation of a hyperbola with vertices $(\pm\sqrt{2}, 0)$ on the x' -axis. Note that the asymptotes for the hyperbola have equations $y' = \pm x'$ in the new system. These correspond to the original x - and y -axes. ■

Example 1 illustrates a method for eliminating a term of an equation that contains the product xy . This method can be used to transform any equation of the form

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

where $B \neq 0$, into an equation in x' and y' that contains no $x'y'$ term. Let us prove that this may always be done. If we rotate the axes through an angle φ , then using (i) of Theorem (12.11) to substitute for x and y gives us

$$\begin{aligned} & A(x' \cos \varphi - y' \sin \varphi)^2 \\ & + B(x' \cos \varphi - y' \sin \varphi)(x' \sin \varphi + y' \cos \varphi) \\ & + C(x' \sin \varphi + y' \cos \varphi)^2 + D(x' \cos \varphi - y' \sin \varphi) \\ & + E(x' \sin \varphi + y' \cos \varphi) + F = 0. \end{aligned}$$

This equation may be written in the form

$$A'(x')^2 + B'x'y' + C'(y')^2 + D'x' + E'y' + F = 0$$

where the coefficient B' of $x'y'$ is given by

$$B' = 2(C - A) \sin \varphi \cos \varphi - B(\cos^2 \varphi - \sin^2 \varphi).$$

To eliminate the $x'y'$ term we must select φ such that

$$2(C - A) \sin \varphi \cos \varphi + B(\cos^2 \varphi - \sin^2 \varphi) = 0.$$

Using double-angle formulas, the last equation may be written

$$(C - A) \sin 2\varphi + B \cos 2\varphi = 0$$

which is equivalent to

$$\cot 2\varphi = \frac{A - C}{B}.$$

This gives us the next result.

Theorem (12.12)

To eliminate the xy -term from the equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

where $B \neq 0$, choose an angle φ such that

$$\cot 2\varphi = \frac{A - C}{B}$$

and use the rotation of axes formulas in Theorem (12.11).

It follows that the graph of any equation in x and y of the type displayed in Theorem (12.12) is a conic, except for certain degenerate cases. (Why?)

Example 2 Discuss and sketch the graph of the equation

$$41x^2 - 24xy + 34y^2 - 25 = 0.$$

Solution Using the notation of Theorem (12.12) we have $A = 41$, $B = -24$, $C = 34$, and

$$\cot 2\varphi = \frac{41 - 34}{-24} = -\frac{7}{24}.$$

Since $\cot 2\varphi$ is negative we may choose 2φ such that $90^\circ < 2\varphi < 180^\circ$, and consequently $\cos 2\varphi = -\frac{7}{25}$. (Why?) We now use the half-angle formulas to obtain

$$\sin \varphi = \sqrt{\frac{1 - \cos 2\varphi}{2}} = \sqrt{\frac{1 - (-7/25)}{2}} = \frac{4}{5}$$

$$\cos \varphi = \sqrt{\frac{1 + \cos 2\varphi}{2}} = \sqrt{\frac{1 + (-7/25)}{2}} = \frac{3}{5}.$$

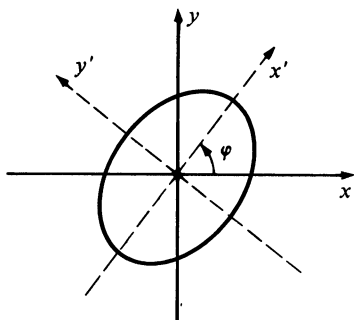


FIGURE 12.25

It follows that the desired rotation formulas (i) of (12.11) are

$$x = \frac{3}{5}x' - \frac{4}{5}y', \quad y = \frac{4}{5}x' + \frac{3}{5}y'.$$

We leave it to the reader to show that after substituting for x and y in the given equation and simplifying, we obtain the equation

$$(x')^2 + 2(y')^2 = 1.$$

Thus the graph is an ellipse with vertices at $(\pm 1, 0)$ on the x' -axis. Since $\tan \varphi = \sin \varphi / \cos \varphi = (\frac{4}{5}) / (\frac{3}{5}) = \frac{4}{3}$, we obtain $\varphi = \tan^{-1}(\frac{4}{3})$. To the nearest minute, $\varphi \approx 53^\circ 8'$. The graph is sketched in Figure 12.25. ■

In some cases, after eliminating the xy -term in Theorem (12.12), the resulting equation will contain an x' or y' term. It is then necessary to translate the axes of the $x'y'$ -coordinate system to obtain the graph. Several problems of this type are contained in the following exercises.

12.5 Exercises

After a suitable rotation of axes, describe and sketch the graph of each of the equations in Exercises 1–14.

1 $32x^2 - 72xy + 53y^2 = 80$

2 $7x^2 - 48xy - 7y^2 = 225$

3 $11x^2 + 10\sqrt{3}xy + y^2 = 4$

4 $x^2 - xy + y^2 = 3$

5 $5x^2 - 8xy + 5y^2 = 9$

6 $11x^2 - 10\sqrt{3}xy + y^2 = 20$

- 7 $16x^2 - 24xy + 9y^2 - 60x - 80y + 100 = 0$
- 8 $x^2 + 2\sqrt{3}xy + 3y^2 + 8\sqrt{3}x - 8y + 32 = 0$
- 9 $5x^2 + 6\sqrt{3}xy - y^2 + 8x - 8\sqrt{3}y - 12 = 0$
- 10 $18x^2 - 48xy + 82y^2 + 6\sqrt{10}x + 2\sqrt{10}y - 80 = 0$
- 11 $x^2 + 4xy + 4y^2 + 6\sqrt{5}x - 18\sqrt{5}y + 45 = 0$
- 12 $15x^2 + 20xy - 4\sqrt{5}x + 8\sqrt{5}y - 100 = 0$
- 13 $40x^2 - 36xy + 25y^2 - 8\sqrt{13}x - 12\sqrt{13}y = 0$
- 14 $64x^2 - 240xy + 225y^2 + 1020x - 544y = 0$
- 15 Prove that, except for degenerate cases, the graph of $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ is
 - (a) a parabola if $B^2 - 4AC = 0$.
 - (b) an ellipse if $B^2 - 4AC < 0$.
 - (c) a hyperbola if $B^2 - 4AC > 0$.
- 16 Use the results of Exercise 15 to determine the nature of the graphs in Exercises 1–14.

12.6 Review: Concepts

Define or discuss each of the following.

- 1 Conic sections
- 2 Parabola
- 3 Focus, directrix, vertex, and axis of a parabola
- 4 Translation of axes
- 5 Ellipse
- 6 Major and minor axes of an ellipse
- 7 Foci and vertices of an ellipse
- 8 Hyperbola
- 9 Transverse and conjugate axes of a hyperbola
- 10 Foci and vertices of a hyperbola
- 11 Asymptotes of a hyperbola
- 12 Rotation of axes

Review: Exercises

In Exercises 1–10, find the foci and vertices and sketch the graph of the conic that has the given equation.

- 1 $y^2 = 64x$
- 2 $y - 1 = 8(x + 2)^2$
- 3 $9y^2 = 144 - 16x^2$
- 4 $9y^2 = 144 + 16x^2$
- 5 $x^2 - y^2 - 4 = 0$
- 6 $25x^2 + 36y^2 = 1$
- 7 $25y = 100 - x^2$
- 8 $3x^2 + 4y^2 - 18x + 8y + 19 = 0$
- 9 $x^2 - 9y^2 + 8x + 90y - 210 = 0$
- 10 $x = 2y^2 + 8y + 3$
- 11 The parabola with focus $F(-4, 0)$ and directrix $x = 4$
- 12 The parabola with focus $(0, -10)$ and directrix $y = 10$
- 13 The parabola with vertex at the origin, symmetric to the x -axis, and passing through the point $(5, -1)$
- 14 The ellipse with vertices $V(0, \pm 10)$ and foci $F(0, \pm 5)$
- 15 The hyperbola with foci $F(\pm 10, 0)$ and vertices $V(\pm 5, 0)$
- 16 The hyperbola with vertices $V(0, \pm 6)$ and asymptotes that have equations $y = \pm 9x$
- 17 The ellipse with foci $F(\pm 2, 0)$ and passing through the point $(2, \sqrt{2})$

Discuss and sketch the graphs of the equations in Exercises 19–23 after making suitable translations of axes.

Find equations for the conics in Exercises 11–18.

- 11 The hyperbola with vertices $V(0, \pm 7)$ and endpoints of conjugate axes $(\pm 3, 0)$
- 12 The parabola with focus $F(-4, 0)$ and directrix $x = 4$
- 13 The parabola with focus $(0, -10)$ and directrix $y = 10$
- 14 The parabola with vertex at the origin, symmetric to the x -axis, and passing through the point $(5, -1)$
- 15 The ellipse with vertices $V(0, \pm 10)$ and foci $F(0, \pm 5)$
- 16 The hyperbola with foci $F(\pm 10, 0)$ and vertices $V(\pm 5, 0)$
- 17 The hyperbola with vertices $V(0, \pm 6)$ and asymptotes that have equations $y = \pm 9x$
- 18 The ellipse with foci $F(\pm 2, 0)$ and passing through the point $(2, \sqrt{2})$
- 19 $4x^2 + 9y^2 + 24x - 36y + 36 = 0$
- 20 $4x^2 - y^2 - 40x - 8y + 88 = 0$
- 21 $y^2 - 8x + 8y + 32 = 0$

- 22 $4x^2 + y^2 - 24x + 4y + 36 = 0$
- 23 $x^2 - 9y^2 + 8x + 7 = 0$
- 24 Find equations of the tangent and normal lines to the hyperbola $4x^2 - 9y^2 - 8x + 6y - 36 = 0$ at the point $P(-3, 2)$.
- 25 Tangent lines to the parabola $y = 2x^2 + 3x + 1$ pass through the point $P(2, -1)$. Find the x -coordinates of the points of tangency.
- 26 Let R be the region that is bounded by the parabola $9y = 5x^2 - 39x + 90$ and an asymptote of the hyperbola $9y^2 - 4x^2 = 36$.
- (a) Find the area of R .
- (b) Find the volume of the solid obtained by revolving R about the y -axis.
- 27 An ellipse having axes of lengths 8 and 4 is revolved about its major axis. Find the volume of the resulting solid.
- 28 A solid has, for its base, the region in the xy -plane bounded by the graph of $x^2 + y^2 = r^2$. Find the volume of the solid if every cross section by a plane perpendicular to the x -axis is half of an ellipse with one axis always of length c .

After making a suitable rotation of axes, describe and sketch the graphs of the equations in Exercises 29 and 30.

29 $x^2 - 8xy + 16y^2 - 12\sqrt{17}x - 3\sqrt{17}y = 0$

30 $8x^2 + 12xy + 17y^2 - 16\sqrt{5}x - 12\sqrt{5}y = 0$

13 Plane Curves and Polar Coordinates

In this chapter parametric and polar equations of curves are discussed. Applications include tangent lines, areas, arc length, and surfaces of revolution.

13.1 Plane Curves

The graph of an equation $y = f(x)$, where f is a function, is often called a *plane curve*. However, to use this as a definition is unnecessarily restrictive, since it rules out most of the conic sections and many other useful graphs. The following statement is satisfactory for most applications.

Definition (13.1)

A **plane curve** is a set C of ordered pairs of the form $(f(t), g(t))$ where the functions f and g are continuous on an interval I .

For simplicity, we shall often refer to a plane curve as a **curve**. The **graph** of the curve C in Definition (13.1) is the set of all points $P(t) = (f(t), g(t))$ in a rectangular coordinate system that corresponds to the ordered pairs. Each $P(t)$ is referred to as a *point* on the curve. We shall use the term *curve* interchangeably with *graph of a curve*. Sometimes it is convenient to imagine that the point $P(t)$ traces the curve C as t varies through the interval I . This is especially true in applications where t represents time and $P(t)$ is the position of a moving particle at time t .

The graphs of several curves are sketched in Figure 13.1 for the case where I is a closed interval $[a, b]$. If, as in (i) of the figure, $P(a) \neq P(b)$, then $P(a)$ and

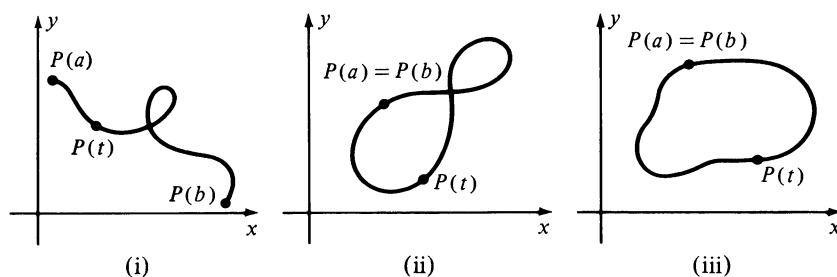


FIGURE 13.1

$P(b)$ are called the **endpoints** of C . Note that the curve illustrated in (i) intersects itself in the sense that two different values of t give rise to the same point. If $P(a) = P(b)$, as illustrated in (ii) of Figure 13.1, then C is called a **closed curve**. If $P(a) = P(b)$ and C does not intersect itself at any other point, as illustrated in (iii) of the figure, then C is called a **simple closed curve**.

It is convenient to represent curves as indicated in the next definition.

Definition (13.2)

Let C be the curve consisting of all ordered pairs $(f(t), g(t))$, where f and g are continuous on an interval I . The equations

$$x = f(t), \quad y = g(t),$$

where t is in I , are called **parametric equations** for C , and t is called a **parameter**.

If we are given the parametric representation (13.2), then as t varies through I , the point $P(x, y)$ traces the curve. Sometimes it is possible to eliminate the parameter and obtain an equation for C that involves the variables x and y . The continuity of f and g implies that a small change in t produces a small change in the position of the point $(f(t), g(t))$ on C . This fact may be used to obtain a rough sketch of the graph by plotting many points and connecting them in the order of increasing t , as illustrated in Example 1.

Example 1 Sketch and identify the graph of the curve C given parametrically by

$$x = 2t, \quad y = t^2 - 1, \quad \text{where } -1 \leq t \leq 2.$$

Solution The parametric equations can be used to tabulate coordinates for points $P(x, y)$ on C as in the following table.

t	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
x	-2	-1	0	1	2	3	4
y	0	$-\frac{3}{4}$	-1	$-\frac{3}{4}$	0	$\frac{5}{4}$	3

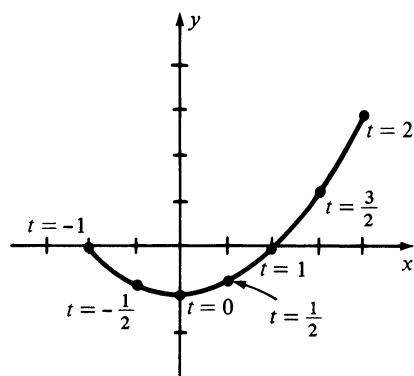


FIGURE 13.2

Plotting points and using the continuity of f and g gives us the sketch in Figure 13.2.

A precise description of the graph may be obtained by eliminating the parameter. To illustrate, solving the first parametric equation for t we obtain $t = x/2$. If we next substitute for t in the second equation the result is

$$y = \left(\frac{x}{2}\right)^2 - 1, \quad \text{or} \quad y + 1 = \frac{1}{4}x^2.$$

The graph of this equation is a parabola with vertical axis and vertex at the point $(0, -1)$. The curve C is that part of the parabola shown in Figure 13.2. ■

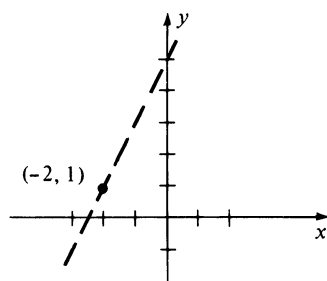
Parametric equations of curves are never unique. The reader should verify that the curve C in Example 1 is given by any of the following parametric representations:

$$x = 2t, \quad y = t^2 - 1; \quad -1 \leq t \leq 2$$

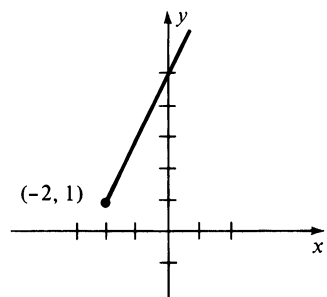
$$x = t, \quad y = \frac{1}{4}t^2 - 1; \quad -2 \leq t \leq 4$$

$$x = t^3, \quad y = \frac{1}{4}t^6 - 1; \quad \sqrt[3]{-2} \leq t \leq \sqrt[3]{4}.$$

The next example illustrates the fact that it is often useful to eliminate the parameter *before* plotting points.



(i) $y - 1 = 2(x + 2)$



(ii) $x = -2 + t^2, \quad y = 1 + 2t^2$

FIGURE 13.3

Example 2 Sketch the graph of the curve C that is given parametrically by

$$x = -2 + t^2, \quad y = 1 + 2t^2$$

where t is in $\mathbb{R}$.

Solution To eliminate the parameter we note, from the first equation, that $t^2 = x + 2$. Substituting for t^2 in the second equation gives us

$$y = 1 + 2(x + 2), \quad \text{or} \quad y - 1 = 2(x + 2).$$

This is an equation of the line of slope 2 through the point $(-2, 1)$, as indicated by the dashes in (i) of Figure 13.3. However, since $t^2 \geq 0$,

$$x = -2 + t^2 \geq -2 \quad \text{and} \quad y = 1 + 2t^2 \geq 1.$$

It follows that the graph of C is that part of the line to the right of $(-2, 1)$, as shown in (ii) of the figure. This fact may also be verified by plotting several points. ■

Example 3 Describe the graph of the curve C having parametric equations

$$x = \cos t, \quad y = \sin t, \quad 0 \leq t \leq 2\pi.$$

Solution We may use the identity $\cos^2 t + \sin^2 t = 1$ to eliminate the parameter. This gives us $x^2 + y^2 = 1$, and hence points on C are on the unit circle with center at the origin. As t increases from 0 to 2π , $P(t)$ starts at the point $A(1, 0)$ and traverses the circle once in the counterclockwise direction. In this example the parameter may be interpreted geometrically as the length of arc from A to P , as illustrated in Figure 1.37. ■

If a curve C is described by means of an equation $y = f(x)$, where f is a continuous function, then an easy way to obtain parametric equations is to let

$$x = t, \quad y = f(t)$$

where t is in the domain of f . For example, if $y = x^3$, then parametric equations are

$$x = t, \quad y = t^3, \quad \text{where } t \text{ is in } \mathbb{R}.$$

We can use many different substitutions for x , provided that as t varies through some interval, x takes on all values in the domain of f . Thus the graph of $y = x^3$ is also given by

$$x = t^{1/3}, \quad y = t, \quad \text{where } t \text{ is in } \mathbb{R}.$$

Note, however, that the parametric equations

$$x = \sin t, \quad y = \sin^3 t, \quad \text{where } t \text{ is in } \mathbb{R}$$

give only that part of the graph of $y = x^3$ that lies between the points $(-1, -1)$ and $(1, 1)$.

Example 4 Find three different parametric representations for the line of slope m through the point (x_1, y_1) .

Solution By the Point-Slope Form, an equation for the line is

$$y - y_1 = m(x - x_1).$$

If we let $x = t$, then $y - y_1 = m(t - x_1)$ and we obtain the parametric equations

$$x = t, \quad y = y_1 + m(t - x_1) \quad \text{where } t \text{ is in } \mathbb{R}.$$

Another pair of parametric equations results if we let $x - x_1 = t$. In this case $y - y_1 = mt$ and hence the line is given parametrically by

$$x = x_1 + t, \quad y = y_1 + mt, \quad \text{where } t \text{ is in } \mathbb{R}.$$

As a third illustration, if we let $x - x_1 = \tan t$ we obtain

$$x = x_1 + \tan t, \quad y = y_1 + m \tan t, \quad \text{where } -\frac{\pi}{2} < t < \frac{\pi}{2}.$$

There are many other ways to represent the line parametrically. ■

A curve C is called **smooth** if it has a parametric representation $x = f(t)$, $y = g(t)$ on an interval I such that the derivatives f' and g' are continuous on I and are not simultaneously zero, except possibly at any endpoints of I . The curve C is **piecewise smooth** if the interval I can be partitioned into sub-intervals such that C is smooth on each subinterval. The graph of a smooth curve has no corners. The curves given in Examples 1–4 are smooth. The curve described in the next example is piecewise smooth.

Example 5 The curve traced by a fixed point P on the circumference of a circle as the circle rolls along a straight line in a plane is called a **cycloid**. Find parametric equations for a cycloid.

Solution Suppose the circle has radius a and that it rolls along (and above) the x -axis in the positive direction. If one position of P is the origin, then Figure 13.4 displays part of the curve and a possible position of the circle.

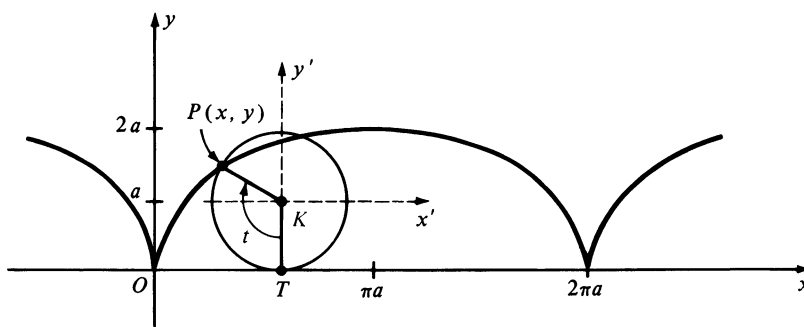


FIGURE 13.4

Let K denote the center of the circle and T the point of tangency with the x -axis. We introduce a parameter t as the radian measure of angle TKP . Since $\overline{OT}$ is the distance the circle has rolled, $\overline{OT} = at$. Consequently, the coordinates of K are (at, a) . If we consider an $x'y'$ -coordinate system with origin at $K(at, a)$ and if $P(x', y')$ denotes the point P relative to this system, then by the translation of axes formulas (12.3), with $h = at$ and $k = a$,

$$x = at + x', \quad y = a + y'.$$

If, as in Figure 13.5, θ denotes an angle in standard position on the $x'y'$ -system, then $\theta = 3\pi/2 - t$. Hence

$$x' = a \cos \theta = a \cos (3\pi/2 - t) = -a \sin t$$

$$y' = a \sin \theta = a \sin (3\pi/2 - t) = -a \cos t,$$

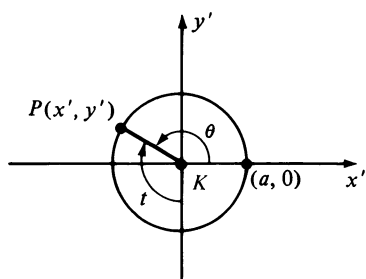


FIGURE 13.5

and substitution in $x = at + x'$, $y = a + y'$ gives us parametric equations for the cycloid, namely

$$x = a(t - \sin t), \quad y = a(1 - \cos t)$$

where t is in $\mathbb{R}$. ■

Differentiating the parametric equations of the cycloid that were obtained in Example 5 gives us

$$\frac{dx}{dt} = a(1 - \cos t), \quad \frac{dy}{dt} = a \sin t.$$

These derivatives are continuous for all t , but are simultaneously 0 if $t = 2\pi n$ for every integer n . Hence the cycloid is not smooth. Note that the points corresponding to $t = 2\pi n$ are the x -intercepts, and the cycloid has a corner at each such point (see Figure 13.4). The graph is piecewise smooth, since it is smooth in each t -interval $[2\pi n, 2\pi(n + 1)]$ where n is an integer.

If $a < 0$, then the graph of $x = a(t - \sin t)$, $y = a(1 - \cos t)$ is the inverted cycloid that results if the circle of Example 5 rolls *below* the x -axis. This curve has a number of important physical properties. In particular, suppose a thin wire passes through two fixed points A and B as illustrated in Figure 13.6, and that the shape of the wire can be changed by bending it in any manner. Suppose further, that a bead is allowed to slide along the wire and the only force acting on the bead is gravity. We now ask which of all the possible paths will allow the bead to slide from A to B in the least amount of time. It is natural to conjecture that the desired path is the straight line segment from A to B ; however, this is not the correct answer. The path that requires the least time coincides with the graph of the inverted cycloid with A at the origin and B the lowest point on the curve. The proof of this result will not be given in this text.

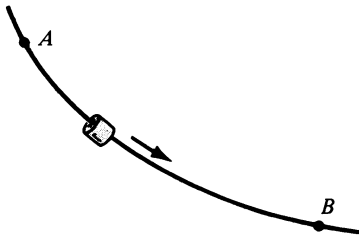


FIGURE 13.6

To cite another interesting property of this curve, suppose that A is the origin and B is the point with x -coordinate $\pi|a|$, that is, the lowest point on the cycloid occurring in the first arc to the right of A . It can be shown that if the bead is released at *any* point between A and B , the time required for it to reach B is always the same!

Variations of the cycloid occur in practical problems. For example, if a motorcycle wheel rolls along a straight road, then the curve traced by a fixed point on one of the spokes is a cycloid-like curve. In this case, the curve does not have sharp corners, nor does it intersect the road (the x -axis) as does the graph of a cycloid. In like manner, if the wheel of a train rolls along a railroad track, then the curve traced by a fixed point on the circumference of the wheel (which extends below the track) contains loops at regular intervals (see Exercise 38). Several other cycloids are defined in Exercises 33 and 34.

13.1 Exercises

In Exercises 1–20, (a) sketch the graph of the curve C having the indicated parametric equations, and (b) find a rectangular equation of a graph that contains the points on C .

- 1 $x = t - 2, y = 2t + 3; 0 \leq t \leq 5$
- 2 $x = 1 - 2t, y = 1 + t; -1 \leq t \leq 4$
- 3 $x = t^2 + 1, y = t^2 - 1; -2 \leq t \leq 2$
- 4 $x = t^3 + 1, y = t^3 - 1; -2 \leq t \leq 2$
- 5 $x = 4t^2 - 5, y = 2t + 3; t \text{ in } \mathbb{R}$
- 6 $x = t^3, y = t^2; t \text{ in } \mathbb{R}$
- 7 $x = e^t, y = e^{-2t}; t \text{ in } \mathbb{R}$
- 8 $x = \sqrt{t}, y = 3t + 4; t \geq 0$
- 9 $x = 2 \sin t, y = 3 \cos t; 0 \leq t \leq 2\pi$
- 10 $x = \cos t - 2, y = \sin t + 3; 0 \leq t \leq 2\pi$
- 11 $x = \sec t, y = \tan t; -\pi/2 < t < \pi/2$
- 12 $x = \cos 2t, y = \sin t; -\pi \leq t \leq \pi$
- 13 $x = t^2, y = 2 \ln t; t > 0$
- 14 $x = \cos^3 t, y = \sin^3 t; 0 \leq t \leq 2\pi$
- 15 $x = \sin t, y = \csc t; 0 < t \leq \pi/2$
- 16 $x = e^t, y = e^{-t}; t \text{ in } \mathbb{R}$
- 17 $x = \cosh t, y = \sinh t; t \text{ in } \mathbb{R}$
- 18 $x = 3 \cosh t, y = 2 \sinh t; t \text{ in } \mathbb{R}$
- 19 $x = t, y = \sqrt{t^2 - 1}; |t| \geq 1$
- 20 $x = -2\sqrt{1 - t^2}, y = t; |t| \leq 1$

In Exercises 21–24, sketch the graph of the curve C having the indicated parametric equations.

- 21 $x = t, y = \sqrt{t^2 - 2t + 1}; 0 \leq t \leq 4$
- 22 $x = 2t, y = 8t^3; -1 \leq t \leq 1$
- 23 $x = (t + 1)^3, y = (t + 2)^2; 0 \leq t \leq 2$
- 24 $x = \tan t, y = 1; -\pi/2 < t < \pi/2$

In Exercises 25 and 26 curves C_1, C_2, C_3 , and C_4 are given parametrically, where t is in $\mathbb{R}$. Sketch the graphs of C_1, C_2, C_3, C_4 , and discuss their similarities and differences.

- 25 $C_1: x = t^2, y = t$
 $C_2: x = t^4, y = t^2$
 $C_3: x = \sin^2 t, y = \sin t$
 $C_4: x = e^{2t}, y = -e^t$
- 26 $C_1: x = t, y = 1 - t$
 $C_2: x = 1 - t^2, y = t^2$
 $C_3: x = \cos^2 t, y = \sin^2 t$
 $C_4: x = \ln t - t, y = 1 + t - \ln t$

The parametric representations in Exercises 27 and 28 give the position of a point $P(x, y)$, where t represents time. Describe the motion of the point during the indicated time interval.

- 27 (a) $x = \cos t, y = \sin t; 0 \leq t \leq \pi$
 (b) $x = \sin t, y = \cos t; 0 \leq t \leq \pi$
 (c) $x = t, y = \sqrt{1 - t^2}; -1 \leq t \leq 1$
- 28 (a) $x = t^2, y = 1 - t^2; 0 \leq t \leq 1$
 (b) $x = 1 - \ln t, y = \ln t; 1 \leq t \leq e$
 (c) $x = \cos^2 t, y = \sin^2 t; 0 \leq t \leq 2\pi$
- 29 If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are distinct points, show that

$$x = (x_2 - x_1)t + x_1, \quad y = (y_2 - y_1)t + y_1,$$

where t is in $\mathbb{R}$, are parametric equations of the line l through P_1 and P_2 . Find three other pairs of parametric equations for the line l . Show that there is an infinite number of different pairs of parametric equations for l .

- 30 What is the difference between the graph of the hyperbola $b^2x^2 - a^2y^2 = a^2b^2$ and the graph of $x = a \cosh t, y = b \sinh t$, where t is in $\mathbb{R}$?
- 31 Show that

$$x = a \cos t + h, \quad y = b \sin t + k,$$

where $0 \leq t \leq 2\pi$ are parametric equations of an ellipse with center at the point (h, k) and semi-axes of lengths a and b .

- 32 Find parametric equations for the parabola that has (a) vertex $V(0, 0)$ and focus $F(0, p)$; (b) vertex $V(h, k)$ and focus $F(h, k + p)$.

- 33 A circle C of radius b rolls on the inside of a second circle having equation $x^2 + y^2 = a^2$, where $b < a$. Let P be a fixed point on C and let the initial position of P be $A(a, 0)$ as shown in the figure. If the parameter t is the angle from the positive x -axis to the line segment from O to the center of C , show that parametric equations for the curve traced by P (called a **hypocycloid**) are

$$x = (a - b) \cos t + b \cos \frac{a - b}{b} t$$

$$y = (a - b) \sin t - b \sin \frac{a - b}{b} t$$

where $0 \leq t \leq 2\pi$. If $b = a/4$ show that

$$x = a \cos^3 t, \quad y = a \sin^3 t$$

and sketch the graph of the curve.

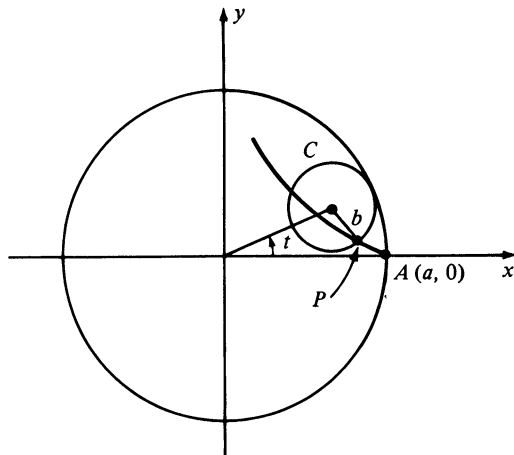


FIGURE FOR EXERCISE 33

- 34 If the circle C of Exercise 33 rolls on the outside of the second circle (see figure) then the curve traced by P is called an **epicycloid**. Show that parametric equations for this curve are

$$x = (a + b) \cos t - b \cos \frac{a + b}{b} t$$

$$y = (a + b) \sin t - b \sin \frac{a + b}{b} t.$$

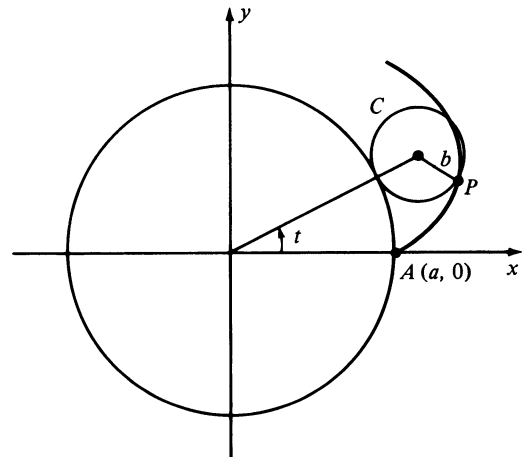


FIGURE FOR EXERCISE 34

- 35 If $b = a/3$ in Exercise 34, find parametric equations for the epicycloid and sketch the graph.
- 36 Problem 17 of the 1982 SAT examination was as follows: The radius of circle A is one-third that of circle B . How many revolutions will circle A make as it rolls around circle B until it reaches its starting point? Use Exercise 35 to show that the answer is 4. (To the embarrassment of the Educational Testing Service of Princeton, NJ, this was not one of the choices given as an answer.)
- 37 If a string is unwound from around a circle and is kept taut in the plane of the circle, then a fixed point P on the string will trace a curve called the **involute of the circle**. If the circle is chosen as in Exercise 33 and the parameter t is the angle from the positive x -axis to the point of tangency of the string, show that parametric equations for the involute are

$$x = a(\cos t + t \sin t), \quad y = a(\sin t - t \cos t).$$

- 38 Generalize the cycloid of Example 5 to the case where P is any point on a fixed line through the center C of the circle. If $b = d(C, P)$, derive the parametric equations

$$x = at - b \sin t, \quad y = a - b \cos t.$$

Sketch a typical graph if $b < a$ (a **curtate cycloid**) and if $b > a$ (a **prolate cycloid**). The term **trochoid** is sometimes used for either of these curves.

13.2 Tangent Lines to Curves

In Example 1 of the preceding section we saw that the curve C having parametric equations

$$x = 2t, \quad y = t^2 - 1, \quad \text{where } -1 \leq t \leq 2$$

can also be described by using an equation of the form $y = k(x)$, where k is a function defined on a suitable interval. As a matter of fact we may take

$$y = k(x) = \frac{1}{4}x^2 - 1, \quad \text{where } -2 \leq x \leq 4.$$

It follows that the slope of the tangent line at any point $P(x, y)$ on C is

$$k'(x) = \frac{1}{2}x \quad \text{or} \quad k'(x) = \frac{1}{2}(2t) = t.$$

We shall now introduce a technique for finding the slope directly from the parametric equations.

Let $x = f(t)$, $y = g(t)$ be parametric equations of a curve C and suppose C can also be represented by a rectangular equation of the form $y = k(x)$, where k is a function. Let us also assume that f , g , and k are differentiable throughout their domains. If $P(f(t), g(t))$ is a point on C , then the coordinates satisfy the equation $y = k(x)$, that is,

$$g(t) = k(f(t)).$$

Applying the Chain Rule,

$$g'(t) = k'(f(t))f'(t) = k'(x)f'(t)$$

and hence the slope of the tangent line at P is

$$k'(x) = \frac{g'(t)}{f'(t)}, \quad \text{provided } f'(t) \neq 0.$$

If $f'(t) = 0$ and $g'(t) \neq 0$ for some t , then C has a vertical tangent line at the point corresponding to t . If $f'(t)$ and $g'(t)$ are simultaneously zero, then further information is required.

The conditions imposed in the preceding discussion are fulfilled if C is smooth and $f'(t) \neq 0$ for all t in a closed interval $[a, b]$. In this case either $f'(t) > 0$ or $f'(t) < 0$ throughout $[a, b]$ (Why?), and it follows from Theorem (7.26) or Theorem (7.27) that f has an inverse function f^{-1} . Since $x = f(t)$ we may write $t = f^{-1}(x)$, where x is in the interval $[f(a), f(b)]$. Hence

$$y = g(t) = g(f^{-1}(x)) = k(x)$$

where k is the composite function $g \circ f^{-1}$.

An easy way to remember the formula for $k'(x)$ is to use the notation given in the next theorem.

Theorem (13.3)

If a smooth curve C is given parametrically by $x = f(t)$, $y = g(t)$, then the slope dy/dx of the tangent line to C at $P(x, y)$ is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \text{provided } \frac{dx}{dt} \neq 0.$$

Example 1 Find the slopes of the tangent and normal lines at the point $P(t)$ on the curve having parametric equations $x = 2t$, $y = t^2 - 1$, where $-1 \leq t \leq 2$.

Solution The curve is the same as that considered in Example 1 of the preceding section (see Figure 13.2). Using Theorem (13.3) with $x = 2t$ and $y = t^2 - 1$, the slope of the tangent line at $P(t)$ is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t}{2} = t.$$

Note that this agrees with the discussion at the beginning of this section. The slope of the normal line is $-1/t$, provided $t \neq 0$. ■

Example 2 If C has parametric equations $x = t^3 - 3t$, $y = t^2 - 5t - 1$, where t is in $\mathbb{R}$, find an equation of the tangent line to C at the point corresponding to $t = 2$. For what values of t is the tangent line horizontal or vertical?

Solution Applying (13.3),

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t - 5}{3t^2 - 3}.$$

Thus at the point P corresponding to $t = 2$, the slope m of the tangent line is

$$m = \frac{2(2) - 5}{3(2^2) - 3} = -\frac{1}{9}.$$

Substituting $t = 2$ in the parametric equations gives us the coordinates $(2, -7)$ of P . Hence an equation of the tangent line is

$$y + 7 = -\frac{1}{9}(x - 2)$$

or, equivalently,

$$x + 9y + 61 = 0.$$

The tangent line is horizontal if $dy/dt = 0$, that is, if $2t - 5 = 0$ or $t = 5/2$.

The tangent line is vertical if $3t^2 - 3 = 0$. (Why?) Thus there are vertical tangent lines at the points corresponding to $t = 1$ and $t = -1$. ■

If the graph of a curve C coincides with the graph of $y = k(x)$, then

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

If y' is a differentiable function of t we can find d^2y/dx^2 by applying Theorem (13.3) to y' as follows:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(y') = \frac{dy'/dt}{dx/dt}.$$

Example 3 Suppose C has parametric equations

$$x = e^{-t}, \quad y = e^{2t}, \quad \text{where } t \text{ is in } \mathbb{R}.$$

- (a) Find d^2y/dx^2 directly from the parametric equations.
- (b) Define a function k which has the same graph as C and check the answer to part (a).
- (c) Discuss the concavity of C .

Solution

- (a) As in the preceding discussion,

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2e^{2t}}{-e^{-t}} = -2e^{3t},$$

$$\text{and} \quad \frac{d^2y}{dx^2} = \frac{dy'}{dx} = \frac{dy'/dt}{dx/dt} = \frac{-6e^{3t}}{-e^{-t}} = 6e^{4t}.$$

- (b) Using the fact that $x = e^{-t} = 1/e^t$, we obtain $e^t = 1/x = x^{-1}$. Since $y = e^{2t} = (e^t)^2$, it follows that C is the graph of the equation

$$y = (x^{-1})^2 = x^{-2}, \quad \text{where } x > 0.$$

The restriction $x > 0$ is necessary because $e^{-t} > 0$ for all t . Thus a function k whose graph coincides with the graph of C may be defined by

$$k(x) = x^{-2}, \quad \text{where } x > 0.$$

Since $k'(x) = -2x^{-3}$,

$$k''(x) = 6x^{-4} = 6(e^{-t})^{-4} = 6e^{4t}$$

which is in agreement with part (a).

- (c) Since $d^2y/dx^2 = 6e^{4t} > 0$ for all t , the curve C is concave upward at every point. ■

13.2 Exercises

Find the slopes of the tangent lines at the point corresponding to $t = 1$ on the curves defined in Exercises 1–10. (See the graphs in Exercises 1–10 of Section 13.1).

- 1 $x = t - 2, y = 2t + 3; 0 \leq t \leq 5$
- 2 $x = 1 - 2t, y = 1 + t; -1 \leq t \leq 4$
- 3 $x = t^2 + 1, y = t^2 - 1; -2 \leq t \leq 2$
- 4 $x = t^3 + 1, y = t^3 - 1; -2 \leq t \leq 2$
- 5 $x = 4t^2 - 5, y = 2t + 3; t \text{ in } \mathbb{R}$
- 6 $x = t^3, y = t^2; t \text{ in } \mathbb{R}$
- 7 $x = e^t, y = e^{-2t}; t \text{ in } \mathbb{R}$
- 8 $x = \sqrt{t}, y = 3t + 4; t \geq 0$
- 9 $x = 2 \sin t, y = 3 \cos t; 0 \leq t \leq 2\pi$
- 10 $x = \cos t - 2, y = \sin t + 3; 0 \leq t \leq 2\pi$
- 11 Let l be the line with parametric equations $x = t/2, y = 2t - 5$. If a curve C is given by $x = t^4 - 7, y = 8t + 3$, where t is in $\mathbb{R}$, find a rectangular equation of the normal line to C that is parallel to l .
- 12 If C has parametric equations $x = 6t + 1, y = t^3 - 2t$, where t is in $\mathbb{R}$, find the points on C at which the tangent line is perpendicular to the line $3x + 5y - 8 = 0$.

- 13 If C is given parametrically by $x = 4t^2, y = 2t - 5$, where t is in $\mathbb{R}$, find a rectangular equation of the line through the point $(4, 1)$ that is tangent to C .
- 14 Find a rectangular equation of the line through the origin that is normal to the curve having parametric equations $x = 3t^2 - 2, y = 2t^3$, where t is in $\mathbb{R}$.

In Exercises 15–18 find the points on the indicated curve at which the tangent line is either horizontal or vertical. Sketch the graph, showing the horizontal and vertical tangent lines. Find d^2y/dx^2 in each case.

- 15 $x = 4t^2, y = t^3 - 12t; t \text{ in } \mathbb{R}$
- 16 $x = t^3 + 1, y = t^2 - 2t; t \text{ in } \mathbb{R}$
- 17 $x = 3t^2 - 6t, y = \sqrt{t}; t \geq 0$
- 18 $x = \sqrt[3]{t}, y = \sqrt[3]{t} - t; t \text{ in } \mathbb{R}$
- 19 What is the slope of the tangent line at the point $P(t)$ on the cycloid $x = a(t - \sin t), y = a(1 - \cos t)$, where t is in $\mathbb{R}$ and $a > 0$? At what points is the tangent line horizontal or vertical? Where does the slope of the tangent line equal 1? Find d^2y/dx^2 and discuss concavity if $0 < t < 2\pi$.
- 20 Answer the questions of Exercise 19 for the hypocycloid $x = a \cos^3 t, y = a \sin^3 t$ where $0 \leq t \leq 2\pi$ and $a > 0$.

13.3 Polar Coordinate Systems

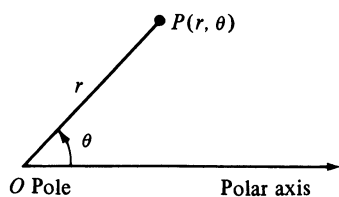


FIGURE 13.7

We have previously specified points in a plane in terms of rectangular coordinates, using the ordered pair (a, b) to denote the point whose directed distances from the x - and y -axes are b and a , respectively. Another important method for representing points is by means of **polar coordinates**. To introduce a system of polar coordinates in a plane we begin with a fixed point O (called the **origin**, or **pole**) and a directed half-line (called the **polar axis**) with end-point O . Next we consider any point P in the plane different from O . If, as illustrated in Figure 13.7, $r = d(O, P)$ and θ denotes the measure of any angle determined by the polar axis and OP , then r and θ are called **polar coordinates** of P and the symbols (r, θ) or $P(r, \theta)$ are used to denote P . As usual θ is considered positive if the angle is generated by a counterclockwise rotation of the polar axis and negative if the rotation is clockwise. Either radians or degrees may be used for the measure of θ .

If the polar axis is the initial side, then there are many angles with the same terminal side. Hence the polar coordinates of a point are not unique. For example, $(3, \pi/4)$, $(3, 9\pi/4)$, and $(3, -7\pi/4)$ all represent the same point (see Figure 13.8). We shall also allow r to be negative. In this event, instead of measuring $|r|$ units along the terminal side of the angle θ , we measure along

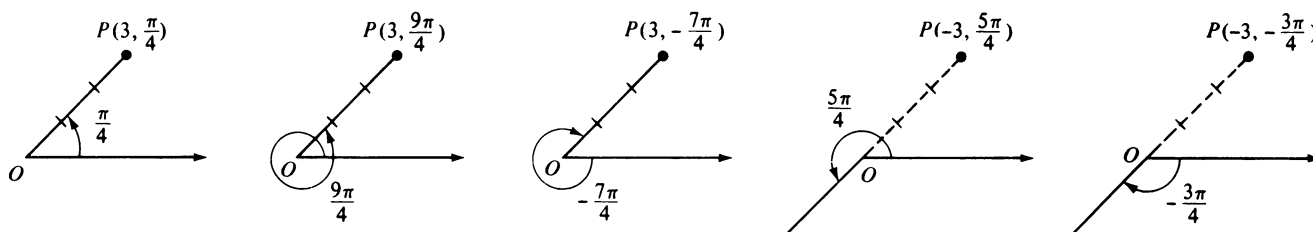


FIGURE 13.8

the half-line with endpoint O that has direction opposite to that of the terminal side. The points corresponding to the pairs $(-3, 5\pi/4)$ and $(-3, -3\pi/4)$ are also plotted in Figure 13.8.

Finally, we agree that the pole O has polar coordinates $(0, \theta)$ for *any* θ . An assignment of ordered pairs of the form (r, θ) to points in a plane will be referred to as a **polar coordinate system** and the plane will be called an **$r\theta$ -plane**.

A **polar equation** is an equation in r and θ . A **solution** of a polar equation is an ordered pair (a, b) that leads to equality if a is substituted for r and b for θ . The **graph** of a polar equation is the set of all points (in an $r\theta$ -plane) that correspond to the solutions.

Example 1 Sketch the graph of the polar equation $r = 4 \sin \theta$.

Solution The following table contains some solutions of the equation.

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
r	0	2	$2\sqrt{2}$	$2\sqrt{3}$	4	$2\sqrt{3}$	$2\sqrt{2}$	2	0

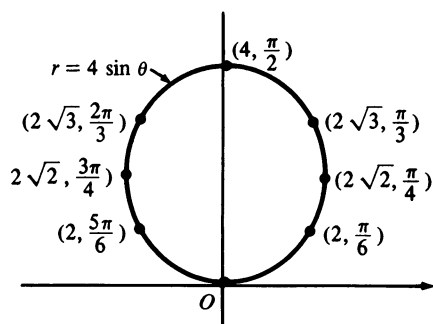


FIGURE 13.9

In rectangular coordinates, the graph of the given equation consists of sine waves of amplitude 4 and period 2π . However, if polar coordinates are used, then the points that correspond to the pairs in the table lie on a circle of radius 2 and we draw the graph accordingly (see Figure 13.9). As an aid to plotting points, we have extended the polar axis in the negative direction and introduced a vertical line through the pole. The proof that the graph is a circle will be given in Example 5. Additional points obtained by letting θ vary from π to 2π lie on the same circle. For example, the solution $(-2, 7\pi/6)$ gives us the same point as $(2, \pi/6)$; the point corresponding to $(-2\sqrt{2}, 5\pi/4)$ is the same as that obtained from $(2\sqrt{2}, \pi/4)$, etc. If we let θ increase through all real numbers we obtain the same points again and again because of the periodicity of the sine function. ■

Example 2 Sketch the graph of the equation $r = 2 + 2 \cos \theta$.

Solution Since the cosine function decreases from 1 to -1 as θ varies from 0 to π , it follows that r decreases from 4 to 0 in this θ -interval. The following table exhibits some solutions of the equation.

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
r	4	$2 + \sqrt{3}$	$2 + \sqrt{2}$	3	2	1	$2 - \sqrt{2}$	$2 - \sqrt{3}$	0

If θ increases from π to 2π , then $\cos \theta$ increases from -1 to 1 , and consequently r increases from 0 to 4 . Plotting points leads to the sketch shown in Figure 13.10, where we have used polar coordinate graph paper which displays lines through O at various angles and circles with centers at the pole. The same graph may be obtained by taking other intervals for θ .

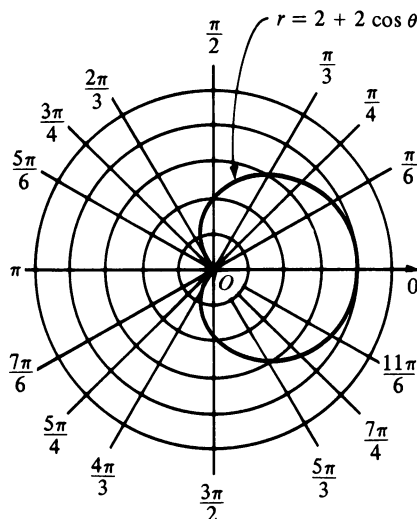


FIGURE 13.10

The heart-shaped graph in Example 2 is called a **cardioid**. In general, the graph of any polar equation of the form

$$r = a(1 + \cos \theta), \quad r = a(1 + \sin \theta)$$

$$r = a(1 - \cos \theta), \quad r = a(1 - \sin \theta)$$

where a is a real number, is a cardioid.

The graph of an equation of the form

$$r = a + b \cos \theta \quad \text{or} \quad r = a + b \sin \theta$$

where $a \neq b$, is called a **limaçon**. The graph is similar in shape to a cardioid; however, there may be an added “loop” as shown in the next example.

Example 3 Sketch the graph of $r = 2 + 4 \cos \theta$.

Solution Some points corresponding to $0 \leq \theta \leq \pi$ are exhibited in the following table.

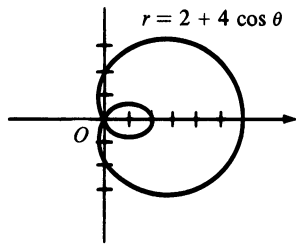


FIGURE 13.11

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
r	6	$2 + 2\sqrt{3}$	$2 + 2\sqrt{2}$	4	2	0	$2 - 2\sqrt{2}$	$2 - 2\sqrt{3}$	-2

Note that $r = 0$ at $\theta = 2\pi/3$. The values of r are negative if $2\pi/3 < \theta \leq \pi$, and this leads to the lower half of the small loop in Figure 13.11. (Verify this fact.) Letting θ range from π to 2π gives us the upper half of the small loop and the lower half of the large loop. ■

Example 4 Sketch the graph of the equation $r = a \sin 2\theta$, where $a > 0$.

Solution Instead of tabulating solutions, let us reason as follows. If θ increases from 0 to $\pi/4$, then 2θ varies from 0 to $\pi/2$ and hence $\sin 2\theta$ increases from 0 to 1. It follows that r increases from 0 to a in the θ -interval $[0, \pi/4]$. If we next let θ increase from $\pi/4$ to $\pi/2$, then 2θ changes from $\pi/2$ to π . Consequently, r decreases from a to 0 in the θ -interval $[\pi/4, \pi/2]$. (Why?) The corresponding points on the graph constitute the first-quadrant loop illustrated in Figure 13.2. Note that the point $P(r, \theta)$ traces the loop in the counterclockwise direction as θ increases from 0 to $\pi/2$.

If we let θ vary from $\pi/2$ to π , then 2θ varies from π to 2π and, therefore, $r = a \sin 2\theta \leq 0$. Thus, the points $P(r, \theta)$ are in the fourth quadrant if $\pi/2 < \theta < \pi$. We shall leave it to the reader to verify that if θ increases from $\pi/2$ to π , then $P(r, \theta)$ traces (in a counterclockwise direction) the loop shown in the fourth quadrant.

Similar loops are obtained for the θ -intervals $[\pi, 3\pi/2]$ and $[3\pi/2, 2\pi]$. In Figure 13.12 we have plotted only those points on the graph that correspond to the largest numerical values of r . ■

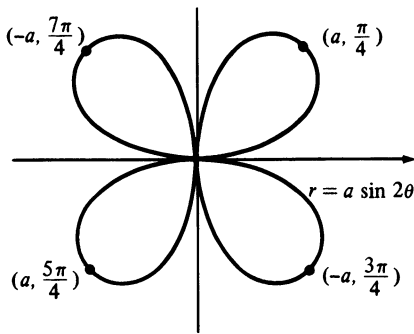


FIGURE 13.12

The graph in Example 3 is called a **four-leafed rose**. In general, any equation of the form

$$r = a \sin n\theta \quad \text{or} \quad r = a \cos n\theta$$

where n is a positive integer greater than 1 and a is a real number, has a graph which consists of a number of loops attached to the origin. If n is even there are $2n$ loops, whereas if n is odd there are n loops (see, for example, Exercises 9, 10, and 22).

Many other interesting graphs result from polar equations. Some are included in the exercises at the end of this section. Polar coordinates are very useful in applications involving circles with centers at the origin or lines that pass through the origin, since equations that have these graphs may be written in the simple forms $r = k$ or $\theta = k$ for some fixed number k . (Verify this fact.)

Let us now superimpose an xy -plane on an $r\theta$ -plane in such a way that positive x -axis coincides with the polar axis. Any point P in the plane may then be assigned rectangular coordinates (x, y) or polar coordinates (r, θ) . It is not difficult to obtain formulas that specify the relationship between the two coordinate systems. If $r > 0$ we have a situation similar to that illustrated

in (i) of Figure 13.13. If $r < 0$ we have that shown in (ii) of the figure where, for later purposes, we have also plotted the point P' having polar coordinates $(|r|, \theta)$ and rectangular coordinates $(-x, -y)$.

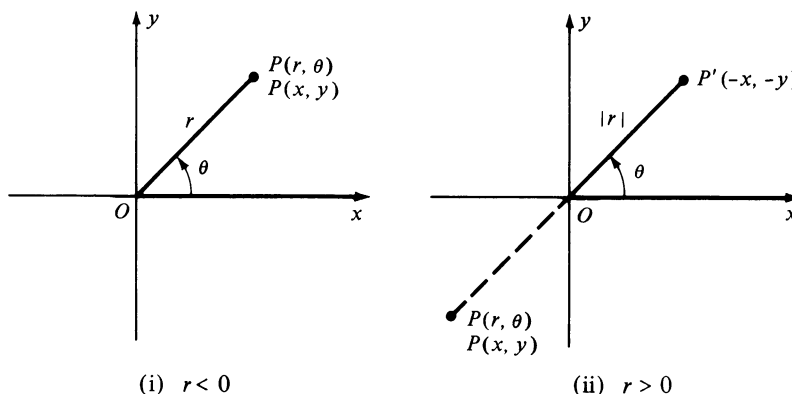


FIGURE 13.13

The following result specifies relationships between (x, y) and (r, θ) . In the statement of the theorem it is assumed that the positive x -axis coincides with the polar axis.

Theorem (13.4)

The rectangular coordinates (x, y) and polar coordinates (r, θ) of a point P are related as follows:

$$(i) \quad x = r \cos \theta, \quad y = r \sin \theta$$

$$(ii) \quad \tan \theta = \frac{y}{x}, \quad r^2 = x^2 + y^2$$

Proof Although we have pictured θ as an acute angle in Figure 13.13, the discussion that follows is valid for all angles. On the one hand, if $r > 0$ as in (i) of Figure 13.13, then

$$x = r \cos \theta, \quad y = r \sin \theta.$$

On the other hand, if $r < 0$ then $|r| = -r$, and from (ii) of Figure 13.13 we see that

$$\cos \theta = \frac{-x}{|r|} = \frac{-x}{-r} = \frac{x}{r},$$

$$\sin \theta = \frac{-y}{|r|} = \frac{-y}{-r} = \frac{y}{r}.$$

Multiplication by r produces (i) of (13.4) and, therefore, these formulas hold whether r is positive or negative. If $r = 0$, then the point is the pole and we again see that the formulas in (i) are true.

The formulas in (ii) follow readily from Figure 13.13. □

We may use Theorem (13.4) to change from one system of coordinates to the other. A more important use is for transforming a polar equation to an equation in x and y , and vice versa. This is illustrated in the next two examples.

Example 4 Find an equation in x and y that has the same graph as the polar equation $r = 4 \sin \theta$.

Solution The equation $r = 4 \sin \theta$ was considered in Example 1. It is convenient to multiply both sides by r , obtaining $r^2 = 4r \sin \theta$. Applying Theorem (13.4) gives us $x^2 + y^2 = 4y$. The last equation is equivalent to $x^2 + (y - 2)^2 = 4$ whose graph is a circle of radius 2 with center at $(0, 2)$ in the xy -plane. ■

Example 5 Find a polar equation of an arbitrary line.

Solution We know that every line in an xy -coordinate system is the graph of a linear equation $ax + by + c = 0$. Substituting for x and y from (i) of (13.4) leads to the polar equation

$$r(a \cos \theta + b \sin \theta) + c = 0. \quad \blacksquare$$

If we superimpose an xy -plane on an $r\theta$ -plane, then the graph of a polar equation may be symmetric with respect to the x -axis, the y -axis, or the origin. It is left to the reader to show that if a substitution listed in the following table does not change the solutions of a polar equation, then the graph has the indicated symmetry.

Tests for Symmetry (13.5)

Substitution	Symmetry
$-\theta$ for θ	the line $\theta = 0$ (x -axis)
$-r$ for r	the pole (origin)
$\pi + \theta$ for θ	the pole (origin)
$\pi - \theta$ for θ	the line $\theta = \pi/2$ (y -axis)

To illustrate, since $\cos(-\theta) = \cos \theta$, the graph of the equation in Example 2 (see Figure 13.10) is symmetric with respect to the x -axis. Since $\sin(\pi - \theta) = \sin \theta$, the graph in Example 1 is symmetric with respect to the y -axis. The graph in Example 4 is symmetric to both axes and the origin. Other tests for symmetry may be stated; however, those listed in (13.5) are among the easiest to apply.

Tangent lines to graphs of polar equations may be found by means of the next theorem.

Theorem (13.6)

The slope m of the tangent line to the graph of $r = f(\theta)$ at the point $P(r, \theta)$ is given by

$$m = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}.$$

Proof If (x, y) are the rectangular coordinates of $P(r, \theta)$, then by Theorem (13.4),

$$x = r \cos \theta = f(\theta) \cos \theta$$

$$y = r \sin \theta = f(\theta) \sin \theta.$$

These may be considered as parametric equations for the graph, where the parameter is θ . Applying Theorem (13.3), the slope of the tangent line at (x, y) is

$$\frac{dy}{dx} = \frac{(dy/d\theta)}{(dx/d\theta)} = \frac{f'(\theta) \sin \theta + f(\theta) \cos \theta}{f'(\theta) \cos \theta - f(\theta) \sin \theta}.$$

This is equivalent to the formula in the statement of the theorem. $\square$

Horizontal tangent lines occur if the numerator in the formula for m is 0 and the denominator is not 0. Vertical tangent lines occur if the denominator is 0 and the numerator is not 0. The case 0/0 requires special attention.

To find the slopes of the tangent lines at the pole, it is necessary to determine the values of θ for which $r = 0$. For such values (and with $r = 0$) the formula in Theorem (13.6) reduces to $m = \tan \theta$. These remarks are illustrated in the next example.

Example 6 Given the cardioid $r = 2 + 2 \cos \theta$, find

- (a) the slope of the tangent line at $\theta = \pi/6$.
- (b) the slope of the tangent line at the pole.
- (c) the points at which the tangent line is horizontal.
- (d) the points at which the tangent line is vertical.

Solution The graph of $r = 2 + 2 \cos \theta$ is sketched in Figure 13.10. If we apply Theorem (13.6), the slope m of the tangent line is given by

$$\begin{aligned} m &= \frac{(-2 \sin \theta) \sin \theta + (2 + 2 \cos \theta) \cos \theta}{(-2 \sin \theta) \cos \theta - (2 + 2 \cos \theta) \sin \theta} \\ &= \frac{2(\cos^2 \theta - \sin^2 \theta) + 2 \cos \theta}{-2(2 \sin \theta \cos \theta) - 2 \sin \theta} \\ &= -\frac{\cos 2\theta + \cos \theta}{\sin 2\theta + \sin \theta}. \end{aligned}$$

(a) At $\theta = \pi/6$,

$$m = -\frac{\cos(\pi/3) + \cos(\pi/6)}{\sin(\pi/3) + \sin(\pi/6)} = -\frac{(1/2) + (\sqrt{3}/2)}{(\sqrt{3}/2) + (1/2)} = -1.$$

(b) To find the slope of the tangent line at the pole we need the values of θ such that $r = 2 + 2\cos\theta = 0$. This gives us $\cos\theta = -1$, or $\theta = \pi$. Substitution in the formula for m produces the meaningless expression $0/0$, and hence we let $r = 0$ in Theorem (13.6), obtaining $m = \tan\theta$. Hence at the pole, $m = \tan\pi = 0$.

(c) To find horizontal tangents we let

$$\cos 2\theta + \cos\theta = 0.$$

This equation may be written as

$$2\cos^2\theta - 1 + \cos\theta = 0,$$

or $(2\cos\theta - 1)(\cos\theta + 1) = 0.$

From $\cos\theta = 1/2$ we obtain $\theta = \pi/3$ and $\theta = 5\pi/3$. The corresponding points are $(3, \pi/3)$ and $(3, 5\pi/3)$. Using $\cos\theta = -1$ gives us $\theta = \pi$. Since the denominator in the formula for m is 0 at $\theta = \pi$, further investigation is required. Indeed, we saw in (b), that there is a horizontal tangent line at the point $(0, \pi)$.

(d) To find vertical tangent lines we let

$$\sin 2\theta + \sin\theta = 0.$$

Equivalent equations are

$$2\sin\theta\cos\theta + \sin\theta = 0,$$

and $\sin\theta(2\cos\theta + 1) = 0.$

Letting $\sin\theta = 0$ and $\cos\theta = -1/2$ leads to the following values of θ : $0, \pi, 2\pi/3$ and $4\pi/3$. We found, in (c), that π gives us a horizontal tangent. The remaining values result in the points $(4, 0)$, $(1, 2\pi/3)$ and $(1, 4\pi/3)$, at which the graph has vertical tangent lines. ■

13.3 Exercises

Sketch the graphs of the polar equations in Exercises 1–24.

1 $r = 5$

2 $\theta = \pi/4$

3 $\theta = -\pi/6$

4 $r = -2$

5 $r = 4\cos\theta$

6 $r = -2\sin\theta$

7 $r = 4(1 - \sin\theta)$

8 $r = 1 + 2\cos\theta$

9 $r = 8\cos 3\theta$

10 $r = 2\sin 4\theta$

11 $r^2 = 4\cos 2\theta$ (lemniscate)

- 12 $r = 6 \sin^2 (\frac{1}{2}\theta)$
 13 $r = 4 \csc \theta$
 14 $r = -3 \sec \theta$
 15 $r = 2 - \cos \theta$
 16 $r = 2 + 2 \sec \theta$ (conchoid)
 17 $r = 2^\theta, \theta \geq 0$ (spiral)
 18 $r\theta = 1, \theta > 0$ (spiral)
 19 $r = -6(1 + \cos \theta)$
 20 $r = e^{2\theta}$ (logarithmic spiral)
 21 $r = 2 + 4 \sin \theta$
 22 $r = 8 \cos 5\theta$
 23 $r^2 = -16 \sin 2\theta$
 24 $4r = \theta$

In Exercises 25–32 find a polar equation that has the same graph as the given equation.

- 25 $x = -3$
 26 $y = 2$
 27 $x^2 + y^2 = 16$
 28 $x^2 = 8y$
 29 $y = 6$
 30 $y = 6x$
 31 $x^2 - y^2 = 16$
 32 $9x^2 + 4y^2 = 36$

In Exercises 33–48 find an equation in x and y that has the same graph as the given polar equation and use it as an aid in sketching the graph in a polar coordinate system.

- 33 $r \cos \theta = 5$
 34 $r \sin \theta = -2$
 35 $r - 6 \sin \theta = 0$
 36 $r = 6 \cot \theta$
 37 $r = 2$
 38 $\theta = \pi/4$
 39 $r = \tan \theta$
 40 $r = 4 \sec \theta$
 41 $r^2(4 \sin^2 \theta - 9 \cos^2 \theta) = 36$
 42 $r^2(\cos^2 \theta + 4 \sin^2 \theta) = 16$
 43 $r^2 \cos 2\theta = 1$
 44 $r^2 \sin 2\theta = 4$
 45 $r(\sin \theta - 2 \cos \theta) = 6$
 46 $r(\sin \theta + r \cos^2 \theta) = 1$

- 47 $r = 1/(1 + \cos \theta)$
 48 $r = 4/(2 + \sin \theta)$
 49 If $P_1(r_1, \theta_1)$ and $P_2(r_2, \theta_2)$ are points in an $r\theta$ -plane, use the Law of Cosines to prove that

$$[d(P_1, P_2)]^2 = r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_2 - \theta_1).$$

 50 Prove that the graphs of the following polar equations are circles and find the center and radius in each case.
 (a) $r = a \sin \theta, a \neq 0$
 (b) $r = b \cos \theta, b \neq 0$
 (c) $r = a \sin \theta + b \cos \theta, ab \neq 0$

In Exercises 51–60 find the slope of the tangent line to the graph of the equation at the indicated value of θ . (Refer to the graphs in Exercises 5–12 and 17–18.)

- 51 $r = 4 \cos \theta, \theta = \pi/3$
 52 $r = -2 \sin \theta, \theta = \pi/6$
 53 $r = 4(1 - \sin \theta), \theta = 0$
 54 $r = 1 + 2 \cos \theta, \theta = \pi/2$
 55 $r = 8 \cos 3\theta, \theta = \pi/4$
 56 $r = 2 \sin 4\theta, \theta = \pi/4$
 57 $r^2 = 4 \cos 2\theta, \theta = \pi/6$
 58 $r = 6 \sin^2 (\frac{1}{2}\theta), \theta = 2\pi/3$
 59 $r = 2^\theta, \theta = \pi$
 60 $r\theta = 1, \theta = 2\pi$
 61 If $\cos \theta \neq 0$, show that the slope of the tangent line to the graph of $r = f(\theta)$ is given by

$$m = \frac{(dr/d\theta) \tan \theta + r}{(dr/d\theta) - r \tan \theta}.$$

- 62 Let $a > 0$ and let C be the set of points in a coordinate plane, the product of whose distances from $P(a, 0)$ and $Q(-a, 0)$ is equal to a^2 . Show that a polar equation for C is $r^2 = 2a^2 \cos 2\theta$. (Hint: Use Exercise 49.)
 63 If the graphs of the polar equations $r = f(\theta)$ and $r = g(\theta)$ intersect at $P(r, \theta)$, prove that the tangent lines at P are perpendicular if and only if

$$f'(\theta)g'(\theta) + f(\theta)g(\theta) = 0.$$

(The graphs are said to be *orthogonal* at P .)

- 64 Use Exercise 63 to prove that the graphs of each of the following pairs of equations are orthogonal at their points of intersection.
 (a) $r = a \sin \theta, r = a \cos \theta$
 (b) $r = a\theta, r\theta = a$

13.4 Polar Equations of Conics

The following theorem provides another method for describing conic sections.

Theorem (13.7)

Let F be a fixed point and l a fixed line in a plane. The set of all points P in the plane such that the ratio $d(P, F)/d(P, Q)$ is a positive constant e , where $d(P, Q)$ is the distance from P to l , is a conic section. Moreover, the conic is a parabola if $e = 1$, an ellipse if $0 < e < 1$, and a hyperbola if $e > 1$.

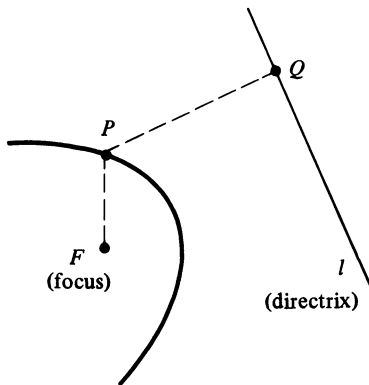


FIGURE 13.14

The constant e is called the **eccentricity** of the conic and should not be confused with the base of the natural logarithms. It will be seen that the point F is a focus of the conic. The line l is called a **directrix**. A typical situation is illustrated in Figure 13.14, where the curve indicates possible positions of P .

We shall prove the theorem if $e \leq 1$ and leave the case $e > 1$ to the reader.

If $e = 1$ in Theorem (13.7), then $d(P, F) = d(P, Q)$ and, according to Definition (12.1), a parabola with focus F and directrix l is obtained.

Suppose next that $0 < e < 1$. It is convenient to introduce a polar coordinate system in the plane with F as the pole and with l perpendicular to the polar axis at the point $D(d, 0)$, where $d > 0$. If $P(r, \theta)$ is a point in the plane such that $d(P, F)/d(P, Q) = e < 1$, then from Figure 13.15 we see that P lies to the left of l . Let C be the projection of P on the polar axis. Since

$$d(P, F) = r \quad \text{and} \quad d(P, Q) = \overline{FD} - \overline{FC} = d - r \cos \theta,$$

it follows that P satisfies the condition in (13.7) if and only if

$$\frac{r}{d - r \cos \theta} = e$$

or

$$r = de - er \cos \theta.$$

Solving for r gives us

$$r = \frac{de}{1 + e \cos \theta}$$

which is a polar equation of the graph. Actually, the same equation is obtained if $e = 1$; however, there is no point (r, θ) on the graph if $1 + \cos \theta = 0$.

The rectangular equation corresponding to $r = de - er \cos \theta$ is

$$\pm \sqrt{x^2 + y^2} = de - ex.$$

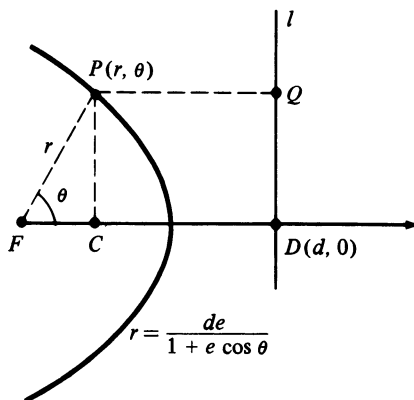


FIGURE 13.15

Squaring both sides and rearranging terms leads to

$$(1 - e^2)x^2 + 2de^2x + y^2 = d^2e^2.$$

Completing the square in the previous equation and simplifying, we obtain

$$\left(x + \frac{de^2}{1 - e^2}\right)^2 + \frac{y^2}{1 - e^2} = \frac{d^2e^2}{(1 - e^2)^2}.$$

Finally, dividing both sides by $d^2e^2/(1 - e^2)^2$ leads to an equation of the form

$$\frac{(x - h)^2}{a^2} + \frac{y^2}{b^2} = 1$$

where $h = -de^2/(1 - e^2)$. Consequently, the graph is an ellipse with center at the point $(-de^2/(1 - e^2), 0)$ on the x -axis and where

$$a^2 = \frac{d^2e^2}{(1 - e^2)^2}, \quad b^2 = \frac{d^2e^2}{1 - e^2}.$$

By Theorem (12.6),

$$c^2 = a^2 - b^2 = \frac{d^2e^4}{(1 - e^2)^2}$$

and hence $c = de^2/(1 - e^2)$. This proves that F is a focus of the ellipse. It also follows that $e = c/a$. A similar proof may be given for the case $e > 1$.

It can be shown, conversely, that every conic that is not a circle may be described by means of the statement in Theorem (13.7). This gives us a formulation of conic sections that is equivalent to the approach used in Chapter 12. Since (13.7) includes all three types of conics, it is sometimes regarded as a definition for the conic sections.

If we had chosen the focus F to the *right* of the directrix, as illustrated in Figure 13.16 (where $d > 0$), then the equation $r = de/(1 - e \cos \theta)$ would have resulted. (Note the minus sign in place of the plus sign.) Other sign changes occur if d is allowed to be negative.

If l is taken *parallel* to the polar axis through one of the points $(d, \pi/2)$ or $(d, 3\pi/2)$, as illustrated in Figure 13.17, then the corresponding equations would contain $\sin \theta$ instead of $\cos \theta$. The proofs of these facts are left to the reader.

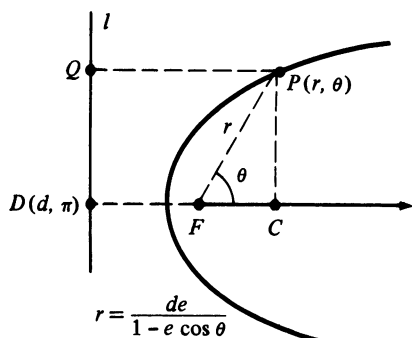


FIGURE 13.16

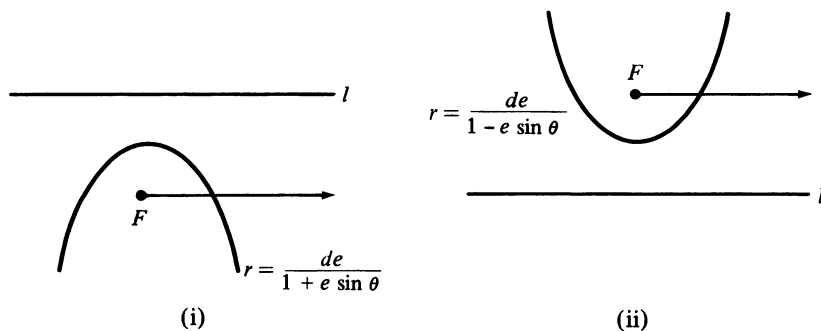


FIGURE 13.17

The following theorem summarizes our discussion.

Theorem (13.8)

A polar equation having one of the four forms

$$r = \frac{de}{1 \pm e \cos \theta}, \quad r = \frac{de}{1 \pm e \sin \theta}$$

is a conic section. Moreover, the conic is a parabola if $e = 1$, an ellipse if $0 < e < 1$, or a hyperbola if $e > 1$.

Example 1 Describe and sketch the graph of the equation

$$r = \frac{10}{3 + 2 \cos \theta}.$$

Solution Dividing numerator and denominator of the given fraction by 3 gives us

$$r = \frac{\frac{10}{3}}{1 + \frac{2}{3} \cos \theta}$$

which has one of the forms in Theorem (13.8) with $e = \frac{2}{3}$. Thus the graph is an ellipse with focus F at the pole and major axis along the polar axis. The endpoints of the major axis may be found by setting θ equal to 0 and π . This gives us $V(2, 0)$ and $V'(10, \pi)$. Hence

$$2a = d(V', V) = 12, \quad \text{or} \quad a = 6.$$

The center of the ellipse is the midpoint of the segment $V'V$, namely $(4, \pi)$. Using the fact that $e = c/a$, we obtain

$$c = ae = 6\left(\frac{2}{3}\right) = 4.$$

Hence

$$b^2 = a^2 - c^2 = 36 - 16 = 20,$$

that is, the semiminor axis has length $\sqrt{20}$. The graph is sketched in Figure 13.18 where, for reference, we have superimposed a rectangular coordinate system on the polar system. ■

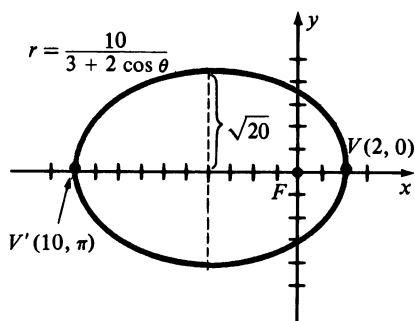


FIGURE 13.18

Example 2 Describe and sketch the graph of the equation

$$r = \frac{10}{2 + 3 \sin \theta}.$$

Solution To express the equation in one of the forms in Theorem (13.8) we divide numerator and denominator of the given fraction by 2, obtaining

$$r = \frac{5}{1 + \frac{3}{2} \sin \theta}.$$

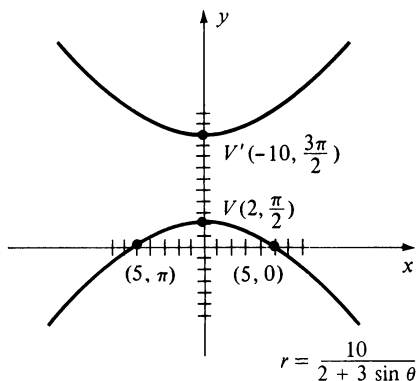


FIGURE 13.19

Thus $e = \frac{3}{2}$ and the graph is a hyperbola with a focus at the pole. The expression $\sin \theta$ tells us that the transverse axis of the hyperbola is perpendicular to the polar axis. To find the vertices we let θ equal $\pi/2$ and $3\pi/2$ in the given equation. This gives us the points $V(2, \pi/2)$, $V'(-10, 3\pi/2)$, and hence

$$2a = d(V, V') = 8, \quad \text{or} \quad a = 4.$$

The points $(5, 0)$ and $(5, \pi)$ on the graph can be used to get a rough estimate of the lower branch of the hyperbola. The upper branch is obtained by symmetry, as illustrated in Figure 13.19. If more accuracy or additional information is desired, we may calculate

$$c = ae = 4(\frac{3}{2}) = 6$$

and

$$b^2 = c^2 - a^2 = 36 - 16 = 20.$$

Asymptotes may then be constructed in the usual way. ■

Example 3 Sketch the graph of the equation

$$r = \frac{15}{4 - 4 \cos \theta}.$$

Solution To obtain one of the forms in (13.8) we divide numerator and denominator of the given fraction by 4, obtaining

$$r = \frac{\frac{15}{4}}{1 - \cos \theta}.$$

Consequently, $e = 1$ and the graph is a parabola with focus at the pole. A rough sketch can be found by plotting the points which correspond to the x - and y -intercepts. These are indicated in the following table.

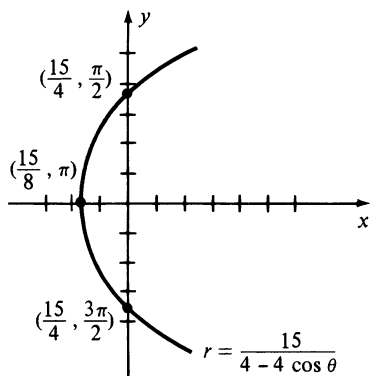


FIGURE 13.20

θ	0	$\pi/2$	π	$3\pi/2$
r	—	15/4	15/8	15/4

Note that there is no point on the graph corresponding to $\theta = 0$. (Why?) Plotting the three points and using the fact that the graph is a parabola with focus at the pole gives us the sketch in Figure 13.20. ■

If only a rough sketch of a conic is desired, then the technique employed in Example 3 is recommended. To use this method we plot (if possible) points corresponding to $\theta = 0, \pi/2, \pi$, and $3\pi/2$. These points, together with the type of conic (obtained from the value of e), readily lead to the sketch.

13.4 Exercises

In Exercises 1–10, identify and sketch the graph of the polar equation.

$$1 \quad r = \frac{12}{6 + 2 \sin \theta}$$

$$2 \quad r = \frac{12}{6 - 2 \sin \theta}$$

$$3 \quad r = \frac{12}{2 - 6 \cos \theta}$$

$$4 \quad r = \frac{12}{2 + 6 \cos \theta}$$

$$5 \quad r = \frac{3}{2 + 2 \cos \theta}$$

$$6 \quad r = \frac{3}{2 - 2 \sin \theta}$$

$$7 \quad r = \frac{4}{\cos \theta - 2}$$

$$8 \quad r = \frac{4 \sec \theta}{2 \sec \theta - 1}$$

$$9 \quad r = \frac{6 \csc \theta}{2 \csc \theta + 3}$$

$$10 \quad r = \csc \theta (\csc \theta - \cot \theta)$$

11–20 Find rectangular equations for the graphs in Exercises 1–10.

In Exercises 21–26 use Theorem (13.7) to find a polar equation of the conic with focus at the pole and the given eccentricity and equation of directrix.

$$21 \quad e = 1/3, r = 2 \sec \theta$$

$$22 \quad e = 2/5, r = 4 \csc \theta$$

$$23 \quad e = 4, r = -3 \csc \theta$$

$$24 \quad e = 3, r = -4 \sec \theta$$

$$25 \quad e = 1, r \cos \theta = 5$$

$$26 \quad e = 1, r \sin \theta = -2$$

27 Find a polar equation of the parabola with focus at the pole and vertex $(4, \pi/2)$.

28 Find a polar equation of an ellipse with eccentricity $2/3$, a vertex at $(1, 3\pi/2)$, and a focus at the pole.

29 Prove Theorem (13.7) for the case $e > 1$.

30 (a) Use Figure 3.16 to derive the formula

$$r = de/(1 - e \cos \theta).$$

(b) Derive the formulas $r = de/(1 \pm e \sin \theta)$ in Theorem (13.8) by using Figure 13.17.

In Exercises 31–36, express the given equation in one of the forms in Theorem (13.8) and then find the eccentricity and an equation for the directrix.

$$31 \quad y^2 = 4 - 4x$$

$$32 \quad x^2 = 1 - 2y$$

$$33 \quad 3y^2 - 16y - x^2 + 16 = 0$$

$$34 \quad 5x^2 + 9y^2 = 32x + 64$$

$$35 \quad 8x^2 + 9y^2 + 4x = 4$$

$$36 \quad 4x^2 - 5y^2 + 36y - 36 = 0$$

In Exercises 37–40, find the slope of the tangent line to the graph of the equation at the point corresponding to the indicated value of θ . (Refer to the graphs in Exercises 1–4.)

$$37 \quad r = \frac{12}{6 + 2 \sin \theta}, \theta = \pi/6$$

$$38 \quad r = \frac{12}{6 - 2 \sin \theta}, \theta = 0$$

$$39 \quad r = \frac{12}{2 - 6 \cos \theta}, \theta = \pi/2$$

$$40 \quad r = \frac{12}{2 + 6 \cos \theta}, \theta = \pi/3$$

41 Given the hyperbola $r = ep/(1 - e \cos \theta)$, where $e > 1$, show that the angles between the line through the polar axis and the asymptotes are given by $\arccos(\pm 1/e)$.

42 A **focal chord** of a conic is a line segment through a focus with endpoints on the conic. Let d_1 and d_2 be the lengths into which a focal chord is divided by the focus. Show that the sum of the reciprocals of d_1 and d_2 is the same for all focal chords of a given conic.

13.5 Areas in Polar Coordinates

Areas of certain regions that are bounded by graphs of polar equations can be found by employing limits of sums of the areas of circular sectors. Suppose R is a region in an $r\theta$ -plane bounded by the lines through O with equations $\theta = a$ and $\theta = b$, where $0 \leq a < b \leq 2\pi$, and by the graph of $r = f(\theta)$, where f is continuous and $f(\theta) \geq 0$ on $[a, b]$. A region of this type is illustrated in (i) of Figure 13.21.

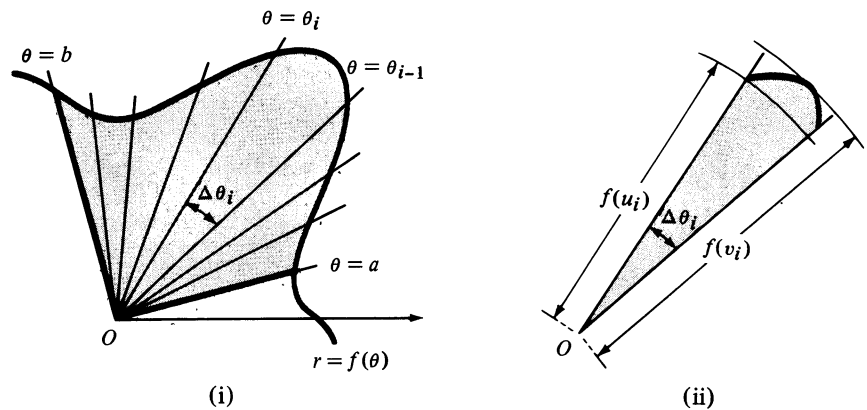


FIGURE 13.21

Let P denote a partition of $[a, b]$ determined by

$$a = \theta_0 < \theta_1 < \theta_2 < \cdots < \theta_n = b$$

and let $\Delta\theta_i = \theta_i - \theta_{i-1}$ for $i = 1, 2, \dots, n$. The lines with equations $\theta = \theta_i$ divide R into wedge-shaped subregions. If $f(u_i)$ is the minimum value and $f(v_i)$ is the maximum value of f on $[\theta_{i-1}, \theta_i]$, then as illustrated in (ii) of Figure 13.21, the area ΔA_i of the i th subregion is between the areas of the inscribed and circumscribed circular sectors having central angle $\Delta\theta_i$ and radii $f(u_i)$ and $f(v_i)$, respectively. Hence, by (2.20),

$$\frac{1}{2}[f(u_i)]^2 \Delta\theta_i \leq \Delta A_i \leq \frac{1}{2}[f(v_i)]^2 \Delta\theta_i.$$

Summing from $i = 1$ to $i = n$, and using the fact that the sum of the ΔA_i is the area A of R , we obtain

$$\sum_{i=1}^n \frac{1}{2}[f(u_i)]^2 \Delta\theta_i \leq A \leq \sum_{i=1}^n \frac{1}{2}[f(v_i)]^2 \Delta\theta_i.$$

The limit of each sum, as the norm $\|P\|$ of the subdivision approaches zero, equals the integral $\int_a^b \frac{1}{2}[f(\theta)]^2 d\theta$. This gives us the following result.

Theorem (13.9)

If f is continuous and $f(\theta) \geq 0$ on $[a, b]$, where $0 \leq a < b < 2\pi$, then the area A of the region bounded by the graphs of $r = f(\theta)$, $\theta = a$, and $\theta = b$ is

$$A = \int_a^b \frac{1}{2} [f(\theta)]^2 d\theta = \int_a^b \frac{1}{2} r^2 d\theta.$$

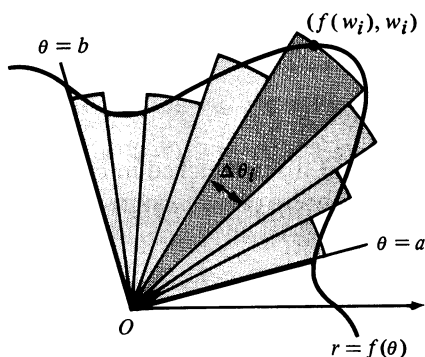


FIGURE 13.22

The integral in Theorem (13.9) may be interpreted as a limit of a sum by writing

$$A = \int_a^b \frac{1}{2} [f(\theta)]^2 d\theta = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n \frac{1}{2} [f(w_i)]^2 \Delta\theta_i$$

where w_i is any number in the subinterval $[\theta_{i-1}, \theta_i]$ of $[a, b]$. The sketch in Figure 13.22 provides a geometric illustration of a typical Riemann sum. It is convenient to think of starting with the area of a circular sector of the type shown in Figure 13.21, and then sweeping out the region R by letting θ vary from a to b while at the same time each $\Delta\theta_i$ approaches zero.

Example 1 Find the area of the region that is bounded by the cardioid $r = 2 + 2 \cos \theta$.

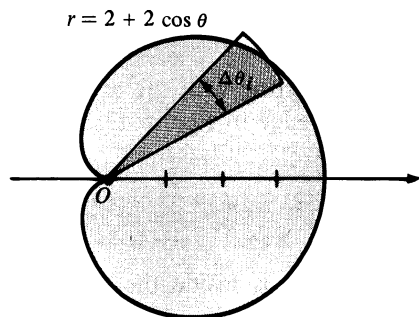


FIGURE 13.23

Solution The graph was sketched in Figure 13.10. It is sketched again in Figure 13.23 together with a typical circular sector of the type discussed previously. Making use of symmetry we shall find the area of the upper half of the region and double the result. The function f of Theorem (13.9) is given by $f(\theta) = 2 + 2 \cos \theta$. To find a and b we observe that the circular sectors will sweep out the upper half of the cardioid if θ varies from 0 to π . Consequently, $a = 0$, $b = \pi$, and

$$\begin{aligned} A &= 2 \int_0^\pi \frac{1}{2} (2 + 2 \cos \theta)^2 d\theta \\ &= \int_0^\pi (4 + 8 \cos \theta + 4 \cos^2 \theta) d\theta. \end{aligned}$$

Using the fact that $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$,

$$\begin{aligned} A &= \int_0^\pi (6 + 8 \cos \theta + 2 \cos 2\theta) d\theta \\ &= \left[6\theta + 8 \sin \theta + \sin 2\theta \right]_0^\pi = 6\pi. \end{aligned}$$

The area could also have been found by using (13.9) with $a = 0$ and $b = 2\pi$. ■

The next result generalizes Theorem (13.9).

Theorem (13.10)

Let f and g be continuous functions such that $f(\theta) \geq g(\theta) \geq 0$ for every θ in $[a, b]$, where $0 \leq a < b < 2\pi$. Let R denote the region bounded by the graphs of $r = f(\theta)$, $r = g(\theta)$, $\theta = a$ and $\theta = b$. The area A of R is

$$A = \frac{1}{2} \int_a^b ([f(\theta)]^2 - [g(\theta)]^2) d\theta.$$

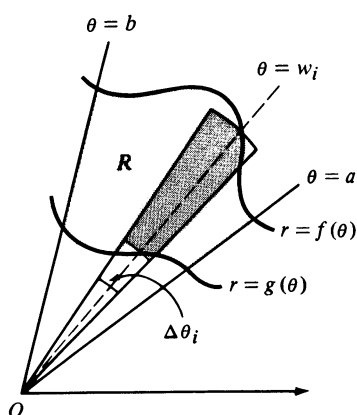


FIGURE 13.24

A region of the type described in Theorem (13.10) is sketched in Figure 13.24. The area A can be found by subtracting the areas of two regions of the type described in (13.9). Specifically,

$$A = \int_a^b \frac{1}{2} [f(\theta)]^2 d\theta - \int_a^b \frac{1}{2} [g(\theta)]^2 d\theta$$

Writing this as one integral gives us Theorem (13.10).

The integral in (13.10) can be expressed as a limit of a sum as follows:

$$A = \lim_{\|P\| \rightarrow 0} \sum_i \frac{1}{2} ([f(w_i)]^2 - [g(w_i)]^2) \Delta\theta_i.$$

The sum may be thought of as the process of starting with the area of a region of the type illustrated by the truncated wedge in Figure 13.24 and then sweeping out R by letting θ vary from a to b .

Example 2 Find the area A of the region which is inside the cardioid $r = 2 + 2 \cos \theta$ and outside the circle $r = 3$.

Solution The region is sketched in Figure 13.25 together with a typical polar element of area. Since the two graphs intersect at the points $(3, -\pi/3)$ and $(3, \pi/3)$, the region will be swept out by the indicated element if θ varies from $-\pi/3$ to $\pi/3$. Using Theorem (13.10),

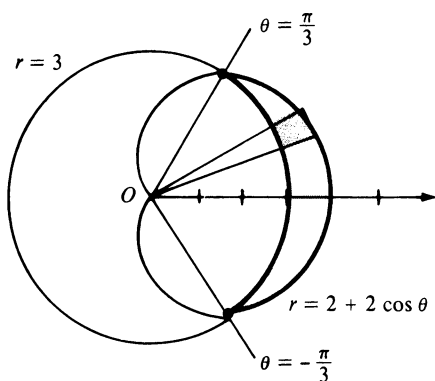


FIGURE 13.25

$$\begin{aligned} A &= \frac{1}{2} \int_{-\pi/3}^{\pi/3} [(2 + 2 \cos \theta)^2 - 3^2] d\theta \\ &= \frac{1}{2} \int_{-\pi/3}^{\pi/3} (8 \cos \theta + 4 \cos^2 \theta - 5) d\theta. \end{aligned}$$

As in Example 1, the integral may be evaluated by letting $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$. It is left to the reader to show that $A = (9\sqrt{3}/2) - \pi$. ■

13.5 Exercises

In Exercises 1–8 sketch the graph of the equation and find the area of the region bounded by the graph.

1 $r = 2 \cos \theta$

2 $r = 5 \sin \theta$

3 $r = 1 - \cos \theta$

4 $r = 6 - 6 \sin \theta$

5 $r = \sin 2\theta$

6 $r^2 = 9 \cos 2\theta$

7 $r = 4 + \sin \theta$

8 $r = 3 + 2 \cos \theta$

In Exercises 9 and 10 find the area of region R .

9 $R = \{(r, \theta): 0 \leq \theta \leq \pi/2, 0 \leq r \leq e^\theta\}$.

10 $R = \{(r, \theta): 0 \leq \theta \leq \pi, 0 \leq r \leq 2\theta\}$.

In Exercises 11–14, find the area of the region bounded by one loop of the graph of the given equation.

11 $r^2 = 4 \cos 2\theta$

12 $r = 2 \cos 3\theta$

13 $r = 3 \cos 5\theta$

14 $r = \sin 6\theta$

In Exercises 15–18 find the area of the region that is outside the graph of the first equation and inside the graph of the second equation.

15 $r = 2 + 2 \cos \theta, r = 3$

16 $r = 2, r = 4 \cos \theta$

17 $r = 2, r^2 = 8 \sin 2\theta$

18 $r = 1 - \sin \theta, r = 3 \sin \theta$

In Exercises 19–22, find the area of the region that is inside the graphs of *both* of the given equations.

19 $r = \sin \theta, r = \sqrt{3} \cos \theta$

20 $r = 2(1 + \sin \theta), r = 1$

21 $r = 1 + \sin \theta, r = 5 \sin \theta$

22 $r^2 = 4 \cos 2\theta, r = 1$

In Exercises 23–26, find the area of the region bounded by the graphs of the equations.

23 $r = 2 \sec \theta, \theta = \pi/6, \theta = \pi/3$

24 $r = \csc \theta \cot \theta, \theta = \pi/6, \theta = \pi/4$

25 $r = 4/(1 - \cos \theta), \theta = \pi/4$

26 $r(1 + \sin \theta) = 2, \theta = \pi/3$

27 A line crosses a circular disc of radius a at a distance b from its center, where $0 < b < a$. Find the area of the smaller of the two circular segments cut from the disc by the line.

28 Let OP be the ray from the pole to the point $P(r, \theta)$ on the spiral $r = a\theta$, where $a > 0$. If the ray makes two revolutions (starting from $\theta = 0$), find the area of the region swept out in its second revolution that was not swept out in the first revolution (see figure).

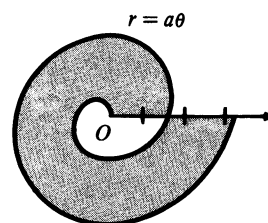


FIGURE FOR EXERCISE 28

13.6 Lengths of Curves

If a curve is the graph of an equation $y = f(x)$, where f is smooth on an interval $[a, b]$, then its arc length may be found by means of Definition (6.15). If the curve is given parametrically or in terms of polar coordinates, then (6.15) is not necessarily applicable. In this section we shall develop a formula that can be used for more general curves.

Suppose a smooth curve C is given parametrically by

$$x = f(t), \quad y = g(t)$$

where $a \leq t \leq b$. Furthermore, suppose C does not intersect itself, that is, different values of t between a and b determine different points on C . Consider a

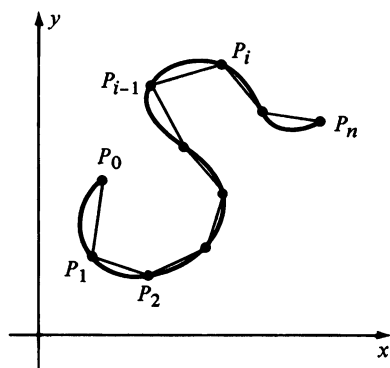


FIGURE 13.26

partition P of $[a, b]$ given by $a = t_0 < t_1 < t_2 < \cdots < t_n = b$. Let $\Delta t_i = t_i - t_{i-1}$ and let $P_i = (f(t_i), g(t_i))$ be the point on C determined by t_i . If $d(P_{i-1}, P_i)$ is the length of the line segment $P_{i-1}P_i$, then the length L_P of the broken line shown in Figure 13.26 is

$$L_P = \sum_{i=1}^n d(P_{i-1}, P_i).$$

In a manner similar to Definition (5.7) we write

$$L = \lim_{\|P\| \rightarrow 0} L_P$$

and call L the **length of C from P_0 to P_n** if, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|L_P - L| < \varepsilon$ for all partitions P with $\|P\| < \delta$.

By the Distance Formula,

$$d(P_{i-1}, P_i) = \sqrt{[f(t_i) - f(t_{i-1})]^2 + [g(t_i) - g(t_{i-1})]^2}.$$

Applying the Mean Value Theorem (4.12), there exist numbers w_i and z_i in the open interval (t_{i-1}, t_i) such that

$$\begin{aligned} f(t_i) - f(t_{i-1}) &= f'(w_i) \Delta t_i, \\ g(t_i) - g(t_{i-1}) &= g'(z_i) \Delta t_i. \end{aligned}$$

Substituting in the formula for $d(P_{i-1}, P_i)$ and removing the common factor $(\Delta t_i)^2$ from the radicand gives us

$$d(P_{i-1}, P_i) = \sqrt{[f'(w_i)]^2 + [g'(z_i)]^2} \Delta t_i.$$

Consequently,

$$L = \lim_{\|P\| \rightarrow 0} L_P = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n \sqrt{[f'(w_i)]^2 + [g'(z_i)]^2} \Delta t_i,$$

provided the limit exists. If $w_i = z_i$ for all i , then the sum is a Riemann sum for the function k defined by $k(t) = \sqrt{[f'(t)]^2 + [g'(t)]^2}$. The limit of this sum is

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt.$$

It can be shown that the limit exists even if $w_i \neq z_i$; however, the proof requires advanced methods and hence is omitted. The next theorem summarizes this discussion.

Theorem (13.11)

If a smooth curve C is given parametrically by $x = f(t)$, $y = g(t)$, where $a \leq t \leq b$, and if C does not intersect itself, except possibly at the end-points of $[a, b]$, then the length L of C is

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

If a curve C is described in rectangular coordinates by an equation of the form $y = k(x)$, where k' is continuous on $[a, b]$, then parametric equations for C are

$$x = t, \quad y = k(t), \quad \text{where } a \leq t \leq b.$$

In this case

$$\frac{dx}{dt} = 1, \quad \frac{dy}{dt} = k'(t) = k'(x), \quad dt = dx$$

and from Theorem (13.11),

$$L = \int_a^b \sqrt{1 + [k'(x)]^2} dx.$$

This is in agreement with the arc length formula given in Definition (6.15).

Example 1 Find the length of one arch of the cycloid that has parametric equations $x = t - \sin t$, $y = 1 - \cos t$.

Solution The graph has the shape illustrated in Figure 13.4, where the radius a of the circle is 1. One arch is obtained if t varies from 0 to 2π . Applying Theorem (13.11),

$$L = \int_0^{2\pi} \sqrt{(1 - \cos t)^2 + (\sin t)^2} dt.$$

Since $\cos^2 t + \sin^2 t = 1$, the integrand reduces to

$$\sqrt{2 - 2 \cos t} = \sqrt{2} \sqrt{1 - \cos t}.$$

Thus

$$L = \int_0^{2\pi} \sqrt{2} \sqrt{1 - \cos t} dt.$$

By a half-angle formula, $\sin^2(t/2) = (1 - \cos t)/2$, or equivalently,

$$1 - \cos t = 2 \sin^2(t/2).$$

Hence $\sqrt{1 - \cos t} = \sqrt{2 \sin^2(t/2)} = \sqrt{2} |\sin(t/2)|$.

The absolute value sign may be deleted since if $0 \leq t \leq 2\pi$, then $0 \leq t/2 \leq \pi$ and hence $\sin(t/2) \geq 0$. Consequently,

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{2} \sqrt{2} \sin(t/2) dt = 2 \int_0^{2\pi} \sin(t/2) dt \\ &= -4 \cos(t/2) \Big|_0^{2\pi} = (-4)(-1) - (-4)(1) = 8. \end{aligned} \quad \blacksquare$$

The next theorem provides a formula for arc length in polar coordinates.

Theorem (13.12)

If a curve C is the graph of a polar equation $r = f(\theta)$, where f' is continuous on $[a, b]$, then the length L of C is

$$L = \int_a^b \sqrt{[f(\theta)]^2 + [f'(\theta)]^2} d\theta.$$

Proof As in the proof of Theorem (13.6), parametric equations for C are

$$x = f(\theta) \cos \theta, \quad y = f(\theta) \sin \theta$$

where $a \leq \theta \leq b$. Hence

$$\frac{dx}{d\theta} = -f(\theta) \sin \theta + f'(\theta) \cos \theta$$

$$\frac{dy}{d\theta} = f(\theta) \cos \theta + f'(\theta) \sin \theta.$$

It is left as an exercise to show that

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = [f(\theta)]^2 + [f'(\theta)]^2.$$

Substitution in Theorem (13.11) with $\theta = t$ gives us the desired formula. $\square$

Example 2 Find the length of the spiral having polar equation $r = e^{\theta/2}$, from $(\sqrt{e}, 1)$ to $(e, 2)$.

Solution A portion of the spiral is sketched in Figure 13.27, together with the points corresponding to $\theta = 1$ and $\theta = 2$. Using Theorem (13.12) with $f(\theta) = e^{\theta/2}$,

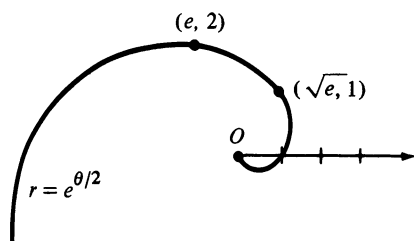


FIGURE 13.27

$$\begin{aligned} L &= \int_1^2 \sqrt{(e^{\theta/2})^2 + (\tfrac{1}{2}e^{\theta/2})^2} d\theta \\ &= \int_1^2 \sqrt{\tfrac{5}{4}e^{\theta}} d\theta = \frac{\sqrt{5}}{2} \int_1^2 e^{\theta/2} d\theta \\ &= \left[\sqrt{5} e^{\theta/2} \right]_1^2 = \sqrt{5}(e - \sqrt{e}). \end{aligned}$$

If an approximation is desired, $L \approx 2.4$. $\blacksquare$

13.6 Exercises

Find the lengths of the curves defined in Exercises 1–7.

1 $x = 5t^2, y = 2t^3; 0 \leq t \leq 1$

2 $x = 3t, y = 2t^{3/2}; 0 \leq t \leq 4$

3 $x = e^t \cos t, y = e^t \sin t; 0 \leq t \leq \pi/2$

4 $x = 3t^2 - 5, y = 2t^2 + 1; 0 \leq t \leq 2$

5 $x = \ln \cos t, y = t; 0 \leq t \leq \pi/3$

6 $x = \cos^3 t, y = \sin^3 t; 0 \leq t \leq \pi/2$

7 $x = 2t, y = t^4 + \tfrac{1}{8}t^{-2}; 1 \leq t \leq 2$

8 Find the length of the curve having polar equation $r = \sin^3(\theta/3)$.

9 Find the length of the spiral $r = e^{-\theta}$ from $\theta = 0$ to $\theta = 2\pi$.

10 Find the length of the spiral $r = \theta$ from $\theta = 0$ to $\theta = 4\pi$.

11 Find the length of the curve $r = \cos^2(\theta/2)$ from $\theta = 0$ to $\theta = \pi$.

12 Find the length of the spiral $r = 2^\theta$ from $\theta = 0$ to $\theta = \pi$.

The parametric equations in Exercises 13–16 give the position (x, y) of a particle at time t . Find the distance the particle travels during the indicated time interval.

13 $x = 4t + 3, y = 2t^2; 0 \leq t \leq 5$

14 $x = \cos 2t, y = \sin^2 t; 0 \leq t \leq \pi$

15 $x = t \cos t - \sin t, y = t \sin t + \cos t; 0 \leq t \leq \pi/2$

16 $x = t^2, y = 2t; 0 \leq t \leq 4$

17 Verify the formula for $(dx/d\theta)^2 + (dy/d\theta)^2$ in the proof of Theorem (13.12).

18 In a manner analogous to the work with plane areas, define the *centroid* $(\bar{x}, \bar{y})$ of a curve and state integral formulas for finding $(\bar{x}, \bar{y})$ when the curve is given parametrically.

19 Find the length of the curve defined by $x = 1/t, y = \ln t; 1 \leq t \leq 2$.

20 Find the length of the cardioid $r = 1 + \cos \theta$.

13.7 Surfaces of Revolution

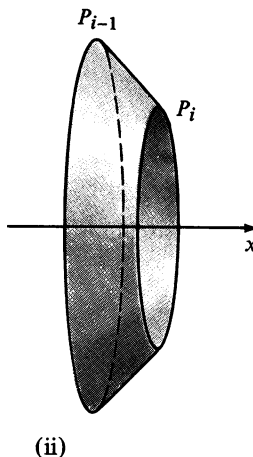
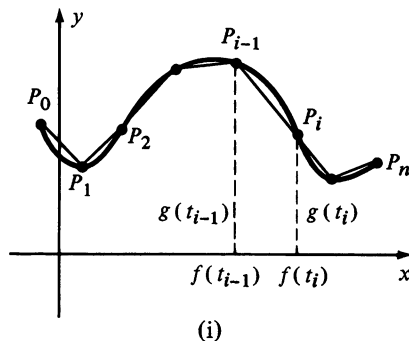


FIGURE 13.28

If a plane curve C is revolved about a line in the plane, a **surface of revolution** is generated. For example, if a circle is revolved about a diameter, the surface of a sphere is obtained. Formulas for surface area may be developed provided C is sufficiently well-behaved. In particular, suppose a smooth curve C is given parametrically by $x = f(t), y = g(t)$, where $a \leq t \leq b$ and $g(t) \geq 0$ for all t . The last condition implies that the graph of C is above the x -axis. We shall use the notation of the previous section, letting P denote a partition of $[a, b]$ and $P_i = (f(t_i), g(t_i))$ denote the point on C that corresponds to t_i , as illustrated in (i) of Figure 13.28. Reasoning intuitively, we observe that if the norm $\|P\|$ is close to zero, then the broken line C' obtained by connecting P_{i-1} to P_i for each i is an approximation to C , and hence the area of the surface generated by revolving C' about the x -axis should approximate the area of the surface generated by C . As illustrated in (ii) of Figure 13.28, each segment $P_{i-1}P_i$ generates the lateral surface of a frustum of a right circular cone having base radii $g(t_{i-1})$ and $g(t_i)$, and slant height $d(P_{i-1}, P_i)$.

It can be shown that the lateral area of a frustum of a right circular cone of base radii r_1 and r_2 and slant height s is given by $\pi(r_1 + r_2)s$. Hence the area of the surface generated by $P_{i-1}P_i$ is

$$\pi[g(t_{i-1}) + g(t_i)]d(P_{i-1}, P_i).$$

Summing terms of this form from $i = 1$ to $i = n$ gives us the area S_P of the surface generated by the broken line C' .

If we use the expression obtained for $d(P_{i-1}, P_i)$ in Section 13.6, then

$$S_P = \sum_{i=1}^n \pi[g(t_i) + g(t_{i-1})] \sqrt{[f'(w_i)]^2 + [g'(z_i)]^2} \Delta t_i$$

where w_i and z_i are in the interval (t_{i-1}, t_i) . Since the length of C is the limit of the length L_P of C' as $\|P\| \rightarrow 0$, it is natural to define the area S of the surface generated by C as

$$S = \lim_{\|P\| \rightarrow 0} S_P.$$

From the form of S_p it is reasonable to expect that this limit is given by the definite integral in the following theorem; however, the proof is beyond the scope of this book.

Theorem (13.13)

Let a smooth curve C be given by $x = f(t)$, $y = g(t)$, where $a \leq t \leq b$, and suppose that $g(t) \geq 0$ for all t . The area S of the surface of revolution obtained by revolving C about the x -axis is

$$S = \int_a^b 2\pi g(t) \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

As an aid to remembering Theorem (13.13), recall that if ds is the differential of arc length, then by Theorem (6.18),

$$(ds)^2 = (dx)^2 + (dy)^2.$$

Assuming that ds and dt are positive, we may write

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

If we regard ds as the slant height of a typical frustum, and if (x, y) represents a point in the corresponding subarc, then *the integrand $2\pi y ds$ in Theorem (13.13) may be regarded as the product of the slant height and the circumference $2\pi y$ of the circle traced by (x, y) .*

It follows in similar fashion that *if the curve C is revolved about the y -axis, and $a \geq 0$, then S is given by*

$$S = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt,$$

where again we may regard $2\pi x$ as the circumference of a circle traced by a point (x, y) on C .

Example 1 Verify that the surface area of a sphere of radius a is $4\pi a^2$.

Solution If C is the upper half of the circle $x^2 + y^2 = a^2$, then the desired surface may be obtained by revolving C about the x -axis. Parametric equations for C are

$$x = a \cos t, \quad y = a \sin t$$

where $0 \leq t \leq \pi$. Applying Theorem (13.13) and using the trigonometric identity $\sin^2 t + \cos^2 t = 1$,

$$\begin{aligned} S &= \int_0^\pi 2\pi a \sin t \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} dt = 2\pi a^2 \int_0^\pi \sin t dt \\ &= -2\pi a^2 \left[\cos t \right]_0^\pi = -2\pi a^2 [-1 - 1] = 4\pi a^2. \end{aligned}$$

■

If C is the graph of an equation $y = f(x)$, where f is smooth on an interval $[a, b]$, then parametric equations for C are

$$x = t, \quad y = f(t)$$

where $a \leq t \leq b$. Using (13.13) and the fact that $t = x$, the area of the surface generated by revolving C around the x -axis is given by (i) of the next theorem.

Theorem (13.14)

If C is the graph of $y = f(x)$, where f' is continuous on $[a, b]$, then

- (i) the area S of the surface generated by revolving C about the x -axis is

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx.$$

- (ii) if $a \geq 0$, the area S of the surface generated by revolving C about the y -axis is

$$S = \int_a^b 2\pi x \sqrt{1 + [f'(x)]^2} dx.$$

Formulas analogous to those in Theorem (13.14) may be stated if C is the graph of an equation $x = g(y)$, where $c \leq y \leq d$.

Example 2 The arc of the parabola $y^2 = x$ from $(1, 1)$ to $(4, 2)$ is revolved about the x -axis. Find the area of the resulting surface.

Solution The surface is illustrated in Figure 13.29. Using (i) of Theorem (13.14) with $y = f(x) = x^{1/2}$ we obtain

$$\begin{aligned} S &= \int_1^4 2\pi x^{1/2} \sqrt{1 + \left(\frac{1}{2x^{1/2}}\right)^2} dx \\ &= \int_1^4 2\pi x^{1/2} \sqrt{\frac{4x + 1}{4x}} dx = \pi \int_1^4 \sqrt{4x + 1} dx \\ &= \frac{\pi}{6} (4x + 1)^{3/2} \Big|_1^4 = \frac{\pi}{6} (17^{3/2} - 5^{3/2}). \end{aligned}$$

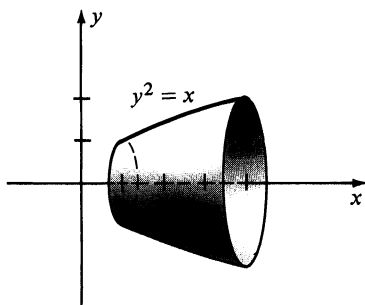


FIGURE 13.29

Formulas for S may also be stated if C is the graph of a polar equation $r = f(\theta)$, where $a \leq \theta \leq b$. In this case parametric equations for C are

$$x = f(\theta) \cos \theta, \quad y = f(\theta) \sin \theta$$

where $a \leq \theta \leq b$. If C is revolved about the polar axis, then substituting in Theorem (13.13) and using the arc length differential obtained in the proof of (13.12) leads to (i) of the next theorem.

Surface Area (Polar Coordinates)
(13.15)

$$(i) \quad S = \int_a^b 2\pi f(\theta) \sin \theta \sqrt{[f(\theta)]^2 + [f'(\theta)]^2} d\theta \quad (\text{around polar axis})$$

$$(ii) \quad S = \int_a^b 2\pi f(\theta) \cos \theta \sqrt{[f(\theta)]^2 + [f'(\theta)]^2} d\theta \quad (\text{around } \theta = \pi/2)$$

The formula in (ii) is found by using $2\pi x$ instead of $2\pi y$. When using (13.15) a and b must be chosen such that the surface does not retrace itself when C is revolved, as would be the case of revolving the graph of $r = \cos \theta$, with $0 \leq \theta \leq \pi$, about the polar axis.

Example 3 The part of the spiral $r = e^{\theta/2}$ from $\theta = 0$ to $\theta = \pi$ is revolved about the polar axis. Find the area of the resulting surface.

Solution The surface is illustrated in Figure 13.30. Using (i) of (13.15) with $f(\theta) = e^{\theta/2}$,

$$\begin{aligned} S &= \int_0^\pi 2\pi e^{\theta/2} \sin \theta \sqrt{(e^{\theta/2})^2 + (\frac{1}{2}e^{\theta/2})^2} d\theta \\ &= \int_0^\pi 2\pi e^{\theta/2} \sin \theta \sqrt{\frac{5}{4}e^\theta} d\theta = \sqrt{5}\pi \int_0^\pi e^\theta \sin \theta d\theta. \end{aligned}$$

Integration by parts leads to

$$S = \frac{\sqrt{5}\pi}{2} e^\theta (\sin \theta - \cos \theta) \Big|_0^\pi = \frac{\sqrt{5}\pi}{2} (e^\pi + 1).$$

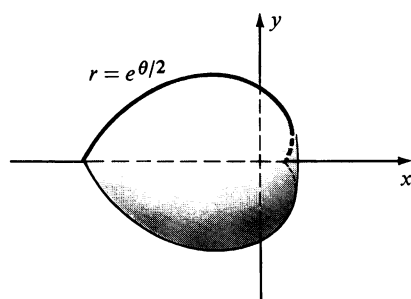


FIGURE 13.30

13.7 Exercises

In Exercises 1–8 find the area of the surface generated by revolving the given curve about the x -axis.

1 $x = t^2, y = 2t; 0 \leq t \leq 4$

2 $x = 4t, y = t^3; 1 \leq t \leq 2$

3 $x = t^2, y = t - \frac{1}{3}t^3; 0 \leq t \leq 1$

4 $x = 4t^2 + 1, y = 3 - 2t; -2 \leq t \leq 0$

5 $y = x^3$ from $x = 1$ to $x = 2$

6 $y = e^{-x}$ from $x = 0$ to $x = 1$

7 $x = a(t - \sin t), y = a(1 - \cos t); 0 \leq t \leq 2\pi$

8 $12y = 4x^3 + (3/x)$ from $x = 1$ to $x = 2$

In Exercises 9–16 find the area of the surface generated by revolving the given arc about the y -axis.

9 $x = 4t^{1/2}, y = \frac{1}{2}t^2 + t^{-1}; 1 \leq t \leq 4$

10 $x = 3t, y = t + 1; 0 \leq t \leq 5$

11 $x = e^t \sin t, y = e^t \cos t; 0 \leq t \leq \pi/2$

12 $x = 3t^2, y = 2t^3; 0 \leq t \leq 1$

13 $x^2 = 16y$ from $(0, 0)$ to $(8, 4)$

14 $8x = y^3$ from $(1, 2)$ to $(8, 4)$

15 $y = \ln x$ from $x = 1$ to $x = 2$

16 $y = \cosh x$ from $x = 1$ to $x = 2$

In Exercises 17–20 find the area of the surface generated by revolving the given curve about the polar axis.

17 $r = 2 + 2 \cos \theta$

18 $r^2 = 4 \cos 2\theta$

19 $r = 2a \sin \theta$

20 $r = 2a \cos \theta$

21 The smaller arc of the circle $x^2 + y^2 = 25$ between the points $(-3, 4)$ and $(3, 4)$ is revolved about the y -axis. Find the area of the resulting surface.

22 If the arc described in Exercise 21 is revolved about the x -axis, find the area of the resulting surface.

23 Prove that the surface area of a right circular cone of altitude a and base radius b is $\pi b \sqrt{a^2 + b^2}$.

24 If the ellipse $b^2 x^2 + a^2 y^2 = a^2 b^2$ is revolved about the y -axis, find the area of the resulting surface.

25 A **torus** is the surface generated by revolving a circle about a nonintersecting line in its plane. Find the surface area of the torus generated by revolving the circle $x^2 + y^2 = a^2$ about the line $x = b$, where $0 < a < b$.

26 The shape of a reflector in a searchlight is obtained by revolving a parabola about its axis. If the reflector is 4 feet across at the opening and 1 foot deep, find its surface area.

27 Parametric equations for the four-cusped hypocycloid $x^{2/3} + y^{2/3} = a^{2/3}$ are $x = a \cos^3 t$, $y = a \sin^3 t$, where $0 \leq t \leq 2\pi$. Find the area of the surface generated by revolving the astroid about the x -axis.

28 Prove that the area of the surface of a sphere that lies between two parallel planes depends only on the distance between the planes and not on their relative positions along a diameter of the sphere.

13.8 Review: Concepts

Define or discuss each of the following.

1 Plane curves

2 Closed curves

3 Parametric equations of a curve

4 Polar coordinate systems

5 Graphs of polar equations

6 Areas in polar coordinates

7 Length of a curve

8 Surfaces of revolution

Review: Exercises

In Exercises 1–3 sketch the graph of the curve, find a rectangular equation of a graph that contains the points on the curve, and find the slope of the tangent line at the point corresponding to $t = 1$.

1 $x = 1/t + 1$, $y = 2/t - t$; $0 < t \leq 4$

2 $x = \cos^2 t - 2$, $y = \sin t + 1$; $0 \leq t \leq 2\pi$

3 $x = \sqrt{t}$, $y = e^{-t}$; $t \geq 0$

4 Let C be the curve with parametric equations $x = t^2$, $y = 2t^3 + 4t - 1$, where t is in $\mathbb{R}$. Find the x -coordinates of the points on C at which the tangent line passes through the origin.

5 Compare the graphs of the following curves, where t is in $\mathbb{R}$.

(a) $x = t^2$, $y = t^3$

(b) $x = t^4$, $y = t^6$

(c) $x = e^{2t}$, $y = e^{3t}$

(d) $x = 1 - \sin^2 t$, $y = \cos^3 t$

In Exercises 6–18, sketch the graph of the equation and find an equation in x and y that has the same graph.

6 $r = 3 + 2 \cos \theta$

7 $r = 6 \cos 2\theta$

8 $r = 4 - 4 \cos \theta$

9 $r^2 = -4 \sin 2\theta$

10 $r = 2 \sin 3\theta$

11 $r(3 \cos \theta - 2 \sin \theta) = 6$

12 $r = e^{-\theta}$, $\theta \geq 0$

13 $r^2 = \sec 2\theta$

14 $r = 8 \sec \theta$

15 $r = 4 \cos^2(\theta/2)$

16 $r = 6 - r \cos \theta$

17 $r = 8/(3 + \cos \theta)$

18 $r = 8/(1 - 3 \sin \theta)$

- 19** Find the slope of the tangent line to the graph of $r = 3/(2 + 2 \cos \theta)$ at the point corresponding to $\theta = \pi/2$.

Find polar equations for the graphs in Exercises 20 and 21.

- 20** $x^2 + y^2 = 2xy$ **21** $y^2 = x^2 - 2x$
- 22** Find a polar equation of the hyperbola which has focus at the pole, eccentricity 2, and equation of directrix $r = 6 \sec \theta$.
- 23** Find the area of the region bounded by one loop of the graph of $r^2 = 4 \sin 2\theta$.
- 24** Find the area of the region which is inside the graph of $r = 3 + 2 \sin \theta$ and outside the graph of $r = 4$.
- 25** The position (x, y) of a particle at time t is given by $x = 2 \sin t$, $y = \sin^2 t$. Find the distance the particle travels from $t = 0$ to $t = \pi/2$.

- 26** Find the length of the spiral $r = 1/\theta$ from $(1, 1)$ to $(\frac{1}{2}, 2)$.
- 27** Find the area of the surface generated by revolving the graph of $y = \cosh x$ from $x = 0$ to $x = 1$ about the x -axis.
- 28** The curve $x = 2t^2 + 1$, $y = 4t - 3$ where $0 \leq t \leq 1$ is revolved about the y -axis. Find the area of the resulting surface.
- 29** The arc of the spiral $r = e^\theta$ from $(1, 0)$ to $(e, 1)$ is revolved about the line $\theta = \pi/2$. Find the area of the resulting surface.
- 30** Find the area of the surface generated by revolving the lemniscate $r^2 = a^2 \cos 2\theta$ about the polar axis.

14 Vectors and Solid Analytic Geometry

Vectors have two different natures, one geometric and the other algebraic. For applications it is necessary to have an understanding of both aspects. Because of this, our development will oscillate between the two approaches, sometimes emphasizing geometry and other times algebra. For simplicity we shall begin with

vectors in a plane. The extension to three dimensions is then easily accomplished by simply taking a third component into account. We shall also discuss some basic topics from solid analytic geometry including lines, planes, cylinders, quadric surfaces, and space curves.

14.1 Vectors in Two Dimensions

In previous work we assigned directions to certain lines, such as the x - and y -axes. In similar fashion, a **directed line segment** is a line segment to which a direction has been assigned. Another name for a directed line segment is a **vector**. If a vector extends from a point A (called the **initial point**) to a point B (called the **terminal point**), it is customary to place an arrowhead at B and use $\overrightarrow{AB}$ to represent the vector (see Figure 14.1). The length of the directed line segment is called the **magnitude** of the vector $\overrightarrow{AB}$ and is denoted by $|\overrightarrow{AB}|$. The vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are considered **equal**, and we write $\overrightarrow{AB} = \overrightarrow{CD}$ if and

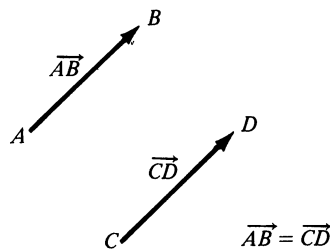


FIGURE 14.1

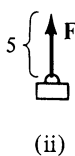
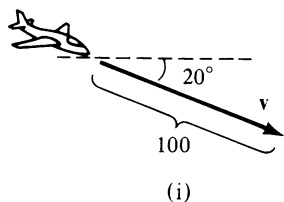


FIGURE 14.2

only if they have the same magnitude and direction, as illustrated in Figure 14.1. Consequently, vectors may be translated from one position to another, provided neither the magnitude nor direction is changed. Vectors of this type are often referred to as **free vectors**.

Many physical concepts may be represented by vectors. To illustrate, suppose an airplane is descending at a constant speed of 100 mph and the line of flight makes an angle of 20° with the horizontal. Both of these facts are represented by the vector in (i) of Figure 14.2, where it is assumed that units have been chosen so that the magnitude is 100. As shown in the figure, we will use a boldface letter such as $\mathbf{v}$ to denote a vector whose endpoints are not specified. For written work, where it is difficult to represent boldface type, a notation such as $\vec{v}$ may be employed. The vector $\mathbf{v}$ in this illustration is called a **velocity vector**.

As a second illustration, suppose a person pulls directly upward on an object with a force of 5 lb, as would be the case in lifting a 5-lb weight. We may indicate this fact by the vector $\mathbf{F}$ in (ii) of Figure 14.2. A vector that represents a pull or push of some type is called a **force vector**.

Another use for vectors is to let $\overline{AB}$ represent the path of a point as it moves along the line segment from A to B . We then refer to $\overline{AB}$ as a **displacement** of the point. As illustrated in (i) of Figure 14.3, a displacement $\overline{AB}$ followed by a displacement $\overline{BC}$ will lead to the same point as the single displacement $\overline{AC}$. By definition, the vector $\overline{AC}$ is called the **sum** of $\overline{AB}$ and $\overline{BC}$, and we write

$$\overline{AC} = \overline{AB} + \overline{BC}.$$

Since we are working with free vectors, any two vectors may be added by placing the initial point of one on the terminal point of the other and then proceeding as in (i) of Figure 14.3.

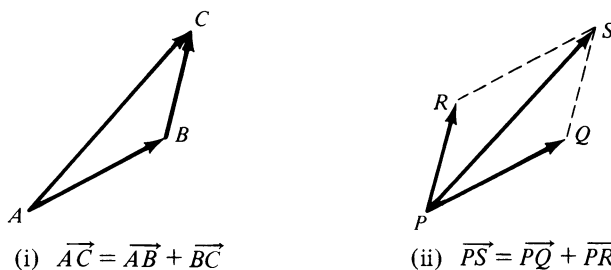


FIGURE 14.3

Another way to find the sum of two vectors is to consider vectors that are equal to the given ones and have the same initial point, as illustrated by $\overline{PQ}$ and $\overline{PR}$ in (ii) of Figure 14.3. If we construct the parallelogram $RPQS$, then since $\overline{PR} = \overline{QS}$, it follows that $\overline{PS} = \overline{PQ} + \overline{PR}$. If $\overline{PQ}$ and $\overline{PR}$ are two forces acting at P , then $\overline{PS}$ is the **resultant force**, that is, the single force that produces the same effect as the two combined forces.

If c is a real number and $\overline{AB}$ is a vector, then $c\overline{AB}$ is defined as a vector whose magnitude is $|c|$ times the magnitude of $\overline{AB}$ and whose direction is the same as $\overline{AB}$ if $c > 0$, and opposite that of $\overline{AB}$ if $c < 0$. Geometric illustrations are given in Figure 14.4. We refer to c as a **scalar** and $c\overline{AB}$ as a **scalar multiple** of $\overline{AB}$.

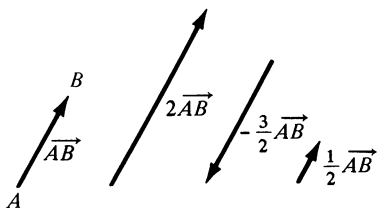
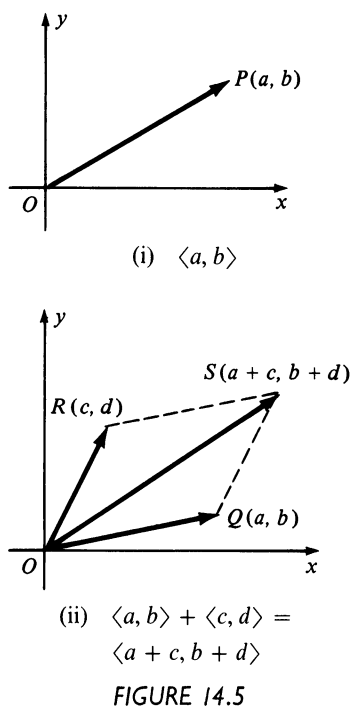


FIGURE 14.4



Let us next introduce a coordinate plane and assume that all vectors under discussion are in that plane. Since the position of a vector may be changed, provided the magnitude and direction are not altered, we may place the initial point of each vector at the origin. The terminal point of a typical vector $\overrightarrow{OP}$ may then be assigned coordinates (a, b) , as shown in (i) of Figure 14.5. Conversely, every ordered pair (a, b) determines the vector $\overrightarrow{OP}$, where P has coordinates (a, b) . We thus obtain a one-to-one correspondence between vectors and ordered pairs. This allows us to regard a vector in a plane as an ordered pair of real numbers instead of a directed line segment. To avoid confusion with the notation for open intervals or points, we shall use the symbol $\langle a, b \rangle$ for an ordered pair that represents a vector.

If, as in (ii) of Figure 14.5, we consider vectors $\overrightarrow{OQ}$ and $\overrightarrow{OR}$ corresponding to $\langle a, b \rangle$ and $\langle c, d \rangle$, respectively, and if we let $\overrightarrow{OS}$ be the vector corresponding to $\langle a + c, b + d \rangle$, it is easy to show, using slopes, that O, Q, S , and R are vertices of a parallelogram. It follows that $\overrightarrow{OQ} + \overrightarrow{OR} = \overrightarrow{OS}$ or, in terms of ordered pairs,

$$\langle a, b \rangle + \langle c, d \rangle = \langle a + c, b + d \rangle.$$

Although we shall not prove it at this time, the ordered-pair formula that corresponds to a scalar multiple $k\overrightarrow{OP}$ of the vector in (i) of the figure is

$$k\langle a, b \rangle = \langle ka, kb \rangle.$$

In this way, vectors take on a dual role, with the algebraic viewpoint of ordered pairs emerging from the previous work involving geometry. It is possible to reverse the order of this development and *begin* with an algebraic definition of vectors and *then* obtain directed line segments as an alternate interpretation. There are several advantages to this approach. One is that proofs of properties of vectors become simple exercises in algebraic manipulations instead of problems in geometry. Another advantage is that the algebraic approach to vectors is easily generalized to vectors in three or more dimensions. With these remarks in mind we shall next state an algebraic definition for vectors. Geometric representations that agree with our previous discussion will be introduced after the definition. We shall continue to use $\langle x, y \rangle$ for an ordered pair that refers to a vector. Such ordered pairs will also be denoted by symbols such as $\mathbf{a}$ or $\vec{a}$. As before, real numbers will be called *scalars*.

Definition (14.1)

The **two-dimensional vector space** V_2 is the set of all ordered pairs $\langle x, y \rangle$ of real numbers, called **vectors**, subject to the following axioms.

- (i) **Addition of vectors.** If $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$ are vectors, then

$$\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2 \rangle.$$

- (ii) **Multiplication of vectors by scalars.** If $\mathbf{a} = \langle a_1, a_2 \rangle$ and c is a scalar, then

$$c\mathbf{a} = \langle ca_1, ca_2 \rangle.$$

It should be clear that (i) and (ii) of Definition (14.1) were motivated by our representation of directed line segments as ordered pairs. The numbers a_1 and a_2 in $\langle a_1, a_2 \rangle$ are called the **components** of the vector. Thus, to add two vectors we add corresponding components.

The **zero vector** $\mathbf{0}$ and the **negative** $-\mathbf{a}$ of a vector $\mathbf{a} = \langle a_1, a_2 \rangle$ are defined as follows:

$$(14.2) \quad \mathbf{0} = \langle 0, 0 \rangle; \quad -\mathbf{a} = -\langle a_1, a_2 \rangle = \langle -a_1, -a_2 \rangle.$$

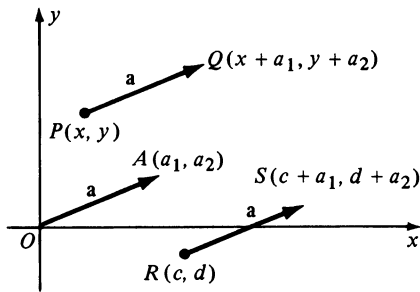


FIGURE 14.6

A nonzero vector $\mathbf{a} = \langle a_1, a_2 \rangle$ will be represented in a coordinate plane by a directed line segment $\overline{PQ}$ with any initial point $P(x, y)$ and terminal point $Q(x + a_1, y + a_2)$. Several representations are shown in Figure 14.6, where the symbol $\mathbf{a}$ is placed next to any directed line segment that represents the vector $\mathbf{a}$ of V_2 . The zero vector $\mathbf{0} = \langle 0, 0 \rangle$ will be represented by any point in the plane. Strictly speaking, a directed line segment $\overline{PQ}$ represents a vector; however, we shall also refer to $\overline{PQ}$ as a *vector*. It should always be clear from the discussion whether the term “vector” refers to an ordered pair or a directed line segment.

The geometric representation for the sum of two vectors $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$ is illustrated in Figure 14.7, where we have chosen an arbitrary point $P(x, y)$ for the initial point of $\mathbf{a}$.

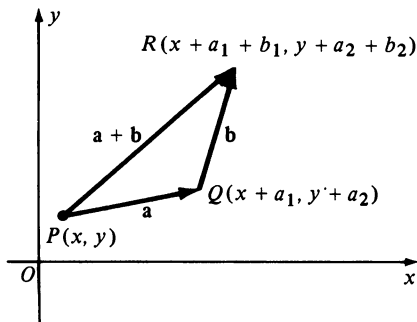


FIGURE 14.7

Example 1 If $\mathbf{a} = \langle -1, 3 \rangle$ and $\mathbf{b} = \langle 4, 2 \rangle$,

- (a) find $\mathbf{a} + \mathbf{b}$, $-\frac{3}{2}\mathbf{b}$, and $2\mathbf{a} + 3\mathbf{b}$.
 (b) represent $\mathbf{a}$, $\mathbf{b}$, $\mathbf{a} + \mathbf{b}$, and $-\frac{3}{2}\mathbf{b}$ geometrically.

Solution (a) Applying Definition (14.1),

$$\mathbf{a} + \mathbf{b} = \langle -1, 3 \rangle + \langle 4, 2 \rangle = \langle -1 + 4, 3 + 2 \rangle = \langle 3, 5 \rangle$$

$$-\frac{3}{2}\mathbf{b} = -\frac{3}{2}\langle 4, 2 \rangle = \langle -6, -3 \rangle$$

$$2\mathbf{a} + 3\mathbf{b} = 2\langle -1, 3 \rangle + 3\langle 4, 2 \rangle = \langle -2, 6 \rangle + \langle 12, 6 \rangle = \langle 10, 12 \rangle.$$

- (b) The vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{a} + \mathbf{b}$, and $-\frac{3}{2}\mathbf{b}$ are represented geometrically in Figure 14.8, where we have chosen the origin for the initial point of each vector. Note that $-\frac{3}{2}\mathbf{b}$ has a direction opposite to that of $\mathbf{b}$ and is $\frac{3}{2}$ as long.

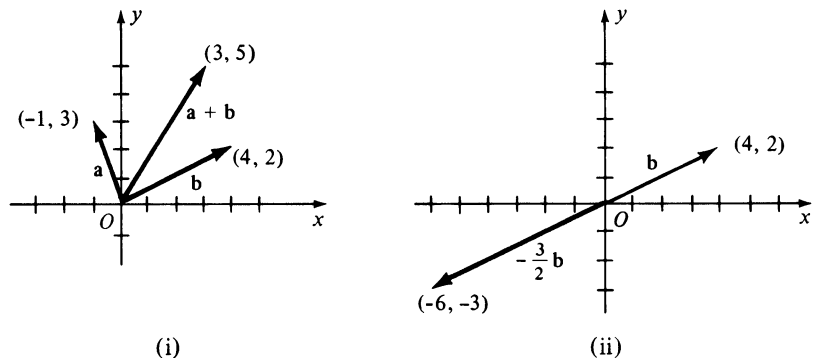


FIGURE 14.8

Definition (14.3)

The **magnitude** $|\mathbf{a}|$ of a vector $\mathbf{a} = \langle a_1, a_2 \rangle$ is $|\mathbf{a}| = \sqrt{a_1^2 + a_2^2}$.

It follows from the Distance Formula that $|\mathbf{a}|$ is the length of any directed line segment used to represent $\mathbf{a}$. Note also that $|\mathbf{a}| \geq 0$, where $|\mathbf{a}| = 0$, if and only if $\mathbf{a} = \mathbf{0}$. The symbol for the magnitude of a vector should not be confused with the symbol $|c|$ for the absolute value of a real number c . As an illustration, if $\mathbf{a} = \langle 3, -2 \rangle$, then

$$|\mathbf{a}| = |\langle 3, -2 \rangle| = \sqrt{9 + 4} = \sqrt{13}.$$

If P is any point in a coordinate plane and O is the origin, then $\overrightarrow{OP}$ is called the **position vector** of P . If $\mathbf{a} = \langle a_1, a_2 \rangle$, then the position vector $\overrightarrow{OA}$ of the point $A(a_1, a_2)$ (see (i) of Figure 14.6) is called the **position vector corresponding to $\mathbf{a}$** . Although $\mathbf{a}$ has many geometric representations, there is precisely one position vector corresponding to $\mathbf{a}$.

Theorem (14.4)

If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are points, then the vector $\mathbf{a}$ in V_2 that has geometric representation $\overrightarrow{P_1P_2}$ is

$$\mathbf{a} = \langle x_2 - x_1, y_2 - y_1 \rangle.$$

Proof If $P_1(x_1, y_1)$ is the initial point of a vector corresponding to $\mathbf{a} = \langle a_1, a_2 \rangle$, then the terminal point is $Q(x_1 + a_1, y_1 + a_2)$. If Q is to coincide with $P_2(x_2, y_2)$, then we must have $x_2 = x_1 + a_1$ and $y_2 = y_1 + a_2$. Consequently, $\mathbf{a} = \langle x_2 - x_1, y_2 - y_1 \rangle$. One illustration of $\overrightarrow{P_1P_2}$ is shown in Figure 14.9, together with the position vector corresponding to $\mathbf{a}$.

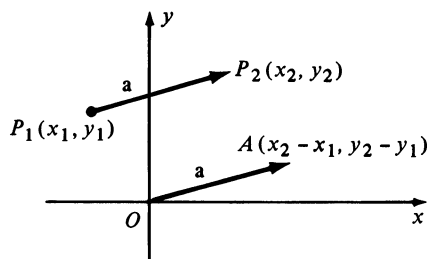


FIGURE 14.9

□

Example 2 Given the points $P(-2, 3)$ and $Q(4, 5)$, find vectors $\mathbf{a}$ and $\mathbf{b}$ that have geometric representations $\overrightarrow{PQ}$ and $\overrightarrow{QP}$, respectively. Sketch $\overrightarrow{PQ}$, $\overrightarrow{QP}$, and the position vectors corresponding to $\mathbf{a}$ and $\mathbf{b}$.

Solution Applying Theorem (14.4), the vectors $\mathbf{a}$ and $\mathbf{b}$ are

$$\mathbf{a} = \langle 4 - (-2), 5 - 3 \rangle = \langle 6, 2 \rangle,$$

$$\mathbf{b} = \langle -2 - 4, 3 - 5 \rangle = \langle -6, -2 \rangle.$$

The sketches are shown in Figure 14.10. Observe that $\mathbf{b} = -\mathbf{a}$.

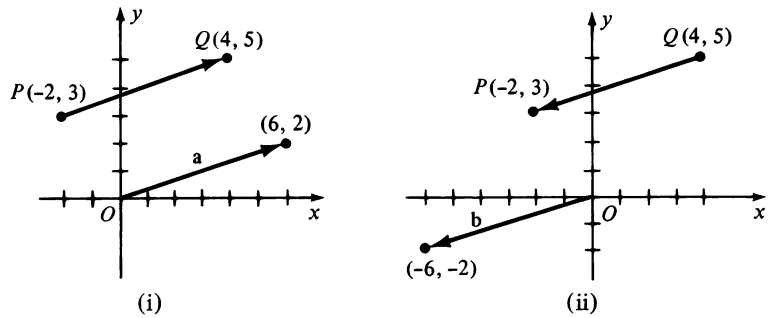


FIGURE 14.10

Four important properties are stated in the next theorem, where $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are any vectors in V_2 .

Theorem (14.5)

- (i) $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$
- (ii) $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$
- (iii) $\mathbf{a} + \mathbf{0} = \mathbf{a}$
- (iv) $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$

Proof For part (i) let $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$. Since $a_1 + b_1 = b_1 + a_1$ and $a_2 + b_2 = b_2 + a_2$,

$$\mathbf{a} + \mathbf{b} = \langle a_1 + b_1, a_2 + b_2 \rangle = \langle b_1 + a_1, b_2 + a_2 \rangle = \mathbf{b} + \mathbf{a}.$$

The remainder of the proof is left as an exercise. $\square$

The operation of **subtraction** of vectors, denoted by $-$, is defined as follows.

Definition (14.6)

Let $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$. The **difference** $\mathbf{a} - \mathbf{b}$ is

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}) = \langle a_1 - b_1, a_2 - b_2 \rangle.$$

The last equality in (14.6) is true because $-\mathbf{b} = \langle -b_1, -b_2 \rangle$. Thus, to find $\mathbf{a} - \mathbf{b}$ we merely subtract the components of $\mathbf{b}$ from the corresponding components of $\mathbf{a}$.

Example 3 If $\mathbf{a} = \langle 5, -4 \rangle$ and $\mathbf{b} = \langle -3, 2 \rangle$, find $\mathbf{a} - \mathbf{b}$ and $2\mathbf{a} - 3\mathbf{b}$.

Solution

$$\mathbf{a} - \mathbf{b} = \langle 5, -4 \rangle - \langle -3, 2 \rangle = \langle 5 - (-3), -4 - 2 \rangle = \langle 8, -6 \rangle$$

$$2\mathbf{a} - 3\mathbf{b} = 2\langle 5, -4 \rangle - 3\langle -3, 2 \rangle = \langle 10, -8 \rangle - \langle -9, 6 \rangle = \langle 19, -14 \rangle$$

■

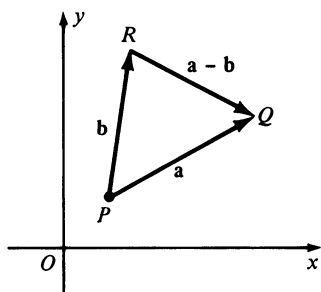


FIGURE 14.11

It is easy to show that if $\mathbf{a}$ and $\mathbf{b}$ are arbitrary vectors, then

$$\mathbf{b} + (\mathbf{a} - \mathbf{b}) = \mathbf{a},$$

that is, $\mathbf{a} - \mathbf{b}$ is the vector that, when added to $\mathbf{b}$, gives us $\mathbf{a}$. This fact leads to a simple geometric interpretation for $\mathbf{a} - \mathbf{b}$. Specifically, if we represent $\mathbf{a}$ and $\mathbf{b}$ by vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ with the same initial point, as illustrated in Figure 14.11, then $\overrightarrow{RQ}$ represents $\mathbf{a} - \mathbf{b}$.

At the beginning of this section the symbol $c\overrightarrow{AB}$ was used to denote a vector having the same direction as $\overrightarrow{AB}$ if $c > 0$ or the opposite direction if $c < 0$. The next definition is the analogue of this concept for vectors in V_2 .

Definition (14.7)

Nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ of V_2 have

- (i) the **same direction** if $\mathbf{b} = c\mathbf{a}$ for some scalar $c > 0$.
- (ii) **opposite directions** if $\mathbf{b} = c\mathbf{a}$ for some scalar $c < 0$.

It is not difficult to prove that if $\mathbf{b} = c\mathbf{a}$, then position vectors corresponding to $\mathbf{a}$ and $\mathbf{b}$ lie on the same line through the origin. Moreover, if $c > 0$, then the terminal points A and B of the position vectors lie in the same quadrant (or on the same positive or negative coordinate axis). If $c < 0$, and A is not on a coordinate axis, then A and B lie in diagonally opposite quadrants.

It is convenient to assume that the zero vector $\mathbf{0}$ has *all* directions.

We say that two vectors $\mathbf{a}$ and $\mathbf{b}$ are **parallel** if and only if $\mathbf{b} = c\mathbf{a}$ for some scalar c . In particular, nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel if they have the same, or opposite, directions. The next theorem brings out the relationship between the magnitudes of $\mathbf{a}$ and $c\mathbf{a}$.

Theorem (14.8)

If $\mathbf{a}$ is a vector and c is a scalar, then $|c\mathbf{a}| = |c||\mathbf{a}|$.

Proof If $\mathbf{a} = \langle a_1, a_2 \rangle$, then $c\mathbf{a} = \langle ca_1, ca_2 \rangle$. Using Definition (14.3) and properties of real numbers,

$$\begin{aligned} |c\mathbf{a}| &= \sqrt{(ca_1)^2 + (ca_2)^2} = \sqrt{c^2(a_1^2 + a_2^2)} \\ &= \sqrt{c^2} \sqrt{a_1^2 + a_2^2} = |c||\mathbf{a}|. \end{aligned}$$

□

Theorem (14.8) implies that the length of a directed line segment that represents $c\mathbf{a}$ is $|c|$ times the length of a directed line segment that represents $\mathbf{a}$. This agrees with the geometric interpretation discussed at the beginning of this section and illustrated in Figure 14.4.

Some properties of scalar multiples of vectors in V_2 are stated in the following theorem.

Theorem (14.9)

- (i) $c(\mathbf{a} + \mathbf{b}) = c\mathbf{a} + c\mathbf{b}$
- (ii) $(c + d)\mathbf{a} = c\mathbf{a} + d\mathbf{a}$
- (iii) $(cd)\mathbf{a} = c(d\mathbf{a}) = d(c\mathbf{a})$
- (iv) $1\mathbf{a} = \mathbf{a}$
- (v) $0\mathbf{a} = \mathbf{0} = c\mathbf{0}$

Proof We shall prove (i) and leave proofs of the remaining properties as exercises. Letting $\mathbf{a} = \langle a_1, a_2 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$, we have

$$\begin{aligned}
 c(\mathbf{a} + \mathbf{b}) &= c\langle a_1 + b_1, a_2 + b_2 \rangle \\
 &= \langle ca_1 + cb_1, ca_2 + cb_2 \rangle \\
 &= \langle ca_1, ca_2 \rangle + \langle cb_1, cb_2 \rangle \\
 &= c\mathbf{a} + c\mathbf{b}.
 \end{aligned}$$

□

There should be no difficulty in remembering the properties in Theorem (14.9) since they resemble familiar properties of real numbers.

The following special vectors $\mathbf{i}$ and $\mathbf{j}$ will be important in future developments.

$$(14.10) \quad \mathbf{i} = \langle 1, 0 \rangle, \quad \mathbf{j} = \langle 0, 1 \rangle$$

We may use $\mathbf{i}$ and $\mathbf{j}$ to obtain another way of denoting vectors in two dimensions, as indicated by the next theorem.

Theorem (14.11)

If $\mathbf{a} = \langle a_1, a_2 \rangle$ is any vector in V_2 , then $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$.

Proof Using (i) and (ii) of Definition (14.1),

$$\langle a_1, a_2 \rangle = \langle a_1, 0 \rangle + \langle 0, a_2 \rangle = a_1\langle 1, 0 \rangle + a_2\langle 0, 1 \rangle = a_1\mathbf{i} + a_2\mathbf{j}. \quad \square$$

The formula $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$ for the vector $\mathbf{a} = \langle a_1, a_2 \rangle$ has an interesting geometric interpretation. Position vectors corresponding to $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{a}$ are

illustrated in (i) of Figure 14.12. Since $\mathbf{i}$ and $\mathbf{j}$ are unit vectors, $a_1\mathbf{i}$ and $a_2\mathbf{j}$ may be represented by horizontal and vertical vectors of magnitudes $|a_1|$ and $|a_2|$, respectively, as illustrated in (ii) of Figure 14.12. The position vector for $\mathbf{a}$ may be regarded as the sum of these vectors. For this reason a_1 is called the **horizontal component** and a_2 the **vertical component** of the vector $\mathbf{a}$.

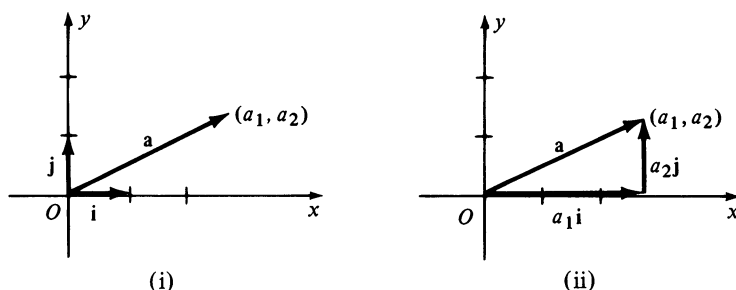


FIGURE 14.12

The vector sum $a_1\mathbf{i} + a_2\mathbf{j}$ is called a **linear combination** of $\mathbf{i}$ and $\mathbf{j}$. Note that if $\mathbf{b} = \langle b_1, b_2 \rangle = b_1\mathbf{i} + b_2\mathbf{j}$, and if c is a scalar, then

$$(a_1\mathbf{i} + a_2\mathbf{j}) + (b_1\mathbf{i} + b_2\mathbf{j}) = (a_1 + b_1)\mathbf{i} + (a_2 + b_2)\mathbf{j},$$

$$(a_1\mathbf{i} + a_2\mathbf{j}) - (b_1\mathbf{i} + b_2\mathbf{j}) = (a_1 - b_1)\mathbf{i} + (a_2 - b_2)\mathbf{j},$$

and

$$c(a_1\mathbf{i} + a_2\mathbf{j}) = (ca_1)\mathbf{i} + (ca_2)\mathbf{j}.$$

These formulas show that linear combinations of $\mathbf{i}$ and $\mathbf{j}$ may be regarded as ordinary algebraic sums.

Example 4 If $\mathbf{a} = 5\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = 4\mathbf{i} - 7\mathbf{j}$, express $3\mathbf{a} - 2\mathbf{b}$ as a linear combination of $\mathbf{i}$ and $\mathbf{j}$.

Solution

$$\begin{aligned} 3\mathbf{a} - 2\mathbf{b} &= 3(5\mathbf{i} + \mathbf{j}) - 2(4\mathbf{i} - 7\mathbf{j}) \\ &= (15\mathbf{i} + 3\mathbf{j}) - (8\mathbf{i} - 14\mathbf{j}) \\ &= 7\mathbf{i} + 17\mathbf{j} \end{aligned}$$

■

A **unit vector** is a vector of magnitude 1. The vectors $\mathbf{i}$ and $\mathbf{j}$ are unit vectors.

Theorem (14.12)

If $\mathbf{a}$ is a nonzero vector, then a unit vector $\mathbf{u}$ having the same direction as $\mathbf{a}$ is

$$\mathbf{u} = \frac{1}{|\mathbf{a}|} \mathbf{a}.$$